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SECTION 01 11 00 SUMMARY OF WORK

PART 1 GENERAL

1.01 BACKGROUND

- A. The Bradbury Dam Outlet Works (Outlet Works) was constructed to release storm water flows and prior water rights flows for downstream users. The outlet works arrangement utilizes a diversion tunnel running through the Monterey formation of the left abutment of the dam. Water is conveyed through a concrete-lined tunnel before transitioning to a 38-inch diameter pipe at the approximate midpoint of the outlet line. Once at the Outlet Works, the water proceeds through a trifurcation and is discharged by two 30-inch fixed-cone valves and a 10-inch butterfly valve. The Outlet Works has been positioned so that discharges from the valves are released into Bradbury Dam's stilling basin.

1.02 LOCATION

- A. Work is located at Bradbury Dam approximately 20 miles northwest of Santa Barbara, California in Santa Barbara County.

1.03 PRINCIPAL COMPONENTS OF WORK

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- A. The work consists of removing and replacing fixed-cone valves and related equipment. Principal components of the work include:
1. Electrical inspection, design, and installation:
 - a. Inspect the power system.
 - b. Complete electrical design based on provided baseline and inspection results. Provide design documentation signed by a U.S. registered Electrical Engineer.
 - c. Perform demolition and installation of electrical equipment.
 - d. Electrical equipment to include panels, fuses, conduit, and conductors.
 2. Remove two (2) existing fixed-cone valves and the following peripheral equipment:
 - a. Valve supports including anchors.
 - b. Pipe spool piece.
 - c. Valve actuators.
 - d. Conductors, cables, and conduit associated with the valve actuators.
 3. Design two (2) valve supports for new fixed-cone valves. Design shall be completed and stamped by a California-registered Professional Engineer (PE).

4. Furnish, install, and test two (2) fixed-cone valves with hoods and the following peripheral equipment:
 - a. Butterfly valves
 - b. Reducer or valve supports grouted and anchored to concrete.
 - c. Reducers.
 - d. Valve actuators.
 - e. Conductors, cables, conduit and grounding associated with the valve actuators.
5. Provide temporary pump to supply water from Lake Cachuma to the Bradbury Dam Spillway during the Outlet Works Outage of the project.

1.04 SPECIFICATIONS REQUIREMENTS

- A. Requirements in Division 1, General Requirements, apply to Divisions 2 through 52.
- B. Imperative statements in these specifications are Contractor requirements, unless otherwise stated.
- C. Where specifications are written in streamlined form, words “shall be” are included by inference where a colon (:) is used within sentence or phrase.

1.05 DEFINITIONS

- A. When specifications use a word or term defined in Federal Acquisition Regulations (FAR), definition of the word or term shall be in accordance with FAR sections in effect at the time solicitation was issued.
- B. A glossary of common terms used by the Bureau of Reclamation is located on the Reclamation website: <https://www.usbr.gov/library/glossary/index.html>
- C. Government: Bureau of Reclamation.
- D. Transferred Works: Facilities owned by the Bureau of Reclamation but operated and maintained by an irrigation district or other entity.
- E. Quality Assurance: Inspection and tests performed by Government to ensure compliance with the terms of the contract.
- F. Quality Control: Activities performed by Contractor to ensure work conforms to contract requirements.
- G. Contractor Quality Testing: Specified tests to be performed by the Contractor.
 1. The Government may use the test results for Quality Assurance.
 2. The Contractor may use the test results as part of Contractor quality control.

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- a. The Government anticipates that these tests will be part of the Contractor's quality control program, however the tests do not relieve the Contractor of providing adequate quality control measures in accordance with the clause at FAR 52.246-12, Inspection of Construction.

1.06 ACRONYMS

A. The following acronyms apply to specifications Divisions 1 through 52:

1. CO: Contracting Officer.
2. COR: Contracting Officer's Representative.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

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SECTION 01 14 10 USE OF SITE

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. Include in prices offered in the Price Schedule for other items of work.

1.02 REFERENCE STANDARDS

A. Bureau of Reclamation (USBR)

1. Cleaning Manual-2012

Inspection and Cleaning Manual for
Equipment and Vehicles to Prevent the
Spread of Invasive Species (Technical
memorandum No. 86-68220-07-05) 2012
Edition

Available online at:

<http://www.usbr.gov/musick/prevention>

1.03 SUBMITTALS

A. Submit the following in accordance with Section 01 33 00 - Submittals.

B. RSN 01 14 10-1, Land Use and Landscape Rehabilitation Plan:

1. For each Contractor use site on Government land.
 - a. Show use location and extent of impact. Uses include but are not limited to the following:
 - 1) Buildings and service areas including offices, shops, warehouses, storage areas, fuel and oil storage areas, and fabrication yards.
 - 2) Parking areas, temporary roads, and haul routes.
 - 3) Utilities including air, power, and water lines; fire hydrants; and compressor station.
 - 4) First-aid and medical facilities.
 - 5) Areas for processing, storing, and disposing of waste materials from construction operations.
 - 6) Temporary fences.
 - b. Describe methods to preserve, protect, and repair, vegetation (such as trees, shrubs, and grass) and other landscape features on or adjacent to

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- jobsite, which are not to be removed and which do not interfere with work required under this contract. Include methods to mark work area limits, protect disturbed areas, and prevent erosion.
- c. Describe methods to protect, and repair if damaged, existing improvements and utilities at or near jobsite.
 - d. Describe methods for removing temporary structures and facilities, cleanup, and rehabilitating site after completion of construction activities.
2. Submit revised drawings of changes in use of Government land made during design and erection stages or after use of Government land is in operation.

1.04 PROJECT CONDITIONS

- A. Government land as shown on drawing 368-208-60010, and referenced in Note 5 of the drawing, may be used for required construction facilities.
 - 1. There is limited space available for lay down and parking at the Outlet Works Control House work area. Additional space is available at Auxiliary Lay Down Areas.
- B. When private land is used for construction facilities, or other construction purposes, make necessary arrangements associated with use of private land.
- C. Location, construction, operation, maintenance, and removal of construction facilities on Government land will be subject to approval of COR.
- D. Do not interfere with work of other contractors or the Government in vicinity, or with reservations made by the Government for use of such land.
- E. Housing for construction personnel will not be permitted on Government land, except housing for guards or watchmen as may be approved by COR.

PART 2 PRODUCTS

2.01 TEMPORARY FENCING

- A. Fencing used for construction shall not have fence posts that penetrate the ground.

PART 3 EXECUTION

3.01 CLEANING

- A. Construction equipment:

1. Before bringing on site, clean construction equipment to remove dirt, vegetation, and other organic material to prevent introduction of noxious weeds, and invasive plant and animal species.
2. Contractor cleaning procedures shall result in equipment being cleaned as well or better than the procedures described in Cleaning Manual.
3. COR will inspect construction equipment following procedures described in Cleaning Manual before allowing equipment onsite.

3.02 RESTORATION

- A. Restore Contractor use areas to pre-construction condition.
- B. After completion of work, regrade and scarify Government land used for construction purposes and not required for completed installation so that surfaces blend with natural terrain and are in a condition that will facilitate revegetation, provide proper drainage, and prevent erosion.

END OF SECTION

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SECTION 01 14 30 INTERRUPTION OF SERVICE

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include cost in prices offered in the Price Schedule for items of work for which interruption of services are required.

1.02 SUBMITTALS

- A. Submit in accordance with Section 01 33 00 - Submittals.
- B. RSN 01 14 30-1, Interruption of Service Request:
1. Area where work will be done (by number or other description).
 2. Outline procedures for accomplishing work including:
 - a. Specific safety precautions to be taken.
 - b. Type and location of barricades.
 - c. Warning signs.
 - d. Protective grounds and devices to be used.
 3. Commencement time of work.
 4. Duration of work.
 5. Number of personnel and their classification.
 6. Description of equipment to be used.
 7. Information indicating that required materials are on site or data indicating shipping dates of materials not on site.

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1.03 PROJECT CONDITIONS

- A. Coordinate and schedule interruption of services with COR:
1. Coordinate work with Government operations.
 2. No specific interruption of service request will be considered unless:
 - a. Request is received at least 30 days prior to the beginning of interrupted service.
 - b. COR has reasonable assurance that materials and equipment required for work will be onsite.

- c. Contractor will be prepared to perform work on date and during period of time requested for specific interruption of service.
- 3. Allowable time and available interruption of service periods will govern interruption of service requests, work scheduling, onsite delivery of materials, and required drawings and data submittals.
- 4. Interruption of service window available to Contractor: Government will unwater penstocks to accommodate Contractor-requested and approved interruption of service only during the following interval.
 - a. December 1 through March 31.
- 5. Facility shall be returned to service within 30 days' notice beginning April 1.
- B. Work shall not begin until all materials and equipment have been received and accepted onsite.
- C. When work cannot be performed during an approved interruption of service period, notify COR that interruption of service is not required.
- D. Show construction interruption of services in construction schedule.

PART 2 PRODUCTS

Not Used

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PART 3 EXECUTION

Not Used

END OF SECTION

SECTION 01 31 22 COORDINATION MEETINGS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost

1. Include in prices offered in the Price Schedule for other items of work.

1.02 COORDINATION MEETINGS

A. Weekly Coordination Meetings:

1. Contractor shall conduct weekly coordination meeting with Government.
2. Purpose: Review work progress, performance, issues, and contract administration with Government representatives.
3. Contractor Attendees: Contractor project manager, superintendent, and other representatives, as needed, who understand the work, schedule, technical requirements, components, and contract administration.
4. The meeting will be convened and led by the Contractor to discuss and provide updates for the following:
 - a. Work Schedule
 - b. Safety and Weekly Safety Walk
 - c. Outages
 - d. Security
 - e. Personnel
 - f. Contract Administration:
 - 1) Modifications
 - 2) Requests for Equitable Adjustment (REAs). (Contractor generated document).
 - 3) Requests for Changes (RFCs). (Contractor generated document).
 - 4) Requests for Information (RFIs). (Contractor generated document).
 - 5) Requests for Proposal (RFPs). (Contractor generated document).
 - 6) Requests for Substitutions (RFSs). (Contractor generated document).
 - 7) Required Submittals

- 8) Quantities and Invoices
- 5. Other outstanding issues.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 IMPLEMENTATION

- A. Contractor shall coordinate meeting time, place, invite list, notifications, and teleconference access with the COR.
- B. Contractor shall prepare and distribute agendas for each meeting.
- C. Contractor shall compile and distribute meeting minutes after each meeting.

END OF SECTION

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SECTION 01 31 30
CONTRACT DOCUMENT MANAGEMENT SYSTEM

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Contract Document Management System:
1. Payment: Lump sum price offered in the Price Schedule.

1.02 DEFINITIONS

- A. CDMS: Contract Document Management System.
1. Contract Documents include but are not limited to:
 - a. Specifications.
 - b. Contract drawings.
 - c. Submittals.
 - d. Approval Drawings (Contractor generated document)
 - e. Shop Drawings (Contractor generated document)
 - f. Final Drawings (Contractor generated document)
 - g. As-built Drawings (Contractor generated document)
 - h. REA: Request for Equitable Adjustment (Contractor generated document).
 - i. RFC: Request for Change (Contractor generated document).
 - j. RFI: Request for Information (Contractor generated document).
 - k. RFP: Request for Proposal (Government generated document).
 - l. RFS: Request for Substitution (Contractor generated document).
 - m. CM: Contract modification.
 - n. Value engineering proposals.
 - o. Government inspection reports.
 - p. Contract schedules.
 - q. Meeting agendas and meeting minutes.
 - r. Letters, memos, and other typical contract correspondence.

1.03 SUBMITTALS

- A. Submit in accordance with Section 01 33 00 - Submittals.

- B. RSN 01 31 30-1, Approval Data:
 - 1. System provider experience, number of systems operating, references.
 - 2. Documentation of system capabilities.
 - 3. Instructions for system use.
- C. RSN 01 31 30-2, Final Data:
 - 1. DVD or CD of documents uploaded to CDMS.

1.04 QUALIFICATIONS

- A. System Provider: At least 5 years successful performance in providing required services.

1.05 CDMS REQUIREMENTS

- A. CDMS shall be capable of generating, storing, tracking, categorizing, and managing Contract Documents.
- B. Access:
 - 1. Web-Based:
 - a. Contract Documents are stored and accessed by authorized individuals via an internet site.
 - b. Compatible with web browsers MS Internet Explorer, Apple Safari, Mozilla Firefox, and Google Chrome.
 - 2. Ability to upload Contract Documents and make available for user download in supported file formats, minimum:
 - a. Adobe Acrobat.
 - b. Microsoft Word.
 - c. Microsoft Excel.
 - d. Microsoft Project.
 - e. Autodesk AutoCAD Civil 3D.
 - 3. Allow Contract Documents to be prepared by the Contractor or by Government.
 - a. Preparing organization will control access of documents. See Document Security and Backup paragraph below.
 - 4. Contractor and Government shall each have a project manager for CDMS.
 - a. Each project manager shall have capabilities to assign users within their organization and to assign user rights which control access to documents based on user class and document type.
 - b. Each organization shall be capable of defining its access hierarchy.

- c. Inclusion of users will be prerogative of organizational project manager.
 - d. Government shall be able to create and modify project properties (i.e. Contractor, project location, description, bid amount, project directory, etc.).
- 5. Email notifications shall be automatically sent to selectable users when new documents are submitted and available for viewing, or alerts are generated.
Examples:
 - a. Updates to project information (e.g. entry of a new submittal);
 - b. Changes to project information (e.g. approval of a submittal);
 - c. Alerts (e.g. submittal under review for greater than 15 days without action).
- 6. Ability to link files; examples:
 - a. RFP linked to a CM.
 - b. Submittal linked to a specification section or drawing.
- C. Reports:
 - 1. Generate reports that list and sort documents by status; examples include:
 - a. Submittals that have been approved.
 - b. RFIs under review.
 - 2. Generate alerts when documents requiring action approach or exceed allowable time.
 - 3. Generate customizable summary reports; examples include:
 - a. Submittal turnaround.
 - b. RFI turnaround.
 - 4. Reports shall be printable, exportable as searchable Portable Document Format (PDF) or printer-friendly HTML, and exportable to Excel, XML, or CSV.
 - 5. Ability to link from an item in a report directly to item by clicking on it. For example, in a submittal report, clicking on a submittal that is 15 days old takes you directly to information on that submittal.
 - 6. Capable of producing a complete and logically organized set of documents within CDMS in both PDF and XML format.
- D. Document Security and Backup:
 - 1. Secure Sockets Layer (SSL) encryption for secured data exchange between browser and server (secure access to documents and information).
 - 2. Allow document access security until authoring party chooses to share it with others.

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- a. Show parties that have access to a document.
 - b. Extent of document sharing shall be determined by authoring party.
 - c. Once documents are shared, track documents so that changes cannot occur without a record of changes.
 3. Backup/archiving to File Transfer Protocol (FTP) site of documents to which an organization has access on a regular (minimum monthly) basis. Backups shall be in PDF format.
 4. Documents shall be downloadable to FTP sites where Government and other organizations can each access their own data confidentially.
- E. General Tracking and Control:
1. Documents shall be grouped into categories (Submittals, RFC, RFI, RFP, Daily Inspection Reports, Meeting Minutes).
 2. CDMS shall track:
 - a. When documents were received or returned, as well as status of documents (e.g. Under Review; or Returned, Approved, or Rejected; etc.).
 - b. Document changes including who entered or changed document and date change was made.
 - c. When a reviewer has opened an item.
- F. Submittal Tracking and Control:
1. Ability to create master list of submittals (“Schedule of Submittals”).
 2. Supply standard submittal form for entering information. CDMS shall allow customizing of standard submittal form. Contractor shall be able to enter submittal summary information (required submittal number, title, description, due date, and specification section shall be required fields) and then upload submittal data in electronic format.
 3. Track date received, date response due, date returned, and approval status.
 - a. Status designations may be modified to fit Government standard designations A/CA/R/NA/Acc/CAcc/NAcc (Approved/Conditionally Approved, Resubmit/Not Approved, Accepted, Conditionally Accepted, Not Accepted).
 4. Supply Submittal Tracking form for Government to enter review comments and action taken. CDMS shall allow customizing of submittal tracking form.
 - a. Allow for Government to route submittal to third parties by generating design-review transmittals with each submittal package item, to track status of individual sub-items within submittal package, and to route each sub-item to appropriate reviewers.
 5. Ability to generate automatic notification after specified days without response.

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G. RFI Tracking and Control:

1. Supply standard RFI form to enter information.
2. Ability to create or attach documents to the RFI form or links to other documents within CDMS.
3. Ability for Government to route RFI to appropriate reviewers and for reviewers to enter comments into standard response form.

H. Inspection Reports Tracking and Control:

1. Ability to attach Government standard inspection and report forms into CDMS and ability to create customizable inspection forms in CDMS. Typical forms:
 - a. Daily Inspection Report (DIR).
 - b. Survey requests.
 - c. Plant inspection reports.
 - d. Testing reports.
2. Ability to link or attach photos to forms and link forms to other documents within CDMS.

I. Contract Modifications:

1. Supply standard Contractor proposal form with ability to attach documents and link to other documents within CDMS.
2. Ability to attach documents to these forms and link to other documents within CDMS.

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J. Correspondence:

1. Ability to generate letters and memos.
 - a. Ability to attach documents to letters and memos, and to reference other documents within CDMS.
 - b. CDMS shall automatically assign sequential numbers to letters and memos.
2. Ability to generate responses to letters and memos.
 - a. System automatically generates memo suffixes, e.g. first response to Memo 32 to be 32.1, second response to be 32.2, etc.
3. Official correspondence from Government to Contractor shall contain Government's official letterhead and corresponding logos.
4. Ability to generate meeting agendas and meeting minutes and to attach documents to minutes. CDMS shall automatically assign sequential meeting numbers.

K. Miscellaneous Tracking and Controls. Ability to:

1. Generate reports tracking RFCs, RFIs, RFPs, Submittals, Memos, DIRs, survey requests, plant inspection reports, and testing reports.
2. Filter logs based on criteria:
 - a. Submittals: Specification Section, Days Out, Status (A/CA/R/NA/Acc/CAcc/NAcc).
 - b. RFIs: Days out, Status (Open/Closed), Specification Section/Dwg No., Responsibility/Ball-In-Court.
3. Sort logs based on a criterion, e.g. for Submittal, Specification Section, Days Out, Status, or other criterion determined by the COR.
4. Keyword search of documents in database.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 IMPLEMENTATION

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- A. The Contractor shall be the administrator of the CDMS site. The Contractor shall maintain and manage CDMS services directly with the software provider. All data and stored information shall belong to the Contractor who has sole responsibility for the CDMS solution.
 - B. The Contractor shall share access to CDMS site and data with the Government for Contract purposes as described herein.
 - C. In accordance with Reclamation Directive and Standard RCD 05-01, all information received, created, or compiled by Government officials, employees, contractors, and other staff (employees) is the property of the Federal Government and must be maintained in accordance with established standards. The CDMS shall not be considered as storage of official records or correspondence.
 - D. CDMS shall be in place and operating before Contractor transmits Contract Documents to Government, except RSN 01 31 30-1, Approval Data.
 1. CDMS operation includes COR, or other authorized Government representative, having program control to assign users and user rights to Government personnel to access appropriate areas of system.
 - E. Contractor shall input Schedule of Submittals before making submittals. Include:
 1. RSNs listed in Table 01 33 00A - List of Submittals.
 2. Scheduled submittal date.

- F. Contractor shall provide a report of submittal status monthly or as requested by COR. Submittal status report shall be in a table format and include:
1. Required Submittal Number (RSN).
 2. Section title.
 3. Submittal title.
 4. Date submitted.
 5. Date of response.
 6. Approval status (Approved/Conditionally Approved, Resubmit/Not Approved, Accepted, Conditionally Accepted, Not Accepted).
 7. Days in review.
 8. Due date.
- G. CDMS shall be operational until final project closeout.

END OF SECTION

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SECTION 01 32 10 CONSTRUCTION PROGRAM

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Costs:
1. Developing Baseline Schedule: Include in lump sum price offered in the Prices Schedule for Mobilization and Preparatory Work.
 2. Updating and Using Construction Program: Include as part of Contractor's overhead.

1.02 DEFINITIONS

- A. Schedule: The Critical Path Method (CPM) of planning and scheduling a construction project where activities are arranged based on activity relationships and network calculations determine when activities can be performed and the critical path of the project.
- B. Project Calendar(s): Cross reference of numerical work days with calendar days. Project calendars serve as the basis for day/date conversion and assign work days and non-workdays.
- C. Resources: Equipment, labor or crews, materials, subcontractors, fabricators, manufacturers, and consultants.
- D. Out of Sequence Work: An activity that starts before its predecessor activities are completed.

1.03 REFERENCE STANDARDS

- A. All references below are to be the current adopted edition at the time of solicitation for this contract. If any conflict arises between installing the new equipment in the existing installation, the contractor is to notify the COR. At which point, the CO will give direction on the course of action.
- B. Associated General Contractors of America (AGC)
1. AGC Manual Construction Planning and Scheduling Manual

1.04 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals.

1. Furnish database files in format compatible with Microsoft Project or Oracle Primavera P6 Professional Project Management.
 2. Furnish schedule and analysis data on CD-ROM disc.
 3. Upon request, provide information and data used to develop and maintain the Construction Program to the CO.
- B. RSN 01 32 10-1, Representative Information:
1. Designation of authorized representative to develop and maintain Construction Program. Include resume with training and experience.
- C. RSN 01 32 10-2, Baseline Schedule:
1. Include:
 - a. Construction Program/Schedule database.
 - b. Definition of project calendars.
 - c. Primavera layout file (.plf).
 - d. Gantt chart (Bar chart) for project on 11 by 17-inch print(s) and in portable document format (.pdf).
 - e. Activity report including logic constraints consisting of predecessors, successors, and constraint dates.
 - f. Tabular listings of resources and resource limitations used to produce baseline schedule. Correlate resources with schedule assumptions and offered price. Resource listings may be independent of schedule database.
 - g. Narrative explanation of project schedule development. Include impacts of resource limitations and weather conditions on project schedule development. Include planned number of work hours per day and hours per day for subcontractors.
- D. RSN 01 32 10-3, Updated Schedule Reports:
1. This submittal is required for a proper invoice for progress payments in accordance with the clauses at FAR 52.232-27 - Prompt Payment for Construction Contracts and WBR 1452.232-82 - Other Invoice Requirements. In accordance with subparagraph (a)(i)(A) of the clause at FAR 52.232-27 - Prompt Payment for Construction Contracts, Government disapproval of this submittal will be considered failure of the Contractor to comply with contract requirements and payment due date may be delayed until 14 days after submittal contents are approved.
 2. Include:
 - a. Construction Program database with updated activity and milestone data.
 - b. Definition of project calendars if revised from baseline calendars.

- c. Primavera layout file (.plf).
- d. Gantt chart for project on 11 by 17-inch print(s) and in portable document format (.pdf).
- e. Narrative report specifically stating status of project.
 - 1) When negative float exists, cite specific actions and conditions which caused "behind schedule" condition and provide proposed course of action to complete the project within the specified delivery time. When float changed from previous month's updated schedule, explain reason for change.
 - 2) List of Contractor-initiated changes to current schedule stating reason for action taken and unresolved issues relating to the Construction Program. Government reserves the right to reject Contractor-initiated changes to current schedule which negatively impact Government action which was initiated on the basis of current schedule.
 - 3) Discuss contract milestones and significant activities that were started, continued, or completed during the update period or scheduled during the next update period.
- f. Submittal Register:
 - 1) List submittals as listed in Table 11-23.00A.
 - 2) Update Submittal Register monthly to include submittals and revised submittals. Register fields shall include RSN name and number, date submitted, date of response, status (e.g., approved, not approved, resubmittal required).
 - 3) This register document is separate from scheduling database.

E. RSN 01 32 10-4, Time Impact Analysis:

- 1. Include:
 - a. Construction Program database with proposed revised activity and milestone data.
 - b. Proposed revised schedule due to change or delay. Highlight or list added, changed, or deleted activities.
 - c. Primavera layout file (.plf).
 - d. Gantt chart for project on 11 by 17-inch print(s) and in portable document format (.pdf).
 - e. Narrative report explaining results and conclusions.

1.05 QUALIFICATIONS

- A. Representative: Experienced in developing and maintaining construction schedules and knowledgeable of activities and progress on-site to develop and maintain accurate and reliable schedules.

1.06 GENERAL

- A. Develop, maintain, and use Construction Program to plan, monitor, evaluate, and report accomplishment of work.

1.07 SCHEDULES

- A. Prepare construction schedules using Critical Path Method outlined in AGC Manual. Use a computer software program to perform mathematical analysis of scheduling data.
 - 1. Prepare schedule based on required sequence and interdependence of activities. Logically sequence activities.
 - 2. Prepare detailed activity network for accomplishing required work organized by Work Breakdown Structure (WBS). Utilize WBS in lieu of activity codes for organization of schedule. If activity codes are utilized in addition to WBS, use unique activity code names assigned as project codes rather than global codes.
 - 3. Activities except “Award” shall have predecessor activities and activities except “Contract Complete” shall have successor activities.
 - 4. Meet contract requirements; milestone(s) in accordance with the clause at FAR 52.211-10, Commencement, Prosecution, and Completion of Work; and relevant clauses and specification sections. Include interim milestone dates, Government interface dates, contract completion date, and other time or seasonal constraints specified in contract documents.
 - 5. Include work of subcontractors, Government interfaces, and contract milestones.
 - 6. Adjust Construction Program/schedule for seasonal weather conditions. Provide a narrative explaining the expected weather and plan for incorporation in schedule. Consider work influenced by high or low ambient temperatures or precipitation. Utilize Primavera P6 calendars to depict historical weather days that impact work. Assign weather calendars to work impacted by weather.
 - 7. Define activities to level of detail resulting in their durations being no greater than 20 workdays.
 - a. Durations for administrative activities (e.g., submittals and reviews) fabrication, manufacturing, and typical level of effort activities (e.g., dewatering, storm water management) will not be subject to workday duration limitation.
 - 8. Include separate activities for submittal preparation and submittal reviews when submittal addresses work that involves significant quantities, long lead times, is on the critical path or near critical path, or as requested by the COR. Assign

Government submittal review activities to a seven-day calendar with durations as specified in Section 01 33 00 - Submittals. Include Required Submittal Number (RSN) in activity description.

9. Include separate activities for fabrication or manufacturing when work involves significant quantities, long lead times, is on the critical path or near critical path, or as requested by the COR
10. Include contract title, contract number, and Contractor's name on transmittal cover sheet and each sheet of Gantt chart.
11. For each activity on Gantt chart, display activity identification number, activity description, planned or original duration, start date, finish date, total float, and calendar identification. Do not display early start and late start fields.
12. Include table of abbreviations used in the schedule, listed and defined alphabetically.
13. Use finish to start logic relationships between activities. Do not utilize start to start, finish to finish, or start to finish logic relationships. Do not use negative lead or lag times.
14. Use durations in units of whole workdays.
15. Provide best estimate of time required to complete activity considering quantity of work, work conditions, location of work, and planned resources for activity.
16. Equate durations of Government reviews and other identified actions to maximum number of calendar days specified in their respective paragraphs.
17. Establish workday calendar(s) and use these in the schedule to translate activity's workday duration into calendar dates. Use unique calendar names; do not use software default calendar names. Save calendars as project calendars, not global calendars.
18. Interruption of service request must be received at least 30 days prior to beginning of interrupted service. See Section 01 14 30, Interruption of Service, for more information.
19. Include 2 days in the schedule for a Government inspection of the pipe after the existing fixed-cone valves have been removed. Contractor will not be able to install new equipment until directed by COR.

B. Baseline Schedule:

1. Represents Contractor's as-planned approach to accomplishing the work. Do not include actual start dates, percent completes, or actual finish dates.

C. Updated Schedule:

1. Meet monthly with COR at Government's project office, or at a location approved by the COR, to review progress made to end date of progress payment period.

Establish dates that activities were started and completed and remaining duration for each activity started but not completed during the period.

- a. Discuss and mutually agree upon changes to the schedule.
 - b. Out-of-sequence activities are not allowed. Revise logic to reflect change in work plan.
 - c. Finalize updated Construction Program database with mutually agreed upon changes.
2. Following receipt of executed contract modification, incorporate activity data stipulated in modification into current schedule for inclusion in next scheduled progress update. Provide appropriate logic relationships for revised activities.
 3. Assign a unique project file name for each schedule update.

1.08 TIME IMPACT ANALYSIS

- A. Provide time impact analysis for contract changes (e.g., change order, proposed modification, or value engineering change proposal) to support a claim or request for equitable adjustment to the contract due to delay or accelerated schedule.
- B. The CO may use time impact analysis to determine if time extension or reduction to contract milestone dates is justified.
 1. Time impact analysis is required for contract changes whether the Contractor's current schedule milestone dates are the same as, earlier than, or later than, those required under the contract.
 2. Changes, additions, or deletions to activities; activity durations; activity time frames; or activity predecessors and successors will not automatically determine that extension or reduction of contract time is warranted or due the Contractor.
 3. Time extensions for performance will be considered only to the extent that the Contractor's current scheduled milestone dates exceed contract milestone dates.
- C. Float is not for exclusive use by or benefit of either the Government or the Contractor.
- D. Perform time impact analyses using data in most recent approved schedule update prior to change or delay event.
 1. Prepare proposed revised schedule and narrative description describing and highlighting changes or delays.
 2. Prepare summary comparing results of two schedule analyses: One using current schedule data from last approved updated schedule prior to event requiring analysis, and one using proposed schedule data incorporating changes or delays.
 - a. Show contract milestones and activities whose periods of performance have shifted as result of change which affects production and/or manufacture schedules, material orders, construction seasons, and labor and/or equipment utilization.

- b. For activities directly affected by change or delay, include the current and proposed items:
 - 1) Activity description.
 - 2) Types and quantities of major pieces of equipment, principal manpower, and pacing materials (materials that affect activity start, duration, or finish).
 - 3) Activity duration.
 - 4) Narrative containing rationale used in developing the proposed logic relationships and activity data.
- c. Data date for impacted schedule used in comparison shall be the same as approved updated schedule data date.
- d. Base schedule comparisons on status of work and available float at time the CO directs or proposes change to the work, the Contractor submits a value engineering change proposal, or when a delay occurs as shown in approved updated schedule.

1.09 REVIEW AND EVALUATION

A. Baseline Schedule:

- 1. Within 21 calendar days after receipt of baseline schedule:
 - a. CO will approve or not approve proposed baseline schedule.
 - b. Upon request from the CO, meet with COR for a joint review of proposed baseline schedule.
 - c. If schedule is not approved, revise and resubmit within 7 calendar days following date of rejection letter.
- 2. Do not proceed with onsite work, except mobilization and surveying, until baseline schedule has been approved by the Government.

B. Updated schedules:

- 1. Submit updated Construction Program schedule monthly.
- 2. Submit updated schedule at least seven days prior to the submission of each invoice request for progress payment.
- 3. If updated schedule is not approved, revise and resubmit updated schedule within 7 calendar days following date of rejection letter.
- 4. The COR will schedule pre-submittal meeting with the Contractor's representative each month to review draft updated schedule prior to the Contractor submitting RSN 01 32 10-3, Updated Schedule.

C. Failure to include elements of the work in schedules will not release Contractor from completing required work under the contract.

- D. Performance will be evaluated by the Government using approved CPM schedules.

1.10 FAILURE TO COMPLY

- A. Failure to comply with requirements of this Section shall be grounds for determination by the CO that the Contractor is not prosecuting the work with sufficient diligence to ensure completion within specified time.
- B. The CO may terminate the Contractor's right to proceed with the work, or separable part of it, in accordance with default terms of this contract.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

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SECTION 01 33 00

SUBMITTALS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include in prices offered in the Price Schedule for other items of work.

1.02 REFERENCE STANDARDS

- A. All references below are to be the current adopted edition at the time of solicitation for this contract. If any conflict arises between installing the new equipment in the existing installation, the contractor is to notify the COR. At which point, the CO will give direction on the course of action.
- B. American Society of Mechanical Engineers (ASME)
1. ASME Y14.1 Decimal-inch Drawing Sheet Size and Format
- C. National Institute of Building Sciences (NIBS)
1. NIBS NCS United States National CAD Standards, Version 6

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1.03 DEFINITIONS

- A. Days: Calendar days.
- B. Required Submittal Number (RSN): Identifies items to be submitted together as a complete submittal.

1.04 SUBMITTAL REQUIREMENTS

- A. In case of conflict between requirements of this section and requirements included elsewhere in these specifications, requirements included elsewhere take precedence.
- B. General:
1. Prepare in English.
 2. Label with contract number and title, and RSN.
 3. Measurement Units: US Customary Units.
 4. Provide a unique transmittal number for each submittal.

C. Drawings:

1. Minimum Identification in Title Block:
 - a. Contract number and title.
 - b. Contractor's or supplier's title and drawing number.
 - c. Date.
2. Reserve 3- by 3-inch space next to title block for review stamps.
3. Size: D size (22 inches by 34 inches) as defined by ASME Y14.1.
4. Draw to scale using computer drafting or drafting equipment, unless otherwise specified.
 - a. Computer drafted drawings:
 - 1) In accordance with NIBS NCS.
 - 2) Electronic file format: Compatible with AUTOCAD, Version 2023 or later.
 - 3) Compile using "eTransmit" utility in AUTOCAD.
 - b. Drawing prepared with drafting equipment, when allowed: Lettering shall be neat.
5. Drawings Designated as "Government Form" in Specifications:
 - a. Computer drafted.
 - b. Government will provide an electronic AutoCAD format template.
 - c. Title block and sheet format:
 - 1) As shown on standard drawing 40-D-7102.
 - 2) Government will provide template.
 - 3) Government will provide specific title block information to be used.
6. Final Drawings:
 - a. Computer drafted.
 - b. Government will provide an electronic AutoCAD format template.
 - c. Show as-built changes, including revision dates, made during installation. Indicate changes by clouding.
7. Electronic Files: On CD or DVD discs.

D. Product Data:

1. Mark manufacturer's data for commercial products or equipment, such as catalog cut sheets.
 - a. Identify manufacturer's name, type, model, size, and characteristics.

- b. Illustrate that product or equipment meets requirements of specifications.
 - c. Mark items to be furnished in a manner that will photocopy (no highlighter).
 - d. Strike through items that do not apply.
- E. Certifications:
 - 1. Certifications by a Registered Professional: Signed and sealed by registered professional.
 - 2. Manufacturer's Certifications: Signed by authorized representative of manufacturer.
- F. Manuals:
 - 1. Copies:
 - a. Printed copies: Bound and indexed.
 - b. Electronic copies: Adobe pdf on CD or DVD discs.
 - 1) Bookmark longer files to assist in navigating file.
 - 2. Contents:
 - a. Parts identification lists, lists of special tools, and accessories.
 - b. Schematics and wiring diagrams.
 - c. Detailed instructions for installing, operating, lubricating, and maintaining equipment.
 - d. As-built drawings, photographs, and test records or reports if required by the specifications.

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1.05 SUBMITTALS PROCEDURES

- A. Submit only checked submittals. Submittals without evidence of Contractor's approval will be returned for resubmission.
- B. Submit complete sets of required materials for each RSN as specified in "Submittals Required" column in Table 01 33 00A - List of Submittals. A complete set includes listed items for RSNs with multiple parts.
- C. Submit sets specified in "Sets to be sent:" columns in Table 01 33 00A - List of Submittals.
 - 1. Submittals identified with "CDMS" in the "Sets to be sent" column shall be submitted electronically in accordance with Section 01 31 30 - Contract Document Management System.
- D. Include the following information in transmittal letters:

1. Contract number and title.
 2. RSN for each attached submittal.
 3. Responsible code.
 4. Number of sets for each RSN.
 5. Identify submittal as initial or resubmittal.
- E. Resubmittal of submittals not approved:
1. Mark changes such that they are readily identifiable and show revision date.
 2. Describe reasons for significant changes in transmittal letter.
 3. Resubmit returned submittals within 14 days after receiving the comments, unless otherwise directed.
 4. Requirements for initial submittals apply to resubmittals.
- F. More than one RSN may be submitted under a transmittal letter, provided responsible code is same.

1.06 REVIEW OF SUBMITTALS

- A. Time Required:
1. Submittal review will require 21 days for review of each submittal or resubmittal, unless otherwise specified.
 2. Time required for review of each submittal or resubmittal begins when complete sets of materials required for a particular RSN are received and extends through return mailing postmark date.
- B. Time in Excess of Specified:
1. CO may extend contract completion date to allow additional time for completing work affected by excess review time.
 - a. Time extension will be to extent that excess review time caused delay to contract completion date.
 - b. Time extension will not exceed time used in excess of specified number of days for review of submittals or resubmittals.
 - c. Concurrent days of excess review time resulting from review of 2 or more separate submittals or resubmittals will be counted only once in extending contract completion date.
 2. No time extension will be allowed if Contractor fails to make complete action submittals in sequence and within time periods specified.
 3. Adjustment for delay will be made only to the extent that:
 - a. Approval was required under the contract, and.

- b. Requests for approval were properly and timely submitted and were approved.
 - 4. Adjustment will be subject to terms of paragraphs (b) and (c) of clause at FAR 52.242-14, Suspension of Work; however, no such delay shall be deemed to be a “suspension order” as term is used in that clause.
- C. Return of Submittals:
 - 1. Return of submittals will be by CDMS response within 21 days.
 - 2. Submittals will be returned either approved, conditionally approved, or not approved with the exception of RSNs 01 32 10-2, 01 32 10-3, 01 35 10-1, 01 35 10-2, 01 35 20-1, 01 35 30-1, 01 35 30-2, and 01 74 00-2 which will be accepted, conditionally accepted, or not accepted.
 - 3. Government will acknowledge acceptance of submittals:
 - a. Submittals will not be returned when they comply with the specifications.
 - b. Submittals that do not comply with the specifications may be returned for resubmittal or additional information may be requested.

1.07 HARD COPY TRANSMITTALS

- A. Addresses for codes listed in Table 01 33 00A - List of Submittals (if required).
 - 1. Contracting Officer Representative (COR), Bureau of Reclamation, Willows Construction Office, Attn: WCO-200, P.O. Box 988, Willows, CA 95988-2615; Express Mail: 1140 West Wood Street, Willows CA 95988-2615.
 - 2. Regional Engineer (RE), Bureau of Reclamation, California-Great Basin Regional Office, Attn: CGB-200, 2800 Cottage Way, Sacramento, CA 95825.
 - 3. Contracting Officer (CO), Bureau of Reclamation, California-Great Basin Regional Office, Attn: CGB-3800, 2800 Cottage Way, Sacramento, CA 95825.
 - 4. Area Manager, Bureau of Reclamation, South-Central California Area Office (SCCAO), Attn: SCCAO-430, 1243 N Street Fresno, CA 93721.
- B. Send original transmittal letter with appropriate number of sets to office listed in “Responsible Code” column in Table 01 33 00A - List of Submittals.
 - 1. Submittals identified with “CDMS” in the “Sets to be sent” column shall be submitted in electronic format in accordance with Section 01 31 30 – Contract Document Management System. However, the following items should be submitted via hardcopy in accordance with Article 1.04 – Submittal Requirements:
 - a. At the discretion of the Government, the Contractor may be required to provide complete sets of hardcopies of all documents which appear unclear or illegible in their electronic format.

- C. Send copy of transmittal letter with appropriate number of sets to offices that are not the responsible code but show “Sets to be sent” in Table 01 33 00A - List of Submittals.
- D. When “Sets to be sent” is 0, send a copy of transmittal letter to that office.
- E. Submittals required by specifications, but not listed in Table 01 33 00A - List of Submittals:
 - 1. Submit in accordance with this section.
 - 2. Submit to Contract Officers Representative, unless otherwise specified.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

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RSN	Clause or Section Title	Submittals required	Due date or delivery time	Responsible code	Sets to be sent:		
					CO	COR	RE
01 14 10-1	Use of Site	Land Use and Landscape Rehabilitation Plan	At least 28 days before use of Government land	COR	CDMS		
01 14 30-1	Interruption of Service	Interruption of Service Request	Submitted at least 30 days before outage is required.	COR	CDMS		
01 31 30-1	Contract Document Management System	Approval Data	Within 7 days of Award	COR	Email		
01 31 30-2	Contract Document Management System	Final Data	Within 14 days of completion of work	COR	Email		
01 32 10-1	Construction Program	Representative Information	Within 7 days after receipt of Notice of Award	COR	CDMS		
01 32 10-2	Construction Program	Baseline Schedule	Within 21 days after receipt of Notice to Proceed	COR	Email		
01 32 10-3	Construction Program	Updated Schedule Reports	With monthly requests for progress payment	COR	CDMS		
01 32 10-4	Construction Program	Time Impact Analysis	Within 28 days after the CO directs a contract change with schedule impacts. Or With proposal for future modification, value engineering proposal, or request or claim for equitable adjustment to the contract.	COR	CDMS		
01 35 10-1	Safety Data Sheets	Complete LHM And SDS	At least 14 days before jobsite delivery of hazardous material	COR	CDMS		
01 35 10-2	Safety Data Sheets	Updated LHM And SDS	At least 14 days before jobsite delivery of hazardous material not previously listed	COR	CDMS		
01 35 20-1	Safety and Health	Safety Program	Submitted and approved before starting onsite work	COR	CDMS		
01 35 20-2	Safety and Health	Monthly Accident Summary Report	Monthly	COR	CDMS		

RSN	Clause or Section Title	Submittals required	Due date or delivery time	Responsible code	Sets to be sent:		
					CO	COR	RE
01 35 30-1	Contractor's Onsite Safety Personnel	Resumes	Prior to employment	COR	CDMS		
01 35 30-2	Contractor's Onsite Safety Personnel	Safety Inspection Reports	At least once each week	COR	CDMS		
01 54 20-1	Use of Government Crane	Crane Operator Qualifications	Submitted and approved before starting associated work	COR	CDMS		
01 55 00-1	Vehicular Access and Parking	Initial Digital Recording	At least 28 days before starting onsite work	COR	CDMS		
01 55 00-2	Vehicular Access and Parking	Post Construction Digital Recording	Within 7 days of completion of onsite work	COR	CDMS		
01 55 00-3	Vehicular Access and Parking	Post Repair Digital Recording	Within 14 days of completion of work	COR	CDMS		
01 51 37-1	Temporary Bypass Pump System	Temporary Bypass Pump System Plan	Submitted and approved before starting associated work	COR	CDMS		
01 51 37-2	Temporary Bypass Pump System	Field Test Results	Within 14 days of completion of work	COR	CDMS		
01 74 00-1	Cleaning and Waste Management	Waste Production and Disposal Plan	At least 28 days before starting onsite work	COR	CDMS		
01 74 00-2	Cleaning and Waste Management	Waste Production and Disposal Records	Within 7 days of waste disposal	COR	CDMS		
01 74 00-3	Cleaning and Waste Management	Environmental Consultant Resume	At least 35 days before beginning environmental assessment	COR	CDMS		
01 74 00-4	Cleaning and Waste Management	Environmental Site Assessment	Within 14 days of completion of work	COR	CDMS		
01 78 30-1	Project Record Documents	Electronic Media Release Form, Upon Request	Submitted with request for CAD files of project drawings	COR	CDMS		
01 78 30-2	Project Record Documents	Final As-built Drawings	Within 14 days of completion of major items of work	COR	CDMS		

RSN	Clause or Section Title	Submittals required	Due date or delivery time	Respon- sible code	Sets to be sent:		
					CO	COR	RE
01 78 30-3	Project Record Documents	Final O&M Manuals	Within 14 days of completion of major items of work	COR	CDMS		
03 62 20-1	Nonshrink Grout for Equipment and Metalwork	Packaged Nonshrink Grout Data	Submitted and approved before starting associated work	RE	CDMS		
05 50 00-1	Metal Fabrications	Shop Drawings	Submitted and approved before starting associated work	RE	CDMS		
09 96 20-1	Coatings	Approval Data	Submitted and approved before starting associated work	RE	CDMS		
09 96 20-2	Coatings	Final Approval Data	Submitted within 14 days after issuing purchase order	RE	CDMS		
09 96 20-3	Coatings	Qualifications	Submitted and approved before starting associated work	CE	CDMS		
09 96 20-4	Coatings	Quality Control Plan	Submitted and approved before starting associated work	RE	CDMS		
09 96 20-5	Coatings	Contractor Quality Testing Report	Submitted within 14 days after application	RE	CDMS		
26 05 10-1	Basic Electrical Materials and Methods	Initial Inspection	Within 14 days of inspection	RE	CDMS		
26 05 10-2	Basic Electrical Materials and Methods	Design Documentation	Within 30 days of approval of RSN 26 05 10-1.	RE	CDMS		
26 05 10-1	Conductors and Cables	Approval Data	Submitted and approved prior to ordering materials.	RE	CDMS		
26 05 10-2	Conductors and Cables	Field Test Reports	Within 14 days of testing.	RE	CDMS		
26 05 20-1	Grounding and Bonding	Approval Data	Submitted and approved prior to ordering materials.	RE	CDMS		
26 05 20-2	Grounding and Bonding	As-built Drawings	Within 14 days of completion of work	RE	CDMS		
26 05 20-3	Grounding and Bonding	Test Reports	Within 14 days of testing.	RE	CDMS		

RSN	Clause or Section Title	Submittals required	Due date or delivery time	Respon- sible code	Sets to be sent:		
					CO	COR	RE
26 05 33-1	Raceways and Boxes	Approval Data and Drawings	Submitted and approved prior to ordering materials.	RE	CDMS		
26 05 33-2	Raceways and Boxes	As-built Drawings	Within 14 days of completion of work	RE	CDMS		
26 05 90-1	Wiring Checkout and Tests	Test Reports	Within 14 days of completion of testing.	RE	CDMS		
26 24 24-1	Distribution Panelboards	Approval Data	Submitted and approved prior to ordering materials.	RE	CDMS		
26 24 24-2	Distribution Panelboards	Factory Test Report	Within 14 days of testing.	RE	CDMS		
26 24 24-3	Distribution Panelboards	Operation and Maintenance Instruction Books	Within 14 days of completion of work	RE	CDMS		
35 22 25-1	Valves	Fixed Cone Valve and Reducer Approval Data	Within 28 days of Notice to Proceed	RE	CDMS		
35 22 25-2	Valves	Butterfly Valve Approval Data	Within 28 days of Notice to Proceed	RE	CDMS		
35 22 25-3	Valves	Fixed Cone Valve Shop Test Data	Within 14 days of completion of work	RE	CDMS		
35 22 25-4	Valves	Butterfly Valve Shop Test Data	Within 14 days of completion of work	RE	CDMS		
35 22 25-5	Valves	Final Data	Within 14 days of completion of work	RE	CDMS		
35 22 25-5	Valves	Final Data – Hard Copies	Within 14 days of completion of work	RE	1 to Area Manager		
35 22 25-5	Valves	Training	Within 14 days of completion of work	RE	CDMS		

END OF SECTION

SECTION 01 33 26 ELECTRICAL DRAWINGS AND DATA

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include in prices offered in the schedule for items requiring electrical drawings and data.

1.02 REFERENCE STANDARDS

- A. All references below are to be the current adopted edition at the time of solicitation for this contract. If any conflict arises between installing the new equipment in the existing installation, the contractor is to notify the COR. At which point, the CO will give direction on the course of action.
- B. American Society of Mechanical Engineers (ASME)
1. ASME Y14.1 Decimal Inch Drawing Sheet Size and Format
- C. Institute of Electrical and Electronic Engineers (IEEE)
1. IEEE 315 Graphic Symbols for Electrical and Electronics Diagrams (Including Reference Designation Letters)
 2. IEEE C37.2 Electrical Power System Device Function Numbers, Acronyms, and Contact Designations
 3. IEEE Y32.9 Graphic Symbols for Electrical Wiring and Layout Diagrams Used in Architecture and Building Construction
- D. National Electrical Manufacturers Association (NEMA)
1. NEMA ICS 1 Industrial Control and Systems: General Requirements

1.03 SUBMITTAL PROCEDURES

- A. In addition to the requirements in Section 01 33 00 – Submittals, prepare electrical drawings and data submittals in accordance with the following.

- B. A qualified electrical engineer shall review the electrical submittals and validate for completeness and accuracy prior to submission.
- C. Drawings, general:
1. Device designations and symbols:
 - a. Conform to IEEE 315, IEEE C37.2, IEEE Y32.9, and NEMA ICS-1.
 - b. As indicated on Standard Drawings 104-D-757.
 2. Size: D-size prints as defined in ASME Y14.1 made from original drawings.
 3. Indicate changes on revised drawings to distinguish them from previous submittals. Describe reasons for significant changes in submittal letters.
- D. Approval Drawings in Government Format:
1. General:
 - a. Furnish with Government title block and in form as indicated on Standard Drawing 40-D-7102.
 - b. Government will provide specific title block information when drawings are submitted.
 - c. Government may revise conductor or device designations on Contractor's drawings when drawings are approved at no charge in contract price or completion time.
 2. Schematic Diagrams:
 - a. In accordance with FIO Drawing 61060, EIM Actuator wiring diagram.
 - b. Format:
 - 1) Vertical ladder diagram form.
 - 2) Label rungs with sequential numbers starting with number 1.
 - 3) Locate rung numbers adjacent and to left of associated rung.
 - 4) Provide unique rung numbers for each circuit.
 - c. Conductor designations: As indicated on drawings.
 - d. Indicate device terminal designations.
 - e. Indicate functional operation.
 - f. Describe unusual or nonstandard operation.
 - g. Indicate ratings and/or values of devices.
 - h. Indicate contacts, including spare contacts, for motor contactors, auxiliary relays, timers, and protective relays.
 - i. Indicate cross-referencing between other schematic and wiring diagrams.

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- j. Conductor designations: As indicated on drawings or as developed by Contractor. Contractor-developed conductor designations shall consist of:
 - 1) Circuit prefix
 - 2) Rung number.
 - 3) Unique rung wire letter such as 5P15C (“5”-Unit Number, “P”-Protection Circuit, “15”-Rung Number, “C”-Unique Rung Wire Letter).
- k. Show switch developments for control, selector, and limit switches.
- l. Show device and terminal block designations.
- 3. Wiring Diagram Drawings:
 - a. Match with schematic diagram drawings.
 - b. Indicate point-to-point wiring in accordance with Standard Drawing 104-D-1165.
 - c. Indicate equipment as mounted on back, side, swing, and door panels. Show panel as viewed from wiring side of panel.
 - d. Identify each item of equipment with designation indicated on schematic diagram.
 - e. Identify control and protective devices with IEEE device number as indicated on specifications drawings or on Standard Drawing 104-D-757.
 - f. Identify components such as fuses and resistors by value.
 - g. Indicate wiring of devices and items of equipment including terminal number.
 - h. Indicate cables, cable and individual wire designations, and connections to external circuits.
 - i. Cable designations: Similar to designations on Standard Drawing 104-D-1165.
- E. Approval Drawings and Data in Manufacturer’s Format:
 - 1. General: Furnish with manufacturer’s standard format, except as noted below.
 - 2. Equipment Layout Drawings:
 - a. Indicate dimensions of equipment.
 - b. Indicate location of devices and items of equipment including nameplates, terminal blocks, conduit entries, and other features in their relative physical location.
 - c. Indicate method of securing equipment to deck, floor, or wall.
 - d. Identify each device and item of equipment with a bill of material reference number.

3. Bill of Material List:
 - a. Provide information on manufacturer, style, type, rating, quantity, and other identifying information for each device or item of equipment.
 - b. Provide unique reference number for each device or item of equipment listed on bill of material.
 4. Nameplate Lists: Provide information on type of material, size, and engraved lettering.
 5. Wiring Diagrams:
 - a. Indicate equipment as mounted on each back panel, side panel, door. Each panel shall be indicated as viewed from wiring side of panel.
 - b. Identify each item of equipment with the device designation indicated on schematic diagrams.
 - c. Indicate point-to-point wiring in accordance with Standard Drawing 104-D-1165.
 - d. Indicate cables, cable and individual wire designations, and connections to all external circuits.
 - e. Indicate no more than one panel per drawing.
 6. Manufacturer's Technical Catalog Data:
 - a. Provide technical data for each device or item of equipment.
 - b. Include manufacturer's name and address; catalog number, type, style, or model number; electrical ratings; and dimensions.
 - c. Where several items are listed on same sheet, indicate items being submitted for approval.
 - d. Indicate items with a red box, highlight, or other clear and obvious visual indication.
 - e. Demonstrate proposed device or item of equipment meets specifications requirements.
 - f. Mark catalog data sheet with appropriate bill of material (BOM) item number.
 - g. Assemble catalog data sheet into one enclosing cover with bill of material (BOM) index,
- F. As-built Drawings:
1. Mark prints of electrical specifications drawings to indicate as-built changes made to equipment and systems during construction.
 2. Provide for grounding, and conduit systems.
 - a. Indicate as-built changes made during construction.

3. Include revision dates.
 4. Mark the drawings in the following colors:
 - a. Red – Additions to original drawings.
 - b. Green – Deletions to original drawings.
 - c. Blue – Notations necessary for explanation of as-built markings.
- G. Final Drawings:
1. Revised to reflect approval comments and as-built condition of installed equipment at time of contract completion.
 2. Provide original plot of schematic and wiring diagrams, equipment layout drawings, bill of materials, and nameplate drawings/lists.
 3. Provide computer drawing files on CD-ROM disc in AutoCAD® (*.dwg).
- H. Test Reports:
1. Typed, 8 1/2- by 11-inch sheets.
 2. Certified.
 3. Submit multi-page reports in bound folder or three-ring binder.
- I. Operation and Maintenance Instructions Books:
1. Print copies:
 - a. Assemble each set of material into one or more books with enclosing covers.
 2. Electronic copies:
 - a. Adobe Acrobat (.pdf) format on CD-ROM discs.
 - b. Conform to print copies.
 - c. Bookmark file(s) to aid in navigating.
 3. Provide following items:
 - a. Index sheet at front of each book which provides page or index tab number information for each device or item of equipment in book.
 - b. Manufacturer's operation and maintenance procedures; installation details, as necessary; and catalog data sheets for each device or item of equipment.
 - c. List of recommended spare parts and components.
 - d. Complete parts lists for all replacement parts.
 - e. Manufacturer's full-size circuit breaker and fuse time-current curves.
 - f. Copies of all factory test reports.
 - g. Copies of bills of materials

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- h. Copies of all drawings.
 - 1) Revised to reflect approval comments.
 - 2) Maximum size: 11- by 17-inch.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

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SECTION 01 35 10

SAFETY DATA SHEETS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include in prices offered in the Price Schedule for other items of work.

1.02 DEFINITIONS

- A. LHM: List of Hazardous Materials.
- B. SDS: Safety Data Sheet.
1. Referred to as Material Safety Data Sheets in the clause at 52.223-3, Hazardous Material Identifications and Material Safety Data - Alternate 1.

1.03 APPLICATION

- A. For the purposes of this contract, “delivered under this contract” in paragraph (b) of the clause at FAR 52.223-3, Hazardous Material Identification and Material Safety Data - Alternate 1, includes:
1. Materials delivered to Government.
 2. Materials incorporated into work.
 3. Materials used by the Contractor during contract performance at the jobsite.

1.04 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals.
- B. RSN 01 35 10-1, Complete LHM and SDS.
- C. RSN 01 35 10-2, Updated LHM and SDS:
1. Comply with paragraph (e) of clause at FAR 52.223-3, Hazardous Material Identification and Material Safety Data - Alternate 1.
 2. In addition to requirements in Table 01 33 00A, submit copies of updated LHM and SDS to Construction Engineer at least 14 days before delivering hazardous materials to job site.

1.05 DELIVERY

- A. Do not deliver hazardous materials to jobsite which are not included on original or previously updated LHM and SDS before receipt of updated LHM and SDS by Construction Engineer.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

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SECTION 01 35 20

SAFETY AND HEALTH

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost: Include in prices offered in the Price Schedule for other items of work.

1.02 REFERENCE STANDARDS

- A. All references below are to be the current adopted edition at the time of solicitation for this contract. If any conflict arises between installing the new equipment in the existing installation, the contractor is to notify the COR. At which point, the CO will give direction on the course of action.
- B. Bureau of Reclamation (Reclamation)
1. Reclamation Safety and Health Standards (RSHS), current version will be provided to the Contractor at the Notice to proceed meeting.
 2. FIST 1-1 Hazardous Energy Control Program available at:
www.usbr.gov/power/data/fist_pub.html
 - a. FHECP Facility Hazardous Energy Control Program.
- C. Occupational Safety and Health Administration (OSHA)
1. 29 CFR Part 1910 OSHA General Industry Standards
 2. 29 CFR Part 1926 OSHA Construction Industry Regulations
 3. Available online at: <https://www.osha.gov/>.
- D. Applicable State Safety and Health Regulations
1. California Code of Regulations (CCR) Title 8 – CAL/OSHA
 - a. Available online at: <https://www.dir.ca.gov/title8/index/T8index.asp>.
 2. California Department of Public Health, Valley Fever Tailgate Training Guide for California Construction Workers
 - a. Available online at:
<https://www.cdph.ca.gov/Programs/CCDPHP/DEODC/OHB/HESIS/CDPH%20Document%20Library/ValleyFeverWorkerTraining.pdf>.

1.03 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals.

B. RSN 01 35 20-1, Safety Program:

1. Written Safety Program in accordance with Section 3 of the RSHS.
2. Cover all aspects of on site and applicable off site operations and activities associated with this contract.
3. Program must follow the format of the outline in Appendix 3C of Reclamation RSHS.
4. Will not be accepted for review by the COR unless it addresses, in order, lettered and numbered per Appendix 3C.
 - a. For items in the outline that do not apply to this contract write “Not Applicable to Contract” next to the number/letter.
5. Provide a list of Job Hazard Analyses (JHA) for each distinct phase of work that will be submitted throughout the project.
6. A generic Company Safety Plan is not acceptable. The Safety Program must be site specific for this contract.
7. The Contractor’s Safety Program must be submitted and accepted by Reclamation prior to commencing on site work, including mobilization.

C. RSN 01 35 20-2, Monthly Accident Summary Report:

1. Form 7-2218 or other acceptable form in accordance with Section 3.5.3 of RSHS.

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1.04 SAFETY AND HEALTH

- A. Do not allow Contractor employees (including subcontractors), Government personnel, or the Public to be exposed to, or work under, conditions which are unsanitary, physically hazardous, or a potential exposure danger to their health or safety during performance of the contract.
- B. Under no circumstances will onsite work, including mobilization, be permitted until the Safety Program has been accepted by the COR.
- C. Fully participate in a Contractor Safety Program Review meeting, according to RSHS Section 3.6.1, prior to mobilization. If applicable, include subcontractor management representatives.
- D. The minimum work crew at any time on the construction site shall consist of no less than two people and shall be in accordance with other contractual obligations.
- E. Develop Job Hazard Analysis for each distinct phase of work under the contract.
 1. Work shall not begin on the phase of work until a JHA is acceptable to COR and shared with all construction employees.

2. Activities involving hazardous materials shall have the appropriate Safety Data Sheet(s) attached to the JHA.
- F. In addition to requirements of the clause at FAR 52.236-13 “Accident Prevention”, fully comply with RSHS.
1. Construction Safety and Health Standards promulgated by the Secretary of Labor may be obtained from any regional or area office of the Occupational Safety and Health Administration of the U.S. Department of Labor.
- G. Be cognizant of and ensure compliance with requirements set forth in paragraphs above.
1. Contractor’s responsibility applies to all operations, including those of the Contractor’s Subcontractors.
 2. When violations of safety and health requirements contained in these specifications or referenced standards are called to the Contractor’s attention by the CO and COR immediately correct the condition to which attention has been directed.
 3. Either oral or written notice shall be deemed sufficient.
- H. When the Contractor fails or refuses to promptly correct a compliance directive, the CO or the COR may issue an order to stop all or any part of the work.
1. When satisfactory corrective action is taken, an order to resume work will be issued.
 2. The Contractor shall not be entitled to extension of time, nor to claim for damage or to additional compensation by reason of either the directive or the stop order.
 3. Failure of the CO and COR to order discontinuance of any or all of the Contractor’s operations shall not relieve the Contractor of the responsibility for the safety of personnel and property.
- I. Maintain an accurate record of, and report to the COR in the manner prescribed by the CO for all near misses, accidents or incidents, occupational diseases, traumatic injury, cases of death to Contractor employees (including subcontractors), Government personnel, or the Public, and any property damage occurring during the performance of work under this contract.
- J. The rights and remedies of the Government provided in this section are in addition to any other rights and remedies provided by law or under this contract.
1. In the event there is a conflict between requirements contained in RSHS, specification paragraphs, Contractor’s approved Safety Program, referenced safety and health codes, and standards, or the U.S. Department of Labor Construction Safety and Health Standards, promulgated under Section 107 of the Contract Work Hours and Safety Standards Act (40 U.S.C. 327 et seq.), as amended, the more protective requirement having jurisdiction shall prevail.

1.05 SPECIAL CALIFORNIA VALLEY FEVER REQUIREMENTS

- A. Coccidioidomycosis, also known as “Valley Fever” or “cocci”, is a disease caused by Coccidioides fungi which infect the lungs. When the fungus spores present in soil are disturbed, the spores may become airborne and can be inhaled.
- B. The Contractor is hereby notified that the spores which cause Valley Fever are endemic to the Counties of Fresno, Kern, Kings, Madera, Merced, Monterey, San Joaquin, San Luis Obispo, Santa Barbara, Tulare, and Ventura.
- C. Activities which disturb soil or expose workers to dust, such as digging, operating earth-moving equipment, driving vehicles, and working in wind-blown areas, may increase the risk of Valley Fever in workers.
- D. Information regarding preventing and recognizing the symptoms of Valley Fever are available from the California Department of Public Health and the California Department of Industrial Relations.
 - 1. The provisions of this section shall be made a part of every subcontract executed pursuant to this contract.
 - 2. State Law (amendment to the California Labor Code) effective May 1, 2020, applies to employers in 11 Counties in the Central Valley Region.
 - a. State Law requires Construction firms and other employers in these California Counties to conduct awareness training, covering 8 required topics, to employees, initially and annually.
 - b. The Valley Fever Tailgate Training Guide for California Construction Workers is available at site described in References above.

1.06 SAFETY AND HEALTH PLAN

- A. Unless waived by the Contracting Officer's Representative (COR), prime and supplementary safety program submittals shall, as a minimum, address all appropriate contractual requirements shown under the program outline. Program items will be so detailed that reviewers can ascertain adequacy. The original program submittal can cover all work phases, or only address the initial work phase with timely supplementary submittals as major work phases occur. Contractors desiring to follow this latter course of action will, in the initial submission, clearly define the original work covered and the work phases to be covered by supplements.
- B. In no case will any phase of work commence until a program for that portion of the work has been accepted by the COR. Original and supplemental submissions covering hazardous operations and/or activities will include a standing operating procedure (SOP) and hazard analysis. The SOP will break down the operation or activity into specific basic steps. The hazard analysis will define the hazards associated with each basic step and proposed method(s) for eliminating or minimizing the hazard. Such methods will

outline, as a minimum, employee training requirements, personal protective equipment requirements, procedural changes, methods for evaluating program effectiveness, etc.

- C. All activities involving use of hazardous and/or toxic materials, work in confined spaces, work at heights over 6 feet, or underwater require an SOP and hazard analysis. Material Safety Data Sheets will be attached to all SOPs governing work that involves use of hazardous or toxic materials.
- D. Prepare a Site Specific Safety and Health Plan with major components labeled in accordance with the following Program Outline:
 - 1. General Requirements
 - a. Statement of Policy
 - b. Statement of Safety and Health Responsibilities
 - c. Statement of Compliance with Regulations, Standards, and Codes
 - d. Statement of Subcontractor Compliance
 - e. Safety Inspection Procedures
 - f. Accident Investigation and Reporting Procedures
 - g. Applicable Emergency Plans
 - h. Confined Space Procedures
 - i. Lockout/Tagout Procedures
 - j. Fire Protection Plans
 - k. Type and location of suppression equipment or systems
 - l. Offsite assistance agreement
 - m. Temporary heating devices
 - 2. Medical:
 - a. Facilities
 - b. Training
 - c. Certifications
 - d. Physician
 - e. Ambulance (Name, location, and telephone number)
 - f. Physical Qualification of Employees
 - g. Records
 - 3. Communications
 - a. Employee Training

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- b. Safety Meetings
- c. Onsite Training
- d. Supervisor Training
- 4. Occupational Health
 - a. Procedures and Equipment to Minimize Hazards
 - b. Testing program for employees and work environments
 - c. Qualified personnel
 - d. Personal protective equipment
 - e. Ventilation plans
- 5. Machinery and Mechanical Equipment
 - a. Procedures and Equipment to Minimize Hazards
 - 1) Testing program for employees and work environments
 - 2) Mobile and stationary equipment
 - b. Inspection Procedures
 - c. Maintenance Procedures
 - d. Operating Personnel
 - e. Protective Safety Devices and Certifications
 - f. Elevators and Aerial Lifts
- 6. Excavation and Demolition
 - a. Tunnels and Shafts
 - 1) Internal combustion engines
 - 2) Ventilation plans
 - 3) Transportation systems and equipment
 - 4) Work environment testing
 - 5) Ground support
 - b. Blasting
 - 1) Blaster certification
 - 2) Written procedures
 - 3) Storage
 - 4) Transportation
 - c. Excavations Other Than Tunnels and Shafts

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- 1) Slide protections
 - 2) Support systems
 - 3) Inspections
 - 4) Access
 - d. High Scaling
 - 1) Definition
 - 2) Personal protective equipment
 - 3) Standing operating procedure
 - e. E. Haulage
 - 1) Haul roads
 - 2) Equipment and Procedures
- 7. Working Surfaces
 - a. Access
 - 1) Ladders
 - 2) Platforms, stairways, and ramps
 - b. Personal Protective Equipment
 - c. Scaffolding
 - d. Safety Nets
- 8. Protection of the Public
 - a. Signs and Barricades
 - b. Flagging Procedures
 - c. Jurisdictional Approvals
- 9. Marine and Diving Operations
 - a. Detailed Plan and Written Procedures
- 10. Electrical Facilities
 - a. Substations
- 11. Required Safety Program Coordination
 - a. 1. Confined Space Program
 - b. 2. Electrical and Lockout/Tagout Program
- 12. Fall Protection
 - a. Worksite Hazard Analysis

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- b. System Selection
 - c. System Installation
 - d. System Inspection
 - e. Rescue Planning
 - f. Fall Protection Training
- E. This outline provides guidance in preparing a safety program and does not cover all material that may be necessary. The Contractor shall review specifications and all safety and health regulations to ensure a comprehensive plan.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

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SECTION 01 35 30
CONTRACTOR’S ONSITE SAFETY PERSONNEL

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include in prices offered in the schedule for other items of work.

1.02 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals.
- B. RSN 01 35 30-1, Resumes:
1. Contractor’s Onsite Safety Representative
 2. Safety Professional.
- C. RSN 01 35 30-2, Safety Inspection Reports:
1. Include a list of noted deficiencies, their abatement dates, and follow-up action for all jobsite activities.
 2. Base inspection report on findings of jobsite walk-through with Government personnel and shall include Government Representative’s name.

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1.03 QUALIFICATIONS

- A. Contractor’s Onsite Safety Representative:
1. Competent person can be supervisory employee, with appropriate level of safety related training and experience. Safety responsibility may be adjunct, collateral or ancillary to their duties. Their main job duties may be leadership, supervision, or management in a craft, trade, or technical specialty.
 2. Has completed at a minimum an OSHA recognized 30-hour Construction Safety Training program and has a minimum of 3 years of experience on similar projects in size and complexity to this contract. Experience should include dealing with the day-to-day operations and functionality of equipment operations not just occasional safety inspection site visits.
- B. Safety Professional:
1. Holds professional status in the safety field by virtue of education, training, certification and experience. Responsibility must be the prevention of harm to people, property, and the environment, rather than responsibility for responding to harmful events. Examples of appropriate qualifications include:

- a. Holds Safety certification from organizations accredited by the National Commission for Certifying Agencies (NCCA), the Council of Engineering and Scientific Specialty Boards (CESB), or a nationally recognized accrediting body which uses certification criteria equal to or greater than that of the NCCA or CESB.
- b. Holds current certification as Certified Safety Professional (CSP), Associate Safety Professional (ASP), Occupational Health and Safety Technologist (OHST), Construction Health & Safety Technician (CHST) or equivalent with a minimum of 3 years safety experience on similar projects in size and complexity to this contract.
- c. Holds a degree in Occupational Safety and Health: Major study – safety or occupational health related fields and a minimum of 3 years safety experience on similar projects in size and complexity to this contract.
- d. The effectiveness of the Contractor's Safety Professional in the administering and implementation of the safety program will be subject to continued review and approval by the Contracting Officer.

1.04 APPLICATION

- A. Designate the Contractor's Onsite Safety Representative prior to start of construction and employ Safety Professional for Full-time on the job.
 1. Safety Professional requirements may be met by retaining appropriate level of services of an acceptable safety consultant.
- B. Contractor's Onsite Safety Representative authorities, duties, and responsibilities:
 1. Responsible for day-to-day implementation of the Contractor's Safety Program
 2. Full authorization to correct unsafe acts on the spot.
 3. Prepare safety inspection reports.
 4. Onsite during any and all construction activities.
- C. Safety Professional duties, and responsibilities:
 1. Review and approve, by signature the Contractor's Safety Program prior to submittal.
 2. Responsible for day-to-day implementation of the Contractor's Safety Program
 3. Full authorization to correct unsafe acts on the spot.
 4. Prepare safety inspection reports.
 5. Attend weekly coordination meetings.
 6. Onsite:
 - a. At minimum once per week.

- b. During all critical lifts.

1.05 QUALITY ASSURANCE

A. Contractor's Onsite Safety Representative and Safety Professional:

1. The effectiveness of the Contractor's Onsite Safety Representative and Safety Professional in prosecuting the safety program will be subject to continued review and approval by the CO.
2. Should the Contractor's safety effort be considered inadequate, the CO has the option to require the Contractor to employ a full-time qualified Safety Professional at no cost to the Government.

B. Safety Program:

1. The effectiveness of the Contractor's Safety Program will be subject to continued review and approval by the CO.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

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SECTION 01 51 00

TEMPORARY UTILITIES

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. Include in prices offered in the Price Schedule for other items of work.

1.02 REFERENCE STANDARDS

- A. All references below are to be the current adopted edition at the time of solicitation for this contract. If any conflict arises between installing the new equipment in the existing installation, the contractor is to notify the COR. At which point, the CO will give direction on the course of action.

B. Institute of Electrical and Electronic Engineers (IEEE)

1. IEEE C2-2017 National Electrical Safety Code (NESC)

C. Occupational Safety and Health Administration (OSHA)

1. 29 CFR 1910.141 General Environmental Controls
2. 29 CFR 1926.51 Sanitation

1.03 TEMPORARY ELECTRICITY

- A. Electric power for small tools may be obtained from Government inside outlet works. This power will be available as single phase, 60 hertz, alternating current at approximately 120 volts. No charge will be made for power obtained from this source.
- B. Provide other electric power required for construction including power for trailer, if required.
- C. Provide generators, transmission lines, distribution circuits, transformers, and other electrical equipment and facilities required for obtaining power and distributing power to points of use.
- D. Comply with IEEE C2 clearances and spacing for temporary communications and supply lines.

1.04 TEMPORARY WATER

- A. Provide water, including potable water, required for construction purposes.

- B. Use water which meets specified requirements for water used in concrete, soil-cement, masonry, grouting, and other permanent work.
- C. Provide means of conveying water to points of use.

1.05 TELEPHONE

- A. Provide reliable telecommunications during construction if cellular phone service is not sufficient.

1.06 SANITARY FACILITIES

- A. Contractor shall provide temporary restrooms for onsite personnel to use in accordance with 29 CFR 1926.51 and its subparts. Facilities shall be and have:
 - 1. Adequate and readily accessible facilities:
 - a. The facilities shall be easily accessible and if the facility is not readily accessible, then the facility is considered to be “not available for service or action.
 - b. A toilet is deemed unusable or not available if it is not adequately sanitized and cleaned for use.
 - 2. Minimum requirement:
 - a. The minimum number of toilets shall be in accordance with 29 CFR 1926.51(c)(1), Minimum number of toilets.
 - 3. Proper basic functions:
 - a. Privacy, as defined by toilets having separate compartments with a door and walls that are “sufficiently high to ensure privacy”
 - b. Proper ventilation and equipment in the portable toilet.
 - c. Adequate lighting inside the toilet and surrounding the facility, especially for nighttime jobs.
 - d. The employer does not have to provide separate toilet facilities for each sex when they will not be occupied by more than one employee at a time, can be locked from the inside, and contain at least one toilet.
 - 4. Handwashing Washing Facilities station:
 - a. Provide adequate washing facilities for employees engaged in the application of paints, coating, herbicides, or insecticides, or in other operations where contaminants may be harmful to the employees.
 - 1) Such facilities shall be in near proximity to the worksite and shall be so equipped as to enable employees to remove such substances.
 - 2) Shall have running hot and cold water, or tepid running water.
 - 3) Washing facilities shall be maintained in a sanitary condition.

- 4) Hand soap or similar cleansing agents shall be provided.
 - 5) A sanitary method to clean hands (paper towels, towel dispenser, or hand-drying air blower).
 - a) When paper towels are use provide garbage can with lid for disposal of wastepaper.
- 5. Servicing schedule:
 - a. Portable toilets shall be serviced regularly to ensure a sanitary condition.
 - b. The Contractor shall establish and implement a schedule to ensure that portable toilets are “maintained in a clean, sanitary, and serviceable condition”.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 REMOVAL

- A. Remove temporary equipment and facilities upon completion of work under this contract.

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SECTION 01 51 37 TEMPORARY BYPASS PUMP SYSTEM

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Payment:
1. Lump sum price offered in the Price Schedule for furnishing, installing, and removing a temporary pump bypass system.

1.02 REFERENCE STANDARDS

- A. All references below are to be the current adopted edition at the time of solicitation for this contract, unless otherwise stated.
- B. Institute of Electrical and Electronics Engineers (IEEE)
1. IEEE C2 National Electrical Safety Code (NESC)
- C. National Fire Protection Association (NFPA)
1. NFPA 70 National Electrical Code (NEC)
- D. National Sanitation Foundation International (NSF)
1. NSF 61 Drinking Water System Components – Health Effects
 2. NSF 372 Drinking Water System Components – Lead Content

1.03 SYSTEM DESCRIPTION

- A. Contractor shall furnish all materials, labor, equipment, power, fuel, maintenance, etc. to implement a temporary pumping system for the purpose of providing the required supply water.
- B. Power will not be available on site. See Article 2.01, Temporary Electricity.
- C. The Temporary Bypass Pump System shall deliver an adjustable flow rate of 9-11 cfs, in increments of 1 cfs, from Lake Cachuma to the Bradbury Dam Spillway.
1. Crest of Bradbury Dam is at El. 766.
 2. Assume Lake Cachuma to be El. 660. Verify current and projected lake elevation with COR prior to submitting RSN 01 51 37-1.
- D. Provide required flow continuously during the outage of the Outlet Works.

1. Including the time required for the Government penstock inspection as described in Section 35 22 25 – Valves.
- E. Temporary pumping system shall be suitable to operate with raw water containing moss, algae, and debris.
- F. The system shall be designed to operate 24 hours per day.
- G. System will be operated by the Government.
- H. System will be maintained by the Contractor.
- I. System shall be metered and log the volume of water delivered.
- J. Screen and position pump intakes for Protected Species protection.
- K. See Section 51 00 00 – Information Available to Offerors, Photo 16, for an overhead view of a potential staging area for the Temporary Bypass Pump System.
- L. Maintain road access across the dam.

1.04 SUBMITTALS

- A. Submit the following in accordance with Section 01 53 00 - Submittals.
- B. RSN 01 57 37-1, Temporary Bypass Pump System Plan:
 1. Detailed plan and description of proposed pumping system. Indicate number and size of pumps. Indicate location, and method of installation for suction and discharge piping and pumps. Indicate staging area for pumps, power equipment, fuel storage, site access point, and expected flow.
 - a. Provide means for road crossing to not disturb vehicle traffic.
 - b. Temporary pipe supports and anchoring required.
 - c. Bypass pump sizes, capacity, number of each size on site and power requirements.
 - d. Screen and position pump intakes for Protected Species protection.
 - e. Method and schedule of screen cleaning.
 - f. Method of noise control for each pump and/or generator.
 - g. Method of protecting discharge structures from erosion and damage.
 - h. Method of oil spill containment.
 - i. Schedule for installation and maintenance of bypass pumping lines.
 - j. Schedule for maintenance of pumping system operation, including inspection, refueling, power supply, and intake screens.

- k. Suction and discharge piping materials.
- C. RSN 01 57 37-2, Field Test Results.
 - 1. Submit test parameters and measured results.

1.05 DELIVERY AND STORAGE

- A. Transport, deliver, handle, and store pipe, fittings, pumps, generators, ancillary equipment, and material to prevent damage and following manufacturers' recommendations.
 - 1. Inspect all material and equipment for proper operation before initiating work.
- B. Repair or replace material found to be defective or damaged at no cost to the Government.

PART 2 PRODUCTS

2.01 TEMPORARY ELECTRICITY

- A. Provide electric power required for bypass pump system.
- B. Provide generators, transmission lines, distribution circuits, transformers, and other electrical equipment and facilities required for obtaining power and distributing power to bypass pump system.
- C. Comply with IEEE C2 clearances and spacing for temporary communications and supply lines.

2.02 TEMPORARY BYPASS PUMP

- A. Fully automatic self-priming units that do not require the use of foot-valves or vacuum pumps in priming system.
- B. Electric or diesel-powered pumps.
- C. Start/stop controls provided for each pump.
- D. Pumps and equipment shall comply with NSF 61 and be certified lead free.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Obtain approval of RSN 01 51 37-1, Bypass Pump System Plan prior to installation.

- B. Shall be in place, tested, and operational prior to Outlet Works outage taking effect.

3.02 TESTING

- A. The Contractor shall coordinate with the COR and provide 5 working days' notice of testing.
- B. The testing for the water supply system shall include, but not be limited to, leakage monitoring and flow calibration.
1. Leakage in the system shall not be pressurized spray, cause erosion, nor exceed 1% of the required flow to the spillway.
 2. The contractor shall also test the adjustable flow rate for a tolerance of a $\pm 2\%$ accuracy across the full operating range of increments.
 3. The flow measurement device to measure flow rate and total flow shall be calibrated with a tolerance of a $\pm 2\%$ accuracy.
- C. Testing approval is required prior to an outage being taken at the Outlet Works.

3.03 TRAINING

- A. The Contractor shall coordinate with the COR and provide 5 working days' notice of training. The training session shall be scheduled with up to 10 operations and maintenance personnel in attendance. Minimum of 10 physical quick reference training pamphlets shall be provided. Instructors must be thoroughly knowledgeable in all aspects of the water supply system and capable of demonstrating the setup and execution of all operational procedures.
- B. The pamphlets shall focus on at least several key areas. One area is Lock-Out-Tag-Out, which involves reviewing appropriate means of isolation and identifying other potential hazards. Another focus shall include equipment operation, including control panels, startup and shutdown procedures, flow adjustment, and verification of proper operation. Additionally, include basic troubleshooting in the pamphlet.
- C. Provide Maintenance Contacts to the Government.

3.04 REMOVAL

- A. Restore areas used for temporary bypass pump system to pre-installation condition as required by Section 01 14 10 – Use of Site.

END OF SECTION

SECTION 01 54 20

USE OF GOVERNMENT CRANE

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include cost in prices offered in the Price Schedule for equipment requiring use of Government crane.

1.02 REFERENCE STANDARDS

- A. American Society of Mechanical Engineers (ASME)
1. ASME B30.2-2011 Overhead and Gantry Cranes (Top Running Bridge, Single or Multiple Girder, Top Running Trolley Hoist)
 2. ASME B30.26-2010 Rigging Hardware
- B. Bureau of Reclamation (USBR)
1. RSHS-2009 Reclamation Safety and Health Standards, including revisions posted at <http://www.usbr.gov/ssle/safety/RSHS/rshs.html>
- C. Code of Federal Regulations (CFR)
1. 29 CFR 1910.179 Occupational Safety and Health Standards - Overhead and Gantry Cranes

1.03 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals.
- B. RSN 01 54 20-1, Crane operator qualifications:
1. Certificates of training and testing.

1.04 PROJECT CONDITIONS

- A. Government-owned crane in outlet works may be used for:
1. Unloading and loading equipment, parts, tools, and supplies at outlet works.
 2. Removing existing equipment from outlet works.

3. Moving equipment, parts, tools, and supplies within outlet works.

B. Crane Capacity: 5 tons.

1.05 CRANE USE

- A. Provide certified crane operators when using Government crane.
- B. Contactor shall provide experienced, competent crane signaler.
- C. Safety: Comply with ASME B30.2, RSHS, and 29 CFR 1910.179. Where there is a conflict between the standards, the more stringent shall prevail.
- D. Provide slings, shackles, and rigging required for use of crane in accordance with ASME B30.26.
- E. Contractor is responsible for proper rigging of equipment and connections to crane.
 1. Comply with ASME B30.2 and 29 CFR 1910.179.
- F. Before use of a crane on each shift, the Contractor's crane operator shall inspect the crane and complete Government's crane inspection form. The crane inspection shall be entered in the crane's logbook and the completed crane inspection form shall be provided to the COR.

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1.06 QUALIFICATIONS

- A. Contractor crane operators shall have appropriate certification and experience for particular crane.
 1. Certification: Valid crane operator certification issued by a crane operator testing organization accredited by a nationally recognized accrediting agency. Operator certifications shall have included successfully passing a written and practical examination.
 2. Experience: Document hours of crane operator experience and crane related experience for particular type of crane. Minimum required hours of experience shown in table below:

Category and Type of Cranes	Hours of Actual Crane Operating Experience	Hours of Crane Related Experience
Overhead Cranes		
(a) Pendant/Remote	40 Hours	40 Hours

Notes:

Hours of actual crane operating experience: Time operator is at controls of crane, has direct control of crane, or combination of operating hours for specific crane type.

Hours of actual crane operator experience over minimum required may be included in hours of crane related experience.

Hours of crane related experience: Time as a signalman/bellman, oiler, crane mechanic, crane inspector, formal classroom training, crane simulator operation, or operating hours on other categories of cranes

3. Drug Testing: Crane operators shall be subject to drug testing. Within 7 days after Government requests random test, submit documentation for each crane operator of a substance abuse test. Within 7 days of an event of a crane operating incident involving personnel injury or death, or damage to the crane or equipment; submit a substance abuse test. Substance abuse testing shall be conducted by a recognized laboratory.
4. Orientation: Contractor crane operators shall participate in orientation of each crane with the Government before operating. Expected length of orientation is 4 hours.

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1.07 SCHEDULING

- A. Coordinate use of Government cranes with COR during weekly coordination meetings.
- B. Government use of cranes will have priority over Contractor use; however the Government will make cranes available for Contractor use to the maximum amount possible while still accomplishing Government maintenance and operations.
- C. Notify COR at least 48 hours before changes to requested crane use during the current week.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

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SECTION 01 55 00 VEHICULAR ACCESS AND PARKING

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. Include in prices offered in the Price Schedule for other items of work.

1.02 REGULATORY REQUIREMENTS

- A. Meet requirements established by jurisdictional authority for use of existing roadways and haul routes; including seasonal or other limitations or restrictions, payment of excess size and weight fees, and posting of bonds conditioned upon repair of damage.
- B. Comply with applicable regulations for haul routes over public highways, roads, or bridges.

1.03 SUBMITTALS

- A. Submit the following in accordance with Section 01 13 00 Submittals.
- B. RSN 01 55 00-1, Initial Digital Recording.
- C. RSN 01 55 00-2, Post Construction Digital Recording.
- D. RSN 01 55 00-3, Post Repair Digital Recording.

1.04 SITE CONDITIONS

- A. Rights-of-way for access to work from existing roads will be established by Government.
1. In accordance with clause at FAR 52.236-10, Operations and Storage Areas, use only established roadways, parking areas, and haul routes; or temporary roadways, parking areas, or haul routes constructed by Contractor when and as authorized by CO.
 2. Subject to clause at FAR 52.249-10, Default (Fixed-Price Construction), unavailability of transportation facilities or limitations thereon shall not become a basis for claims for damages or extension of time for completion of work.
- B. Access to the Bradbury Dam Outlet Works requires crossing the spillway on a narrow, 240-foot-long bridge rated for H-20 loading. Horizontal clearances are approximately 9'-9" curb-to-curb measured at the bridge deck and 10'-7" between fence posts.

PART 2 PRODUCTS

2.01 MATERIALS

- A. Materials to maintain and repair existing roadways, parking areas, and haul routes: In accordance with requirements of jurisdictional authority.
- B. Materials to construct, maintain, and repair temporary roadways, parking areas, and haul routes: As approved by COR.
- C. Materials to maintain roadways and parking areas constructed under this contract and used by Contractor for construction work: In accordance with specified requirements for construction of those roadways and parking areas.

PART 3 EXECUTION

3.01 EXAMINATION

- A. Investigate condition of available public or private roads for clearances, restrictions, bridge-load limits, bond requirements, and other limitations that affect or may affect access and transportation operations to and from jobsite.

3.02 RECORDS

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- A. Make preconstruction, post construction and post repair digital recordings of access road from Highway 154 to Auxiliary Lay Down Area and Outlet Works Control House.
 - 1. DVD format with a minimum 720p resolution.
 - 2. Label:
 - a. Contract number and title.
 - b. Contractor's name.
 - c. Date recording is made.
 - d. Preface video with this information.
- B. COR will be present during recording. Notify COR at least 3 days prior to recording.

3.03 ESTABLISHED ROADWAYS AND PARKING AREAS

- A. Established roadways and parking areas are available for Contractor's use subject to existing restrictions and approval of the COR.
- B. Existing onsite streets and driveways may be used for construction traffic. Tracked vehicles are not allowed.
- C. Designated lay down areas may be used by construction personnel for parking.

- D. Do not allow heavy vehicles or construction equipment in established parking areas.

3.04 HAUL ROUTES

- A. Perform work on rights-of-way established by Government as necessary to construct and maintain any roads, bridges, or drainage structures required for establishment and use of haul routes for construction operations.
- B. Use existing available public highways, roads, or bridges as haul routes subject to applicable local regulations.
- C. Minimize interference with or congestion of local traffic.
- D. Provide barricades, flaggers, and other necessary precautions for safety of public where haul routes cross public highways or roads.

3.05 MAINTENANCE

- A. Maintain roadways, parking areas, and haul routes in a sound, smooth condition.
- B. Maintain roadbed, side slopes, structures, and surfacing of roads and parking areas until completion and acceptance of all work under this contract. As approved by COR, defer until latest practicable date within specified completion period, placement of surfacing on roads or parking areas subject to heavy and deterioration caused by Contractor's construction operations or equipment.
- C. Maintain surfacing of gravel-surfaced roads and parking areas in a smooth condition until completion and acceptance of work under this contract.

3.06 REPAIR

- A. Promptly repair ruts, broken pavement, potholes, low areas with standing water, and other deficiencies to maintain road surfacing and drainage in original or specified condition.

3.07 REMOVAL

- A. Remove materials used to construct temporary roadways, parking areas, and haul routes prior to contract completion. Recycle salvageable materials as approved by COR.

END OF SECTION

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SECTION 01 56 10
PROTECTION OF EXISTING INSTALLATIONS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include in prices offered in the Price Schedule for other items of work, except as specified.
 2. Costs for repair of installations damaged by the Contractor's operations are the Contractor's responsibility.

1.02 PROJECT CONDITIONS

- A. Drawings included in these specifications show items of existing materials and equipment but may not show all equipment and materials existing at the jobsite.
- B. Obtain the location of embedded conduit, pipe, cable, ground mat, and other buried items before performing any drilling or cutting of concrete.

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PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 PROTECTION

- A. Provide protection for personnel and existing facilities from harm due to the Contractor's operations. Protection shall be subject to approval of the Government.
- B. Arrange protective installations to permit operation of existing equipment and facilities by the Government while work is in progress.
- C. Do not discharge anything but clear water into building drainage system.
- D. Prevent dust from entering ventilating systems.

3.02 REMOVAL OF PROTECTIVE INSTALLATIONS

- A. Remove protective installations after purpose has been served. Materials furnished by the Contractor to provide protection remain property of the Contractor.

3.03 REPAIR

- A. Repair, at Contractor's expense, damage to existing installations due to Contractor's operations or Contractor's failure to provide proper protection. At the Government's option, damage may be repaired by the Government, and the Contractor will be backcharged repair costs.

END OF SECTION

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SECTION 01 60 00 PRODUCT REQUIREMENTS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. When the Price Schedule includes a separate item for furnishing a material, include cost of furnishing, hauling, storing, and handling in the price offered in the Price Schedule for the item.
 2. When the Price Schedule does not include a separate item for furnishing a material, include cost of furnishing, hauling, storing, and handling in the price offered in the Price Schedule for work for which the material is required.

1.02 DEFINITIONS

- A. Essential Characteristics: As used in these specifications, the term "essential characteristics" is synonymous with the term "salient characteristics."
- B. Salient Characteristics: Those qualities of an item that are essential to ensure that the intended use of the item can be satisfactorily realized.

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1.03 REFERENCE STANDARDS

- A. All references below are to be the current adopted edition at the time of solicitation for this contract. If any conflict arises between installing the new equipment in the existing installation, the contractor is to notify the COR. At which point, the CO will give direction on the course of action.
- B. American Society of Mechanical Engineers (ASME)
1. ASME B1.1 Unified Inch Screw Threads, (UN and UNR Thread Form)
 2. ASME B1.20.1 Pipe Threads, General Purpose, Inch
- C. Bureau of Reclamation (USBR)
1. RSHS Reclamation Safety and Health Standards, including revisions posted at <http://www.usbr.gov/ssle/safety/RSHS/rshs.html>

1.04 DELIVERY, STORAGE, AND HANDLING

- A. Transport and handle manufactured products in accordance with manufacturer's instructions.
- B. Store and protect manufactured products in accordance with manufacturer's instructions and RSHS. Obtain instructions from the manufacturer before delivery of materials to jobsite. Maintain a copy of instructions at jobsite.
- C. Protect materials from adverse effects of moisture, sunlight, ultraviolet light, or weather during storage at jobsite.
- D. Remove and replace damaged items with new items.
- E. Store curing compounds, sealants, adhesives, paints, coatings, sealers, joint compounds, grouts, and similar products at the temperature and environmental conditions recommended by manufacturer.

1.05 MAINTENANCE

- A. Extra Materials:
 - 1. Furnish additional maintenance materials specified as "extra materials" in the specifications. Provide maintenance material identical to installed material and provide from the same manufacturer's production lot as installed material.
 - 2. Package extra materials for storage and label with complete product identification on packaging.
 - 3. Deliver extra materials to the Government at jobsite and place in storage as directed by the COR.

PART 2 PRODUCTS

2.01 MATERIALS

- A. Provide materials required for completion of work.
- B. Provide type and quality described in these specifications. Make diligent effort to procure specified materials from all available sources.
- C. Furnish new materials conforming to referenced standards unless otherwise specified.
- D. For materials not covered by these or referenced specifications, furnish materials of standard commercial quality.

- E. If materials to be used deviate from or are not covered by recognized specifications and standards, submit, for approval, justification for and exact nature of the deviation, and complete specifications for materials proposed for use.
- F. Make parts accurately to standard gauge where possible.
 - 1. Use unified screw threads conforming to ASME B1.1 or B1.20.1 for threads, including but not limited to those of bolts, nuts, screws, taps, pipes, and pipefittings.
 - 2. For internal connections only, the Contractor may deviate from ASME standards, provided a complete set of taps and dies are furnished as required to facilitate repair or replacement.
- G. Permanently mark fasteners with a symbol identifying the manufacturer and with symbol(s) indicating grade, class, type, and other identifying marks in accordance with reference or applicable standard.

2.02 SUBSTITUTIONS

- A. If materials required by these specifications become unavailable, because of Government priorities or other causes, substitute materials may be used.
- B. Obtain written approval to use substitute materials from the COJ State in the request for approval the amount of the adjustment, if any, to be made is favor of the Government.
- C. The Government's determination as to whether substitution will be permitted and as to what substitute materials may be used, shall be final and conclusive.
- D. If approved substitute materials are of less value to the Government or involve less cost to the Contractor than specified material, a contract adjustment will be made in favor of the Government. Where the amount involved or the importance of substitution warrants, a deductive modification to the contract will be issued.
- E. No payments in excess of prices offered in the Price Schedule will be made because of substitution of one material for another or because of use of one alternate material in place of another.

2.03 WORKMANSHIP

- A. Accurately manufacture and fabricate materials in accordance with best modern practice and requirements of these specifications, notwithstanding minor errors or omissions therein.
- B. Use liberal factors of safety and adequate shock-absorbing features in designs, especially for parts subjected to variable stress or shock, including alternating or vibrating stress or shock.

- C. Include provisions which prevent components from loosening for shock-absorbing features and parts subject to vibration.
- D. For rotating parts of motors and exciters, maximum unit stress due to runaway speed shall not exceed 2/3 of the yield point.

2.04 GOVERNMENT CONTRACT SOURCE QUALITY ASSURANCE

- A. Materials will be subject to inspection in accordance with clause at FAR 52.246-12 “Inspection of Construction” at following locations, as determined by the CO:
 - 1. At place of production or manufacture.
 - 2. At shipping point.
 - 3. At jobsite.
- B. To allow sufficient time to provide for inspection, submit at time of issuance, copies of purchase orders, including drawings and other pertinent information, covering material on which inspection will be made as advised by the CO, or submit other evidence if such purchase orders are issued verbally or by letter.
- C. Inspection of materials at locations specified above or waiving of inspection shall not be construed as being conclusive as to whether materials and equipment conform to contract requirements under the clause at FAR 52.246-12 “Inspection of Construction,” nor shall the Contractor be relieved thereby of the responsibility for furnishing material meeting the requirements of these specifications.
- D. Acceptance of materials will be made only at the jobsite.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Install products in accordance with manufacturer’s recommendations unless otherwise specified.

3.02 GOVERNMENT CONTRACT FIELD QUALITY ASSURANCE

- A. Final inspection and acceptance of materials will be made only at the jobsite after installation and testing.

END OF SECTION

SECTION 01 74 00 CLEANING AND WASTE MANAGEMENT

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include in prices offered in the schedule for other items of work except as specified.
 2. Cost of environmental site assessments are the Contractor's responsibility.

1.02 REFERENCE STANDARDS

- A. All references below are to be the current adopted edition at the time of solicitation for this contract. If any conflict arises between installing the new equipment in the existing installation, the contractor is to notify the COR. At which point, the CO will give direction on the course of action.
- B. Bureau of Reclamation (USBR)
1. RSHS
Reclamation Safety and Health Standards,
including revisions posted at
<http://www.usbr.gov/ssle/safety/RSHS/rshs.html>
- C. Code of Federal Regulations (CFR)
1. 40 CFR 261.3
Definition of Hazardous Waste
 2. 49 CFR 171-179
Transportation - Hazardous Waste
Regulations

1.03 DEFINITIONS

- A. Hazardous waste: Defined as hazardous by 40 CFR 261.3; or by other Federal, State, or local laws or regulations.

1.04 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals.
- B. RSN 01 74 00-1, Waste Production and Disposal Plan:
1. For each type of waste, list estimated quantity and planned disposal location.
- C. RSN 01 74 00-2, Waste Production and Disposal Records.

1. For each type of waste, list quantity and disposal location.
 2. Include certifications that waste was properly disposed.
 3. Hazardous wastes manifests.
- D. RSN 01 74 00-3, Environmental Consultant Resume:
1. Describe experience on similar project.
- E. RSN 01 74 00-4, Environmental Site Assessment.

1.05 QUALIFICATIONS

- A. Environmental consultant: Minimum 2 years experience in conducting environmental site assessments for similar construction.

1.06 REGULATORY REQUIREMENTS

- A. Comply with Federal, State, and local laws and regulations.
- B. Comply with RSHS.
- C. Conform to most stringent requirement in cases of conflict between specifications, regulatory requirements, and RSHS.

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1.07 PROJECT CONDITIONS

- A. Report waste materials discovered at jobsite to COR.
1. Cease work in areas where waste may be hazardous until waste materials are investigated by the Government.
 2. If waste is hazardous, the CO may order delays in time of performance or changes in work, or both.
 3. If such delays or changes are ordered, an equitable adjustment will be made in the contract in accordance with applicable clauses of the contract.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 PROGRESS CLEANING

- A. Keep work and storage areas free from accumulations of waste materials and rubbish.

3.02 FINAL CLEANUP

- A. Remove temporary plant facilities, buildings, concrete footings and slabs, rubbish, unused materials, concrete forms, and other similar materials which are not part of permanent work.
- B. Leave premises "broom clean."
- C. Clean interior and exterior glass surfaces exposed to view; remove temporary labels, stains and foreign substances; polish transparent and glossy surfaces; vacuum carpeted and soft surfaces.
- D. Clean equipment and fixtures to a sanitary condition.
- E. Clean filters of operating equipment.
- F. Clean debris from roofs, gutters, downspouts and drainage systems.
- G. Sweep paved areas and rake clean landscaped areas.

3.03 DISPOSAL

- A. Nonhazardous waste materials:
 - 1. Reuse at jobsite or reuse or recycle waste materials removed from jobsite whenever possible.
 - 2. Dispose of materials not reused at jobsite by removal from jobsite.
 - 3. Dispose of nonhazardous waste materials that are not reused or recycled at appropriately permitted disposal facilities.
- B. Hazardous Waste Disposal:
 - 1. Dispose by removal from jobsite.
 - 2. Recycle hazardous waste whenever possible.
 - 3. Dispose of hazardous waste materials that are not recycled at appropriately permitted treatment or disposal facilities.
 - 4. Transport hazardous waste in accordance with 49 CFR 171-179.
- C. Certification: Certify that wastes are disposed of in accordance with Federal, State, and local regulations.

3.04 SITE ASSESSMENT

- A. Upon completion of work, perform site assessment at following areas for work done under these specifications:
 - 1. Hazardous waste accumulation areas.

2. Petroleum dispensing and storage areas where aggregate storage of petroleum at jobsite was over 110 gallons.
 3. Hazardous material storage areas.
- B. Employ qualified environmental consultant to perform assessments.
- C. Demonstrate and document by appropriate analytical sampling that site contamination is less than State action cleanup levels.

3.05 RECORDS

- A. Keep records of types and amounts of waste materials produced.
- B. Keep records of waste material disposal.

END OF SECTION

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SECTION 01 78 30 PROJECT RECORD DOCUMENTS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. Include in prices offered in the Price Schedule for other items of work.

1.02 UPDATED RECORD DRAWINGS

- A. Update record drawings to document work or services performed for progress payments in accordance with the clauses at FAR 52.232-27 - Prompt Payment for Construction Contracts. The CO will retain from progress payments in accordance with WBR 1452.232-83 Payment for Technical Data if project drawings are not updated.

1. At a minimum, meet monthly with COR to review and approve progress made to full-sized record drawings prior to the submittal for progress payment. Meeting location to be approved by COR.
2. Verify with the COR that mark-ups to the drawings are current with the progress of the work prior to submitting invoice for progress payment.
3. CO will retain ___% from progress payments (exclusive of any payments for mobilization or preparatory work) until the Contractor receives final acceptance of the contract.

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1.03 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals.

- B. RSN 01 78 30-1, Electronic Media Release Form, Upon Request.

1. Refer to Article 1.04 Electronic Drawings below.

- C. RSN 01 78 30-2, Final As-built Drawings:

1. Certified marked sets: As-built drawings that have been certified by Contractor as up to date and accurate in the description of work.
2. Submit to COR for approval prior to commissioning, training, and/or Government possession.
3. For Information Only (FIO) Drawings: Revision date and details.

- D. RSN 01 78 30-3, Final O&M Manuals

1. Provide O&M Manuals.

- a. In both hardcopy and electronic form.
- b. Number of hard copies as indicated in Section 01 33 00 – Submittals.

1.04 ELECTRONIC DRAWINGS

- A. CADD Files, will be distributed upon request, after:
1. Notice to Proceed.
 2. Contractor signs form in Section 51 00 50 – Electronic Media Release.

1.05 RECORD DRAWINGS

- A. Maintain 2 sets of full-size prints of contract drawings marked to show accurate and complete records of as-built conditions.
1. Keep drawings at the jobsite and mark as work progresses.
 2. If FIO drawings require changes, mark with revisions.
- B. Use the following to represent changes.
1. Supplementary notes, legends, and details necessary to clearly portray as-built construction.
 2. In lieu of referencing, incorporate clarifications from other documents such as RFI's, where as-built construction is identified.
 3. Include pictures, sketches or other representations to further explain as-built construction. Do not provide in lieu of as-built drawings.
 4. Mark and dimension to show variations between actual construction and that indicated or specified in contract documents.
 - a. Include buried or concealed construction and utilities.
 - b. Include existing items, topographic features, and utility lines revealed during construction which differ from those shown on contract drawings.
 5. Mark to define construction actually provided where choice of materials or methods is permitted in specifications, or where variations in scope or character of work from that of the original contract are authorized.
- C. Mark drawings in the following colors:
1. Red: Additions.
 2. Green: Deletions.
 3. Blue: Notations necessary for explanation of markings.
- D. Allow the Government to review the drawings at all times.

- E. Following successful commissioning of each unit or Government possession, submit As-built drawings with changes resulting from commissioning and training to COR for review and approval.
- F. Upon completion of work, sign marked as-built drawings.
 - 1. Sign and date each as-built drawing as certified correct. If no revisions were necessary to illustrate as-built conditions mark the drawing with “As-Built, No Changes.”
 - 2. Only certify FIO drawings which require changes.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

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SECTION 03 62 20
NONSHRINK GROUT FOR EQUIPMENT AND METALWORK

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include in prices offered in the Price Schedule for items of work requiring nonshrink grout.

1.02 REFERENCE STANDARDS

- A. ASTM International (ASTM)
1. ASTM C150/C150M-19a Portland Cement
 2. ASTM C1107/C1107M-17 Packaged Dry, Hydraulic-Cement Grout (Nonshrink)

1.03 SUBMITTALS

- A. Submit the following in accordance with Section 01 13 00 Submittals.
- B. RSN 03 62 20-1, Packaged Nonshrink Grout Data:
1. Manufacturer's product data.
 2. Manufacturer's certification that product meets specifications.
 3. Manufacturer's instructions:
 - a. Surface preparation, mixing, installation, finishing, curing, and protection recommendations.
 - 1) Include specific recommendations based on anticipated jobsite temperatures and environmental conditions.
 - 2) Include manufacturer's recommended methods for float finishing grout.
 - 3) Include manufacturer's recommended minimum temperature and protection period for protecting grout from freezing and damage after placement.

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PART 2 PRODUCTS

2.01 PACKAGED NONSHRINK GROUT

- A. Pre-mixed commercial grout mixture:
 - 1. ASTM C1107.
 - 2. Cement in mixture: ASTM C150, Type II.
- B. Water:
 - 1. Clean.
 - 2. Free of detrimental quantities of silt, organic matters, salts, or other impurities.
 - 3. As recommended by grout manufacturer.
- C. Mix: In accordance with manufacturer's instructions to consistency recommended by manufacturer for application.

PART 3 EXECUTION

3.01 PREPARATION

- A. Clean and prepare base concrete surface to which nonshrink grout will be bonded in accordance with manufacturer's instructions or as directed by CORA.
 - 1. Roughen surface.
 - 2. Remove dirt, laitance, loose or defective concrete, curing compound, coatings, and other foreign material.
 - 3. Wash with water.
- B. Lubricate concrete surface by washing with water immediately before placing nonshrink grout when delay occurs between washing and placing nonshrink grout.

3.02 PLACING

- A. Place nonshrink grout in accordance with manufacturer's instructions to completely fill space to be grouted.
- B. Use forms, where required, to confine nonshrink grout.

3.03 CURING

- A. Cure exposed surface of packaged nonshrink grout by method recommended by manufacturer.

- B. Do not apply loads sooner than 72 hours after placement and only after nonshrink grout has attained compressive strength of at least 3,000 lb/in².

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SECTION 05 50 00 METAL FABRICATIONS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include cost of furnishing and installing valve or reducer support, including anchor bolts, in the Price Schedule for Valve Removal and Installation.

1.02 REFERENCE STANDARDS

- A. All references below are to be the current adopted edition at the time of solicitation for this contract. If any conflict arises between installing the new equipment in the existing installation, the contractor is to notify the COR.
- B. American Institute of Steel Construction (AISC)
1. AISC 325 Manual of Steel Construction
- C. American Society of Mechanical Engineers (ASME)
1. ASME B18.2.1 Square and Hex Bolts and Screws, Inch Series
 2. ASME B18.21.1 Washers: Helical Spring-Lock, Tooth Lock, and Plain Washers (Inch Series)
- D. ASTM International (ASTM)
1. ASTM A 36 Carbon Structural Steel
 2. ASTM A 193 Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service
 3. ASTM A 194 Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High Pressure or High Temperature Service, or Both
 4. ASTM A 240 Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications
 5. ASTM A 992 Steel for Structural Shapes for Use in Building Framing
 6. ASTM F 844 Washers, Steel, Plain (Flat), Unhardened for General Use

- E. American Welding Society, Inc. (AWS)
 - 1. AWS D1.1/D1.1M Structural Welding Code - Steel
- F. Commercial Item Description (CID)
 - 1. CID A-A-1923A Shield, Expansion (Lag, Machine and Externally Threaded Wedge Bolt Anchors)

1.03 DEFINITIONS

- A. Miscellaneous metalwork: Where either shown on the drawings or specified elsewhere in this Section or these specifications means metal fabrications as used in this Section.

1.04 SUBMITTALS

- A. RSN 05 50 00-1, Shop Drawings:
 - 1. Shop drawings of supports including all dimensions, materials, all applicable standards specifications, center of gravity, and total weight.

1.05 QUALIFICATIONS

- A. Qualify welders in accordance with AWS D1.1 using procedures, materials, and equipment of the type required for the work.
- B. Make welders certifications available to COR as needed.

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1.06 DELIVERY, STORAGE, AND HANDLING

- A. Protect from corrosion, deformation, and other types of damage.
- B. Store items in an enclosed area free from contact with soil and weather.
- C. Remove and replace damaged items with new items.

PART 2 PRODUCTS

2.01 MATERIALS

- A. Arc-Welding Electrodes:
 - 1. Filler metal and required shielding gases or fluxes: AWS D1.1.
 - 2. Filler metal for steel, minimum tensile strength: 70,000 psi.
- B. Bolts, Nuts, And Washers:
 - 1. Nuts: ASTM A 194, Grade 8 stainless steel.
 - 2. Washers:

- a. Material:
 - 1) For use with ASTM A 193 bolts: ASTM A 240, Type 304.
 - 2) ASTM F844.
- b. Dimensions: ASME B18.21.1
- 3. Bolts:
 - a. ASTM A 193, except anchor bolts and studbolts.
 - b. Grade B8 stainless steel.
 - c. Length of bolt threads: ASME B18.2.1.
 - d. Thread class: 2 free-fit, American National coarse-thread series.
- C. Supports for Reducers and Valves:
 - 1. Fabricate either a reducer support or a valve support based on Fixed-Cone Valve manufacturer's recommendation. Only one support per pipeline is required.
 - 2. Fabricate supports as shown on the drawings.
- D. Structural steel:
 - 1. Shapes except wide flange sections: ASTM A 36.
 - 2. Wide flange sections: ASTM A 992

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2.02 ANCHORS

- A. Expansion Anchors:
 - 1. CID A-A-1923A, Type 4, 304 or 316 stainless.
 - 2. Bolt length: As shown on drawings. If not shown, provide bolt length with 4-1/2 inch minimum embedment.

2.03 FABRICATION

- A. Fabricate metalwork in accordance with AISC 325 and these specifications.
 - 1. Perform welding and related work in accordance with AWS D1.1.
- B. If straightening is necessary, use methods that will not injure the metal.
- C. After shop work completion and before galvanizing, if required, clean material of rust, loose scale, dirt, oil, grease, slag from welded areas, and other foreign substances.

PART 3 EXECUTION

3.01 PREPARATION

- A. Where locations and dimensions of miscellaneous metalwork shown on drawings are dependent upon existing equipment and/or equipment furnished, confirm locations and dimensions prior to fabrication of miscellaneous metalwork.

3.02 INSTALLATION

A. Anchors:

1. Contractor may use suitable expansion anchors meeting requirements of this Section in lieu of embedded concrete inserts shown on drawings.
2. Drill holes for anchors straight and true and of diameter recommended by anchor manufacturer.
3. Install anchors in accordance with manufacturer's recommendations.
4. Follow manufacturer's recommendations when embedded steel or reinforcement is encountered during drilling for anchors.
5. When drilling water is used, clean surfaces of concrete to remain exposed immediately to prevent discoloration.

B. Grout:

1. Where shown on drawings, fill spaces under metalwork completely as required with nonshrink grout in accordance with Section 03 62 20 – Nonshrink Grout For Equipment and Metalwork.
2. Clean surfaces of metalwork to be in contact with or embedded in concrete or grouting mortar in accordance with method B of Section 09 96 20 - Coatings.

C. Holes in Metalwork:

1. Drill, or drill and tap as required, holes in metalwork required for installation.

D. Painting:

1. Paint surfaces of miscellaneous metalwork exposed after installation; except aluminum, galvanized steel, and cast iron; in accordance with Section 09 96 20 - Coatings.

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END OF SECTION

SECTION 09 96 20 COATINGS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. General:

1. Payment for various items to be coated includes the following costs:
 - a. Containment, safety, and health activities for coatings work.
 - b. Surface preparation, such as abrasive blasting, cleaning, and grinding.
 - c. Coating application and remedial work.

B. Cost:

1. Include cost of coating the following in applicable lump sum prices offered in the Price Schedule for Valve Removal and Installation:
 - a. Exterior surfaces of new 24-inch fixed-cone valves.
 - b. Interior and exterior surfaces of new pipe reducer.
 - c. Valve or reducer support.
 - d. Interior and exterior surfaces of new 30-in butterfly valves.

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1.02 REFERENCE STANDARDS

- A. All references below are to be the current adopted edition at the time of solicitation for this contract. If any conflict arises between installing the new equipment in the existing installation, the contractor is to notify the COR. At which point, the CO will give direction on the course of action.

B. ASTM International (ASTM)

- | | | |
|----|---------------|---|
| 1. | ASTM C920 | Elastomeric Joint Sealants |
| 2. | ASTM D870 | Testing Water Resistance of Coatings Using Water Immersion |
| 3. | ASTM D2794-93 | Resistance of Organic Coatings to the Effects of Rapid Deformation (Impact) |
| 4. | ASTM D4060 | Abrasion Resistance of Organic Coatings by the Taber Abraser |
| 5. | ASTM D4285-83 | Indicating Oil or Water in Compressed Air |

- | | | |
|----|---------------|--|
| 6. | ASTM D4541 | Pull-Off Strength of Coatings Using Portable Adhesion Testers |
| 7. | ASTM D4587-11 | Conducting Tests on Paint and Related Coatings and Materials Using a Fluorescent UV-Condensation Light- and Water-Exposure Apparatus |
| 8. | ASTM D5894 | Cyclic Salt Fog/UV Exposure of painted Metal, (Alternating Exposures in a Fog/Dry Cabinet and a UV/Condensation Cabinet) |
| 9. | ASTM G8-96 | Cathodic Disbonding of Pipeline Coatings |
- C. International Organization for Standardization (ISO)
- | | | |
|----|------------|--|
| 1. | ISO 8502-3 | Assessment of dust on Steel Surface Prepared for Painting (Pressure-Sensitive Tape Method) |
|----|------------|--|
- D. The Society for Protective Coatings (SSPC)/NACE International (NACE)
- | | | |
|-----|---------------------------|--|
| 1. | SSPC-AB1 | Mineral and Slag Abrasives |
| 2. | SSPC-AB2 | Cleanliness of Recycled Ferrous Metallic Abrasives |
| 3. | SSPC-AB3 | Newly Manufactured or Re-Manufactured Steel Abrasives |
| 4. | SSPC-Guide 15 | Field Methods for Retrieval and Analysis of Soluble Salts on Substrates and Other Nonporous Substrates |
| 5. | SSPC-PA1 | Shop, Field, and Maintenance Painting of Steel |
| 6. | SSPC-PA2 | Measurement of Dry Paint Thickness with Magnetic Gages |
| 7. | SSPC-QP1 | Evaluating Painting Contractors (Field Application to Complex Industrial Structures) |
| 8. | SSPC-QP3-10/AISC 420 | Evaluating the Qualification of Shop Painting Contractors |
| 9. | SSPC-SP1 | Solvent Cleaning |
| 10. | SSPC-SP5/NACE 1 | White Metal Blast Cleaning |
| 11. | SSPC-SP5 (WAB)/NACE WAB 1 | White Metal Wet Abrasive Blast Cleaning |
| 12. | SSPC-SP10/NACE 2 | Near-White Blast Cleaning |

- | | | |
|-----|----------------------------|--|
| 13. | SSPC-SP10 (WAB)/NACE WAB 2 | Near-White Metal Wet Abrasive Blast Cleaning |
| 14. | SSPC-SP11 | Power Tool Cleaning to Bare Metal, Level 3 minimum requirement |
| 15. | SSPC-SP WJ-4/NACE WJ-4 | Surface Preparation Standard – Waterjetting Cleaning of Metals – Light Cleaning |
| 16. | SSPC-TR2/NACE 6G198 | Wet Abrasive Blast Cleaning |
| 17. | SSPC-VIS1 | Guide and Reference Photographs for Steel Surfaces Prepared by Abrasive Blast Cleaning |
| 18. | SSPC-VIS3 | Visual Standard for Power- and Hand-Tool Cleaned Steel |
| 19. | SSPC-VIS4/NACE-VIS7 | Guide and Reference Photographs for Steel Surfaces Prepared by Waterjetting |
| 20. | SSPC Paint Manual | Volume 1, Good Painting Practices, Fifth Edition 2016 |
| 21. | NACE SP 0188 | Discontinuity (Holiday) Testing of New Protective Coatings on Conductive Substrates |
| 22. | NACE SP 0287 | Field Measurement of Surface Profile of Abrasive Blast Cleaned Steel Surfaces Using a Replica Tape |
| 23. | NACE SP 0178 | Detail, Finish, and Design Considerations for Immersion Service. |

1.03 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals.
- B. RSN 09 96 20-1, Approval Data:
 - 1. For each coating material:
 - a. Manufacturer's product data, application, and SDS sheets.
 - b. Supplier's name, address, and phone number.
 - c. Identification of "Items to be coated" including sub-letter and sub-number listed in Coating Tabulations.
 - 2. For each caulk and filler:
 - a. Manufacturer's product data, application, and MSDS sheets.
 - b. Supplier's name, address, and phone number.

- c. Identification of areas of use including sub-letter and sub-number listed in Coating Tabulations.
- 3. For each substituted coating material “equal” to specified product in Coating Tabulation:
 - a. List of not less than 3 similar successful material applications:
 - 1) Project name, location, type of structure, and application dates.
 - 2) Owner’s name, address, and telephone number.
 - b. Certification that material meets or exceeds specified requirements:
 - 1) Manufacturer’s certification including technical representative’s signature and date of signature.
 - 2) Independent laboratory’s certified test reports.
- C. RSN 09 96 20-2, Final Approval Data:
 - 1. For each coating material:
 - a. Purchase orders including order number, date, manufacturer’s designated product name, batch number(s), and quantity purchased.
- D. RSN 09 96 20-3, Qualifications:
 - 1. Coating Contractor:
 - a. Current SSPC-QP1 and/or SSPC-QP3 certifications.
 - b. List not less than five comparable completed jobs for each coating type including owner’s name, address, and telephone numbers completed within the last five years.
 - 2. Coating Applicators:
 - a. Current Coating Application Specialist (CAS)
 - b. List of not less than five comparable jobs for each coating type including owner’s name, address, and telephone number completed within the last five years.
 - 3. Coating Contractor Inspectors:
 - a. Qualified to inspect specified coating materials by coating experience or certifications. Provide the following:
 - 1) Successfully completed training in the application and inspection of coating materials similar to materials and conditions specified. Provide documentation.
 - 2) Letter from contractor listing the personnel performing QC inspections, certified at a minimum as NACE Coating Inspector Level 1 (CIP) or as a SSPC Protective Coatings Inspector (PCI).

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- E. RSN 09 96 20-4, Quality Control Plan:
 - 1. Quality Control Manager:
 - a. Name and title.
 - 2. Quality Control Procedure:
 - a. Quality control plan.
 - 1) Procedure for addressing identified non-conformities.
- F. RSN 09 96 20-5, Contractor Quality Testing Report:
 - 1. Comprehensive report of daily quality control test reports.

1.04 DELIVERY, STORAGE, HANDLING

- A. Deliver materials to jobsite in original, undamaged, unopened containers labeled with manufacturer's name, designated product name, batch number, date of manufacture, and any special instructions.
- B. Deliver materials in containers not larger than 5-gallons as packaged by manufacturer unless suitable equipment is provided at jobsite to handle and thoroughly mix materials in larger containers.
- C. Store coating materials and thinners in accordance with the manufacturer's recommendations, and as a minimum, store off the ground, under cover, and at 50 to 95 degrees Fahrenheit.
- D. Do not use coating material which has exceeded manufacturer's specified storage stability period (shelf life).

1.05 AMBIENT CONDITIONS

- A. Maintain environmental conditions to meet coating manufacturer's requirements during surface preparation, coating application, and curing period. Provide heating or cooling, and dehumidification as needed. Complete curing before placing the coating systems into service.
- B. Coatings shall be applied when the substrate surface temperature is 5 degrees F or more above the dew point temperature and for at least 8 hours after application.

1.06 PROJECT CONDITIONS

- A. Existing coatings may contain hazardous materials. Use caution when disturbing or removing coatings. Refer to Sections 02 82 20, 02 83 30, 02 83 33, 02 84 40.
- B. Galvanic isolation of dissimilar metals shall be required. Coatings in this specification section shall not serve as the primary galvanic isolation.

- C. Contractor shall comply with applicable federal, state, and local air pollution and environmental control regulations for surface preparations, blast cleaning, disposal of spent materials and debris, and coatings application.

PART 2 PRODUCTS

2.01 EQUIPMENT

- A. Surface preparation and spray application equipment:
1. Sized in accordance with coating manufacturer's recommendations.
 2. Provide:
 - a. Pressure gauges and pressure regulators.
 - b. Air supply lines free from oil and moisture.

2.02 ABRASIVES

- A. Mineral and slag abrasives: Meets SSPC-AB1, type I (natural minerals) and type II (slags), class A.
- B. SSPC-AB2 for recycled cleanliness.
- C. SSPC-AB3 Class I (steel) or II (iron) for angular shaped grain. Shot abrasives are not acceptable.
- D. Commercially available products or processes to render the hazardous waste stream non-hazardous and not exceeding toxicity threshold limits.
1. Additive stabilizers for leachable toxic metal wastes are allowed, except elemental iron is not permitted.

2.03 COATINGS

- A. Specified in Coating Tabulations:
1. Apply only one coating system per option in Coating Tabulations.
 2. Apply compatible products of same manufacturer for coating system components.
- B. Volatile Organic Compounds (VOC): Do not exceed maximum permitted by Federal, State, and local air pollution control regulations as supplied in container or by addition of thinner material.
- C. Factory color or tint only. Do not color or tint at jobsite.
- D. Use manufacturer's recommended thinners for coating materials.
- E. Do not use accelerator products unless approved by COR.

2.04 FILLERS AND CAULKS

A. Caulks:

1. Coating manufacturer's standard or compatible caulk material.
2. Polyurethane, silicone, or silicone acrylic caulk material: Meets or exceeds ASTM C920 type S or M, grade NS, class 25, suitable for water immersion service.

B. Fillers:

1. Carboguard 501, as manufactured by Carboline, www.carboline.com.
2. Devmat 142HB, as manufactured by International Paint, www.international-pc.com.
3. PPG Amercoat 114 A, as manufactured by PPG, www.ppgpmc.com.
4. Steel Seam FT910, as manufactured by Sherwin Williams, www.sherwin-williams.com.
5. Surfacing Epoxy Series 215 was manufactured by Tnemec, www.tnemec.com.
6. Provide one of the above fillers, or equal with following essential characteristics:
 - a. Coating manufacturer's standard filler material.
 - b. Approved use by manufacturer.

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2.05 MISCELLANEOUS MATERIALS

A. Deionized Water.

B. Chlor-Rid as manufactured by Chlor-Rid, www.Chlorid.com,

C. HoldTight 102 as manufactured by Holdtight Solutions, www.holdtight.com,

D. Chlor-Gone, as manufactured by Aquaron, www.aquaron.com

E. Provide one of the above salt removers, or equal with following essential characteristics:

1. Commercially available additive for soluble salt removal.
2. Water soluble.
3. Biodegradable.
4. Zero volatile organic compounds.

PART 3 EXECUTION

3.01 SHOP PAINTING

- A. Shop coated and lined equipment shall be primed and finish in the shop and touched up in the field with identical coating materials after installation and in accordance with the coating's manufacturers standard surface preparation and application instructions.

3.02 GALVANIC AND DIELECTRIC ISOLATION

- A. Galvanic isolation of dissimilar metals shall be required.
- B. Coatings shall not serve as the primary galvanic isolation.

3.03 PROTECTION OF ADJACENT SURFACES

- A. Remove, mask, or otherwise protect items or surfaces not coated from contamination and damage during cleaning and coating operations.
- B. Examples include protecting mating and machined surfaces, mechanical and electrical equipment (open or enclosed), instruction and similar plates, and wet and newly coated surfaces.

3.04 SURFACE PREPARATION

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- A. Remove or repair surface irregularities and impart profile as specified:
 - 1. Welds:
 - a. Welds: Smooth and continuous.
 - b. Weld spatter, protrusions, laminations, slag, buckshot, and slivers. Remove and grind smooth in accordance with NACE SP 0178.
 - c. Undercuts and porosity: Fill with weld material and grind smooth.in accordance with NACE SP 0178.
 - d. Projections, sharp edges, high points, fillets, and corners: Grind smooth to a radius of at least 1/16-inch.
 - 2. Substrate defects:
 - a. Pitting, gouges, scratches, porosity, and similar:
 - 1) Grind smooth.
 - 2) Defects exceeding 20-percent reduction in wall thickness require weld repairs.
 - 3. Grind full length of edges and corners smooth to a radius of at least 1/16-inch.
- B. Surface preparation method:

1. See Coating Tabulations.
 - a. Method A: SSPC-SP1.
 - b. Method C: SSPC-SP10/NACE 2.
 - c. Method C-1: Repair of defective or damaged coated areas.
 - 1) For metal substrate:
 - a) SSPC-SP1.
 - b) SSPC-SP10/NACE 2.
 - c) SSPC-SP11 where abrasive blasting is impractical.
 - i. Level 3 shall be required with no amount of residue permitted in pit bottoms and surface a profile of 1-mil minimum unless a high value is specified.
 - 2) For surrounding intact coating (nominal 2- to 3-inch margin):
 - a) SSPC-SP1.
 - b) Feather abrupt edges and around repair area by hand or power tool with non-woven pad into adjacent tightly bonded coatings.
 - c) Roughen or abrade surface in accordance with manufacturer's recommendations to achieve matted or lusterless finish.
 2. Compare prepared steel surfaces to following visual reference photographs for allowable visible contaminants and stains:
 - a. SSPC-VIS1 for abrasive blast cleaning.
 - b. SSPC-VIS3 for power and hand tool cleaning.
 3. Assessment of dust contaminants:
 - a. Perform clear tape test method in accordance with ISO 8502-3.
 - b. Perform three tests per shift.
 - c. Acceptance Criteria: Degree of cleanliness shall meet Class 3.
- C. Surface profile:
1. Prepare as specified in Coating Tabulations.
 2. Where not specified, prepare in accordance with manufacturer's instructions for metallic substrates or existing coating surfaces and service environment.
 3. Where manufacturer's instructions do not specify a surface profile, prepare blasted surfaces to following profile:
 - a. Atmospheric Service Environments: 1-mil or greater angular profile and less than specified millage of first applied coat.

- b. Burial and Immersion Service Environments: Angular profile 2-mil or greater and less than specified millage of first applied coat.
 - 4. Inspect surface profile in accordance with NACE SP 0287:
 - a. Use replica tape suitable for surface profile depth range.
 - b. Perform at least two tests per 1,000-square feet.
- D. Soluble salts:
 - 1. Measure concentration of chlorides, nitrates and sulfates in accordance with SSPC Guide 15.
 - 2. Perform at least two tests per 1,000-square feet of surface area. Include corrosion pit areas.
 - 3. Acceptance criteria: Do not exceed 3-micrograms per square centimeter for chlorides and 5-micrograms per square centimeter for nitrates and sulfates.]
- E. Re-clean or perform additional surface preparation of contaminated surfaces before coating application.

3.05 COMPRESSED AIR EQUIPMENT

- A. Keep air compressor and spray application equipment lines free of oil and moisture during work.
- B. Compressed air quality:
 - 1. Perform tests in accordance with ASTM D4285.
 - 2. Perform test at the beginning of every shift.

3.06 FILLERS AND CAULK APPLICATION

- A. Caulk areas including crevices, gaps, plate seams, skip welds, and conduit penetrations.
- B. Fill areas including prepared pits and/or porosity, plate seams, and gaps at bolts and rivets.
- C. Apply in a uniform texture, neatness, and color matched appearance.
- D. Apply in accordance with manufacturer's instructions and achieve tight bond to substrate or previous coats.

3.07 COATING APPLICATION

- A. Coating shall be done in a workmanlike manner to produce an even film of uniform thickness. The finished surfaces shall be free from runs, drops, ridges, waves, laps, sags, curtains, brush marks, and variations in color, texture, and finish.

- B. Stripe coat:
1. Brush apply stripe coat to each coat applied.
 2. Apply stripe coat prior to spray application of the primer, when practical.
 3. Apply to edges, corners, interior angles, pits, plate seams, crevices, rivets, nuts, threads, bolts, washers, welds, and similar features.
- C. Apply number of coats and coating thickness specified in Coating Tabulations.
1. Alternate coat colors to differentiate between coats.
 2. Apply in accordance with manufacturer's instructions.
 3. Apply in accordance with SSPC PA1.
- D. Prevent overspray or dry spray. When such conditions occur, clean surfaces prior to application of subsequent coats.
- E. Re-clean contaminated surfaces between coats.
- F. Dry Film Thickness (DFT):
1. After each coat, inspect hardened coating system for DFT compliance in accordance with SSPC PA2.
 2. Acceptance Criteria: no single spot measurement less than 80 percent minimum or in excess of 120 percent maximum specified.
- G. Discontinuity (Holiday) Testing:
1. Inspect in accordance with NACE SP 0188:
 - a. Use maximum test voltage for DFT as recommended by coating manufacturer to prevent coating damage.
- H. Coating Adhesion:
1. Test in accordance with ASTM D4541 at the direction of the COR.

3.08 CONTRACTOR FIELD QUALITY TESTING

- A. Inspection Equipment:
1. Calibrate inspection devices and equipment in accordance with the manufacturer's instructions.
 2. Verify calibration prior to every use.
- B. Qualified personnel shall conduct tests and sign reports.
1. Perform tests in presence of the COR:
 2. Provide rigging for COR to observe inspections.

- C. Make daily reports for review by the COR, including:
 - 1. Date of work.
 - 2. Description of areas and work performed.
 - 3. Environmental conditions.
 - 4. Results from the following quality control tests:
 - a. Soluble salt.
 - b. Surface profile.
 - c. Visual comparison of prepared surfaces.
 - d. Assessment of dust contaminants.
 - e. Compressed air quality.
 - f. Dry film thickness (DFT).
 - g. Discontinuity (Holiday) testing.
 - h. Coating adhesion testing (when directed by the COR).
- D. Final inspection will take place after repairs, but before placing into service.

3.09 REPAIR AND REMEDIAL WORK

- A. Repair defective or damaged areas of coatings in accordance with coating manufacturer's recommendations.

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3.10 COATING TABULATIONS

Tabulation No. 1		
Items to be coated: - Exterior surfaces of new 24-inch fixed cone valves - Interior and exterior surfaces of the new reducers - Valve or reducer supports - Interior and exterior surfaces of new 30-inch butterfly valves		
Notes: - Do not paint aluminum-brass, bronze, and stainless steel metalwork unless specified. - Do not coat seals, mating surfaces, and machined surfaces where paint could interfere with proper operation or fit. Coat exposed machined surfaces after installation. - Coat miscellaneous metal work not otherwise tabulated.		
Coating materials – Option 1 Epoxy	Number and thickness of coats	Surface preparation method
Carboline: Carboguard 891 PPG: Amerlock 2 or 400 or Amercoat 235 Sherwin Williams: Duraplate 235 or Macropoxy 646 Tnemec: Epoxoline Series N69 or V69 Color and gloss: sky blue and gloss	3 or more base coats, apply at 6 to 8 mils DFT per coat, plus stripe coats. Total base coat system excluding stripe coats 18-mil DFT, minimum 24-mil DFT, maximum	SSPC-SP1 followed by: for steel: SSPC-SP5/NACE 1 or SP10/NACE 2 for wet abrasive blast cleaning with soluble salt remover: SSPC-TR2/NACE 6G198 2-4 mil surface profile Follow manufacturer's surface preparation and application instructions to apply subsequent coats
Coating materials – Option 2 Moisture-cured urethane	Number and thickness of coats	Surface preparation method
Sherwin Williams: Corrothane I Miozinc Corrothane I Coal Tar	1 or more prime coats, apply at 3 to 5 mils DFT per coat, plus stripe coats.	for steel: SSPC-SP5/NACE 1 or SP10/NACE 2

Tabulation No. 1		
Wasser: MC Miozinc 2.8 or 100 MC Tar 2.8 or 100	2 or more coats of tar, apply at 5 to 7 mils DFT per coat. Total base coat system excluding stripe coats: 13-mil DFT, minimum 19-mil DFT, maximum	for wet abrasive blast cleaning with soluble salt remover: SSPC-TR2/NACE 6G198 2 mil surface profile Follow manufacturer's surface preparation and application instructions to apply subsequent coats
PERFORMANCE REQUIREMENTS:		
Testing Protocol		Acceptance Criteria
Fresh/deionized water immersion test, ASTM D 870:		passes, 2 years test with water at ambient temperature, no blisters on scribed or unscribed sides.
Dilute Harrison immersion test, Modified ASTM D 870:		passes, 2 years test with water at ambient temperature, no blisters on scribed or unscribed sides.
Abrasion resistance, ASTM D 4060, CS-17 wheel, 1,000 cycles, 1-kg load:		Epoxy: 100 milligram loss or less MC Urethane: 250 milligram loss or less
Direct impact, ASTM D 2794:		30 inch-pounds
Pulloff adhesion, ASTM D 4541, annex A2, type II tester:		1000 psi or greater
Cyclic testing salt fog/UV, ASTM D 5894:		1/4 inch or less undercutting at 3000 hrs
Cathodic disbondment, ASTM G 8:		1/4 inch or less disbondment 120 day test

END OF SECTION

PART 1 GENERAL

A. General Electrical System:

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A. All references below are to be the current adopted edition at the time of solicitation for this contract. If any conflict arises between installing the new equipment in the existing installation, the contractor is to notify the COR. At which point, the CO will give direction on the course of action.

- Basic Electrical Materials and Methods
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- 2. ASTM A153/A153M Zinc Coatings (Hot-Dip) on Iron and Steel Hardware
- D. Institute of Electrical and Electronics Engineers (IEEE)
 - 1. IEEE C2 National Electrical Safety Code (NESC)
- E. InterNational Electrical Testing Association, Inc. (NETA)
 - 1. NETA ATS Acceptance Testing Specifications
- F. National Fire Protection Association (NFPA)
 - 1. NFPA 70 National Electrical Code (NEC)

1.03 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 – Submittals.
- B. RSN 26 05 02-1, Initial Inspection.
- C. RSN 26 05 02-2, Design Documentation.

1.04 QUALIFICATIONS

- A. Welder Qualifications: In accordance with ASME B16.34.

1.05 MAINTENANCE

- A. Provide special tools and accessories required for installation or normal operation and maintenance of equipment and installations. Include studs, bolts, and nuts other than United States Standard used in equipment or installation.
- B. Deliver special tools and spare parts for equipment to project location as approved by COR. Provide shipments with item list(s).
- C. Special tools and spare parts shall be onsite prior to installation of equipment.

1.06 ADMINISTRATIVE REQUIREMENTS

- A. Perform the following sequence of activities in order where the subsequent activity cannot begin until the preceding activity has finished.
 - 1. Bid electrical work depicted in baseline scope.
 - a. Assume the following quantities for bidding the installation work:
 - 1) Flex Conduit: 20 feet
 - 2) Rigid Conduit, 1 inch: 400 feet
 - 3) Rigid Conduit, 2 inch: 50 feet

- 4) Panel, 12 circuit, 200 amps, 240kV: 1
- 5) Panel, 34 circuit, 200 amps, 240kV: 1
2. Inspect Outlet Works power supply system to the limits shown in drawings and Section 51 00 00 – Information Available to Offerors.
3. Submit an updated design of the power supply system stamped by a qualified Professional Electrical Engineer.
4. Submit product data for approved design.
5. Install new design and perform testing.
6. Submit as-built drawings stamped by a qualified Professional Electrical Engineer.

PART 2 PRODUCTS

2.01 SYSTEM DESCRIPTION

A. Design Requirements

1. New power supply system shall provide power to valves installed as part of this project as well as all other existing equipment.
2. New power supply system shall meet all applicable codes.
3. Two new panels shall be included in the design: one outside and one inside as shown in 51 00 00 – Information Available to Offerors.
4. Design documentation shall cite relevant code as justification for any deviation from what is shown in the Government's baseline design.

2.02 MATERIALS AND EQUIPMENT

- A. Conform to NFPA 70 and IEEE C2, and Section 01 60 00 – Product Requirements.
- B. Provide galvanized steel mounting brackets, support structures, bolts, nuts, and washers for electrical equipment in accordance with ASTM A 123 and A 153, unless otherwise specified.
- C. Provide galvanized steel or non-corrosive metal for mounting bolts, nuts, and washers for minor items of electrical equipment and lighter weight items. Do not use cadmium-plated mounting hardware.
- D. Change designs as required where Contractor-furnished electrical equipment and materials differ in size, type, ratings, or other physical properties from designs in these specifications. Contracting Officer will approve changes at Contractor's expense, unless Contractor can demonstrate that changes are necessary regardless of manufacturer.
- E. Provide special tools and appliances furnished by manufacturer for maintenance and adjustment of manufacturer's electrical equipment.

PART 3 EXECUTION

3.01 INITIAL INSPECTION

- A. Inspect equipment to be removed or replaced per baseline design as shown in drawings and Section 51 00 00 – Information Available to Offerors.
- B. Coordinate with COR to confirm operational status of existing equipment in the Outlet Works. Submit operational status as part of inspection report.
- C. Submit proposed list of deviations from the baseline design to ensure final design meets applicable codes and is sufficient for providing power to all existing and new equipment. Include applicable codes that require deviation from the baseline design.

3.02 DESIGN DOCUMENTATION

- A. Design Documentation to include:
 - 1. Updated single line diagram, signed by a Professional Electrical Engineer.
 - 2. List of electrical equipment to be furnished and installed as part of the design.
 - 3. Testing plan for the electrical equipment in the final design.
 - 4. Cable, conductor, and conduit schedule.

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3.03 INSTALLATION

- A. Do not remove existing equipment or install electrical equipment until approval drawings and data and factory test reports for associated equipment have been approved by Government and equipment is onsite, unless otherwise approved by the COR.
- B. Electrical Installations, Assembly Operations, and Adjustments: Comply with NFPA 70 and IEEE C2.
- C. Make electrical installations complete and ready for service.
- D. Install electrical equipment in accordance with directions furnished by manufacturer's instruction books.
- E. Tighten nuts used in electrical equipment assembly with torque wrenches to torque values recommended by equipment manufacturers.
- F. Drill holes in bolted steel structures and provide fastenings required for mounting or installing electrical equipment and materials.
- G. Installation of electrical equipment includes:
 - 1. Leveling and grouting channel bases.
 - 2. Drilling holes, furnishing hardware, and assembling components to each other.

3. Furnishing materials for and making all connections correctly in accordance with final wiring diagrams.
 4. Tagging wires and cables at each end.
 5. Testing.
- H. Grout electrical equipment installed on concrete foundations in place to provide full and even bearing.

3.04 REPAIR

- A. Repair damaged painted surfaces of equipment to match original finish.

3.05 CONTRACTOR FIELD QUALITY CONTROL AND TESTING

- A. In accordance with requirements of FAR 52.246-12 Inspection of Construction, IEEE C2, and the various equipment Sections.

3.06 FIELD QUALITY ASSURANCE

- A. Government will inspect equipment during construction. Acceptance of equipment will be made after equipment is operational.
- B. Government will witness equipment field tests. Provide proof of Government witness on field test reports.
1. Inform COR at least 5 working days in advance of testing. Failing to inform COR may require retesting if tests were performed with no government witness.

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SECTION 26 05 10 CONDUCTORS AND CABLES

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include in price offered in the schedule for General Electrical System.
 - a. Includes cost of conductors, cables, mounting equipment, installation and making electrical connections to equipment and devices.

1.02 REFERENCE STANDARDS

- A. All references below are to be the current adopted edition at the time of solicitation for this contract. If any conflict arises between installing the new equipment in the existing installation, the contractor is to notify the COR. At which point, the CO will give direction on the course of action.
- B. ASTM International (ASTM)
1. ASTM B 8 Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft
- C. National Electrical Manufacturers Association (NEMA)
1. NEMA WC 70 Power Cables Rated 2000V or Less for the Distribution of Electrical Energy
- D. National Fire Protection Association (NFPA)
1. NFPA 70 National Electrical Code (NEC)
- E. Underwriters Laboratories, Inc. (UL)
1. UL 486B Wire Connector

1.03 DESIGN REQUIREMENTS

- A. Determine conductor sizes, except where shown or specified, in accordance with NFPA 70. In the event of conflict between design requirements and NFPA 70, the more stringent requirement applies.

1.04 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals and Section 01 33 26 - Electrical Drawings and Data.

- B. RSN 26 05 10-1, Approval Data:
 - 1. Manufacturer's technical catalog data.
- C. RSN 26 05 10-2, Field Test Reports.

1.05 DEFINITIONS

- A. Cable: Cable, cables, wire, or wires of one or more insulated conductors.
- B. Power Cable: For power loads including receptacle outlets, motors, alternating- and direct-current distribution circuits, HVAC and lighting circuits, and cable used for HVAC lighting equipment.
- C. Multiconductor Control Cable: For control, metering, indication, annunciation, and relaying circuits; and circuits not identified as power circuits.
- D. Instrumentation Cable: For RTD and transducer output circuits.

1.06 PROJECT CONDITIONS

- A. Locations of existing cables indicated on drawings are approximate.
- B. Contractor to field verify locations, lengths and size of conductor/cables required for this project.

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PART 2 PRODUCTS

2.01 CABLE

- A. Manufactured no more than 24 months prior to Notice to Proceed.
- B. Round, except for 2-conductor cable with parallel conductors.
- C. Conductors:
 - 1. Copper in accordance with ASTM B 8, class B or C.
 - 2. Stranded conductors.
- D. AWG or kcmil designation.
- E. Coverings or insulation: Suitable vertical position without injury to covering or deformation of insulation when supported in accordance with NFPA 70.

2.02 PHASE IDENTIFICATION

- A. Power Cable Phase Conductors:
 - 1. 120/240, single-phase: Black.

2. 120/208- or 240-volt, 3-phase: Black, red, blue. Colored phase tape may be used in lieu of colored insulation.
 3. 277/480-volt, 3-phase: Black with brown, orange, yellow phase tape.
- B. Neutral (Grounded) Conductors: In accordance with NFPA 70.
- C. Equipment Grounding Conductors:
1. 6 AWG and smaller: Green insulation.
 2. Larger than 6 AWG: Green tape at both ends and visible points including junction boxes.

2.03 COLOR CODING

- A. Control and Instrumentation Cable:
1. Colored insulation or jacket compound. Do not apply color coatings to insulation or jacket surface.
 2. In accordance with NEMA WC 57, Table E-2.

2.04 POWER CABLE AND MULTICONDUCTOR CONTROL CABLE

- A. 600-volt rated insulation unless specified otherwise.
- B. Type:
1. Single-conductor or multi-conductor, non-shielded type.
- C. Suitable For:
1. Installation in raceways.
- D. Insulation Type:
1. Single-conductor cable: THHN/THWN or XHHW-2.
 2. Multi-conductor cable: THHN/THWN or XHHW-2 for 14, 12, 10 AWG. TFFN or TFN for 18 and 16 AWG.
- E. Multiconductor Overall Jacket:
1. Type: PVC (Polyvinyl chloride) or better, flame-retardant.
 2. Jacket thickness:
 - a. 16 AWG conductors: 45 mils for 12 conductor or less. 60 mils for greater than 12 conductors.
 - b. 14 AWG conductors: 45 mils for 9 conductor or less. 60 mils for greater than 9 conductors.

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- F. Single-conductor power cable in accordance with NEMA WC 70. Multiconductor cable in accordance with NEMA WC 57. Cable UL listed and labeled.
- G. Provide multi-conductor power cable with integral ground conductor. Size ground conductor in accordance with NEMA WC 70.

2.05 INSTRUMENTATION CABLE

- A. Purpose: Twisted pair for transducer output circuits and twisted triad for RTD circuits.
- B. Type: Single-pair or triad shielded or multiple individual shielded pairs or triads with overall shield.
- C. Suitable for installation in raceways and cable trays.
- D. Conductor size: No. 16 AWG, or as shown.
- E. Insulation: 600 volts, 90 degrees C, minimum.
- F. Single-Pair or Triad Shield: Aluminum-mylar tape providing 100 percent coverage with a tinned, stranded copper drain wire.
- G. Multiple Pair or Triad Shield:
 - 1. Pair or triad shield: Aluminum-mylar tape providing 100 percent coverage with a tinned, stranded copper drain wire.
 - 2. Cable shield: Aluminum-mylar tape providing 100 percent coverage with a tinned, stranded copper drain wire.
 - 3. Each pair or triad sequentially numbered for identification.
- H. Jacket: Black PVC with rip-cord.

2.06 ACCESSORIES

- A. Single Conductor Markers:
 - 1. Type: Heat-shrink, machine printable sleeve.
 - 2. Color: White with black text.
 - 3. Text size: As large as practical but at least 7 point (0.06-inch).
 - 4. 1-inch in length, minimum. Diameter suitable for installation on intended conductor.
- B. Cable Tags:
 - 1. Type: Non-conductive, heat resistant and non-corroding.

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- a. Engraved Valve Tags, as manufactured by R&R Identification, www.rridentification.com, or Valve Tags, as manufactured by Brady, www.bradyid.com, or equal with the following essential characteristics:
 - 1) Minimum 2-ply laminate, engravable.
 - 2) Round shape with attachment hole suitable for tie wraps or similar.
2. Color: White with black text.
3. Text size: As large as practical but at least 16 point (0.16-inch).
- C. Terminal connectors for control, instrumentation, and telemetry cable:
 1. Heavy-duty, nylon insulated sleeve, pressure-crimp-type with ring tongues.
 2. Tin-plated copper.
 3. Serrated inner barrel.
 4. UL listed.
 5. Compatible with conductor size and type of cable for which it is used.

PART 3 EXECUTION

3.01 INSTALLATION

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- A. When using cable ties, do not over tighten, to ensure tie does not cut conductor insulation or cable outer jacket. Do not use cable ties to support cables.
- B. Block cable opening in sleeves under equipment or passing through blockouts, with silicone-foam, fire-retardant type material.
- C. Cable Splicing:
 1. Do not splice power, control, or instrumentation cables.
- D. Cable Installed in Raceways:
 1. Do not pull cable into conduit until they have been cleaned and are free from obstructions and sharp corners.
 2. Draw a clean, dry, tight-fitting rag through conduit immediately before installing cable.
 3. Install cable using gradual and uniform pulling stresses. Do not exceed allowable pulling tensions and sidewall pressures as recommended by cable manufacturer.
 4. Install cable to prevent cuts or abrasions in insulation or protective covering and kinks in cable.
 5. When a lubricant is needed as an aid to pulling, use only soapstone or other suitable lubricant that is not injurious to cable insulation.

6. When mechanical cable pulling equipment is used, use a tension meter release mechanism to ensure pulling tension remains below tensile strength of the cable.
 7. Remove and replace any cable damaged during installation.
 8. Where conductors carrying alternating current are installed in ferrous metal enclosures or metal raceways, arrange to avoid heating surrounding ferrous metal by induction. Group phase conductors together.
- E. Cable Installed in Vertical or Inclined Plane:
1. Install in locations shown. If not shown, install in accordance with NFPA 70.
 2. Clamp cable entering enclosures, junction boxes, or equipment with commercial cable clamps.
- F. Grounding.
1. Provide one nonactive conductor grounded at both ends in each Multiconductor control cable.
 2. Ground shielding of each shielded control cable on both ends directly to equipment ground bus by method indicated on Standard Drawing 40-D-4335, Figure 8.
 3. Ground shielding of instrumentation cable only to source equipment ground bus using pressure crimp type connectors with ring tongues on drain wire.
- G. Wiring methods:
1. Terminate control and instrumentation cable with terminal connectors.
 - a. Make terminations with electric, hydraulic, or ratcheted type pressure crimper.
 - b. Do not use plier crimpers.
 2. At termination point of multiconductor cable, form conductors into neat packs and tie conductors with self-locking cable ties.
 3. Do not combine alternating- and direct-current circuits, or current and potential transformer circuits in same multiconductor cable.
 4. Clamp cable entering equipment with commercial type cable clamps.
 5. Leave sufficient length cable ends to make connections conveniently to equipment and devices.
 6. Retain spare single conductors at each end of a multiconductor cable in a length equal to longest single conductor of multiconductor cable.
 7. Terminate spare conductors of multi-conductor cable at terminal blocks. Ground spare conductors by using wire jumpers between terminations and equipment ground bus.

3.02 CABLE AND CONDUCTOR IDENTIFICATION

A. Power Cable:

1. Provide a single-line marker on the load end of feeder and branch-circuit conductors that originate at panelboards, switchboards, and motor control centers. The marker to include designation of source equipment and circuit number.

B. Conductor Markers:

1. Text to be machine or computer generated. Handwritten text is not acceptable.
2. Provide individual conductors of multiconductor control cable and instrumentation cable with conductor marker. Print conductor designation, as shown on approved schematic diagram, on first line. Print conductor destination and device terminal number on second line.
3. A single line conductor marker will be acceptable provided the conductor designation is enclosed in parenthesis followed by the conductor destination.
4. Do not heat shrink marker onto conductor. Allow marker to rotate about the conductor.
5. Provide markers for spare conductors of multi-conductor cables. The marker to include cable designation and the word "SPARE". Numerically sequence the spare conductor markers e.g. (1CSA-CS3-SPARE1, 2CSA-CS3-SPARE2, etc.).

C. Cable Tags:

1. Print tags with cable designations as shown on approved wiring and interconnection diagrams. Cable designation to be a single line of text that includes cable number, originating equipment designation, and destination equipment designation. Develop cable designations when not shown. Examples of cable designations are:
 - a. 1MCA-DV1 – First cable from motor control center MCA to discharge valve No. 1).
 - b. 3MCB-ACP – Third cable from motor control center MCB to auxiliary control panel ACP.
 - c. Note, if the second part of the cable designation (destination equipment) would not fit on the tag, it can be eliminated for the tag marking.
2. Securely attach cable tags to multi-conductor cables near ends and single-conductor power cables, as directed. Attach to cable with standard nylon self-locking tie wraps or cable ties. Use two wraps, one firmly around cable, second used to attach tag to tie wrap on cable so tag can dangle freely. Pull tag tie wrap so that loop is approximately ½-inch in diameter.

3.03 CONTRACTOR FIELD QUALITY TESTING

- #### **A. Perform testing after installation, but before connection to equipment.**

- B. Notify Contracting Officer's Representative, in writing, of date, time, and cables to be tested at least 5 working days before testing. Testing dates to be mutually agreeable between Contractor and Contracting Officer's Representative.
- C. Perform following tests:
1. Continuity test: Test continuity of each conductor prior to performing insulation resistance test.
 2. Insulation resistance tests for power, control, and instrumentation cables:
 - a. Measure insulation resistance between each conductor and station ground with all other conductors in same cable or conduit grounded.
 - b. Test voltage: 1,000 volts direct-current, minimum.
 - c. Minimum acceptable insulation resistance: $R \text{ in } M\Omega = (\text{rated voltage in kV} + 1) \times 1000 / \text{length in ft.}$
 3. Record following test data on test report for each conductor tested:
 - a. Date of test.
 - b. Name or names of Contractor's personnel who performed test.
 - c. Name of Government inspector who witnessed test.
 - d. Conductor identification.
 - e. Megohms between conductor and ground.
 - f. Total length of conductor.
- D. If any conductor fails continuity or insulation resistance testing, remove, and replace cable and conductor.

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END OF SECTION

SECTION 26 05 20 GROUNDING AND BONDING

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include in price offered in the schedule for General Electrical System.
 - a. Includes cost of items requiring grounding and bonding for all electrical connections to equipment and devices.

1.02 REFERENCE STANDARDS

- A. All references below are to be the current adopted edition at the time of solicitation for this contract. If any conflict arises between installing the new equipment in the existing installation, the contractor is to notify the COR. At which point, the CO will give direction on the course of action.
- B. ASTM International (ASTM)
1. ASTM B 3
 2. ASTM B 8
- C. Institute of Electrical and Electronic Engineers (IEEE)
1. IEEE 81
 2. IEEE 837
 3. IEEE C2
- D. National Fire Protection Association (NFPA)
1. NFPA 70
- E. Underwriters Laboratories (UL)
1. UL 467

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Soft or Annealed Copper wire
Concentric-Lay-Stranded Copper
Conductors, Hard, Medium-Hard, or Soft

Measuring Earth Resistivity, Ground
Impedance, and Earth Surface Potentials of
a Ground System

Qualifying Permanent Connections Used in
Substation Grounding

National Electrical Safety Code (NESC)

National Electric Code (NEC)

Grounding and Bonding Equipment

1.03 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals and 01 33 26 - Electrical Drawings and Data.

- B. RSN 26 05 20-1, Approval Data
 - 1. Manufacturer’s technical catalog data
- C. RSN 26 05 20-2, Testing Plan:
 - 1. Submit proposed testing plan to demonstrate relevant equipment is grounded.
 - 2. For the purposes of this submittal
- D. RSN 26 05 20-3, As-built Drawings:
 - 1. Specifications drawings indicating As-built changes made to grounding systems.
- E. RSN 26 05 20-4, Test Reports.

1.04 PROJECT CONDITIONS

- A. Locations of existing ground conductors indicated on drawings are approximate.
- B. Existing ground conductor (shown on the interior wall) interfaces to the existing grounding system. Refer to Photo 1 in specification Section 51 00 00. The photo illustrates a possible interface point(s) between the existing grounding system and the contractor’s electrical conduit, junction boxes to name a few. Refer to Section 26 05 33 Raceways and Boxes.

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PART 2 PRODUCTS

2.01 MATERIALS

- A. Ground Cable:
 - 1. Annealed bare-copper cable:
 - a. ASTM B 8, class B.
 - b. Solid wires used in forming the copper cable: ASTM B 3.
 - 2. Concentric stranded.
 - 3. Sizes:
 - a. As indicated on drawings or matching size of the tie in.
 - b. Where not shown, minimum sizes:
 - 1) Equipment grounding conductors:
 - a) Major equipment: Equal in size to electrode conductors.
 - b) Minor equipment: No. 4 AWG
 - 2) Electrode interconnecting conductors: Equal in size to electrode conductors.

2.02 ACCESSORIES

- A. Cable Connectors:
1. Welded, bolted solderless or compression type.
 2. Current-carrying capacity: Equal to cable with which they are used.
 3. Connectors for ground cables, including fittings, lugs, bolts, nuts, and washers: Copper alloy containing not more than 4 percent zinc.
 4. Bolted solderless or compression type connectors: IEEE 837.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Install in accordance with NFPA 70, IEEE C2, and this Section.
- B. Connect equipment, metal conduit, metal pull box covers, exposed metal surfaces, and other miscellaneous metalwork to grounding system as indicated on Standard Drawings 40-D-4334, 40-D-4335 and as shown on drawings.
- C. Ground equipment and miscellaneous metalwork that is required to be grounded but is not indicated on drawings to grounding system.
- D. Connections:
1. Number of grounding connections for equipment may be one, two, or more in some cases.
 2. Remove paint, enamel, scale, oil, grease, or other foreign nonconductive material from point of contact on metal surfaces before making ground connections.

3.02 CONTRACTOR FIELD QUALITY TESTING

- A. Test resistance of grounding system:
1. Notify COR at least 5 working days in advance of time of test in order that test may be witnessed by COR.
 2. Measured resistance: Less than 5 ohms.

END OF SECTION

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SECTION 26 05 33 RACEWAYS AND BOXES

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include in price offered in the schedule for General Electrical System.
 - a. Includes cost of conduit, fittings, and accessories to make electrical connections to equipment and devices.

1.02 REFERENCE STANDARDS

- A. All references below are to be the current adopted edition at the time of solicitation for this contract. If any conflict arises between installing the new equipment in the existing installation, the contractor is to notify the COR. At which point, the CO will give direction on the course of action.
- B. American National Standards Institute (ANSI)
1. ANSI C80.1 Electric Rigid Steel Conduit (ERSC)
- C. National Electrical Manufacturers Association (NEMA)
1. NEMA FB 1 Fittings, Cast Metal Boxes, And Conduit Bodies For Conduit And Cable
 2. NEMA RN 1 PVC Externally Coated Galvanized Rigid Steel Conduit
- D. National Fire Protection Association (NFPA)
1. NFPA 70 National Electric Code (NEC)
- E. Underwriters Laboratories (UL)
1. UL 360 Liquid-Tight Flexible Metal Conduit

1.03 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals and 01 33 26 - Electrical Drawings and Data.
- B. RSN 26 05 33-1, Approval Data and Drawings
1. Manufacturer's technical catalog data.
 2. Conduit layout drawing(s)

C. RSN 26 05 33-2, As-built Drawings:

1. Marked prints of conduit layout drawing(s) indicating “as-built” changes made to electrical conduit during construction.

PART 2 PRODUCTS

2.01 CONDUIT

A. General:

1. UL listed.
2. Sizes as shown on drawings.
3. Rate for intended purposes or labeled by nationally recognized testing laboratory (NRTL) accredited by Federal Occupational Safety and Health Administration (OSHA) when applicable.

B. Rigid steel conduit and attached fittings:

1. Zinc-coated and in accordance with ANSI C80.1.

C. Liquid tight Flexible Metal Conduit (LFMC):

1. Interlocked steel construction with flame-retardant PVC jacket.
2. Sunlight resistant.
3. Color: Grey
4. Conform to NFPA 70 and UL 360.
5. Fittings conform to NEMA FB 1.

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D. Plastic-coated rigid steel conduit:

1. In accordance with rigid steel conduit specified above.
2. External PVC coating: In accordance with NEMA RN 1.
 - a. Apply equal bonded coating to elbows, fittings, couplings, hangers, and other accessories.
3. Sleeve fittings and couplings to provide watertight joint. Use solvent to make fittings and couplings watertight as recommended by manufacturer.

2.02 CONDUIT ACCESSORIES

A. UL Listed.

B. Conform to NEMA FB 1 as applicable.

C. Steel materials and hardware: Galvanized.

- D. Provide fittings required to complete electrical conduit systems. Includes, but not limited to, caps, connectors, couplings (excluding expansion and expansion-deflection couplings), nipples, reducers, elbows, pipe plugs, locknuts, bondnuts, bushings, seals (excluding wall penetration seals), and any other fittings.
- E. Provide devices required to complete electrical conduit systems and to fasten, clamp, attach, and support each conduit in place.
1. Includes threaded joint compound; protective sealant; supports and clamps.
 2. Complete with bolts, washers, and nuts.
- F. Protective Sealant:
1. Water repellant.
 2. Resistant to peeling and cracking.
- G. Exposed Outlet Bodies and Boxes: Cast iron, malleable iron, or cast aluminum.
- H. Steel Outlet, Junction, and Pull Boxes: Pressed galvanized steel, minimum of four inches, square, with galvanized cover or extension ring as required.
- I. Fabricated Sheet Steel Boxes:
1. Large junction and conduit boxes, excluding outlet boxes, to contain wiring devices or to accommodate lighting fixtures.
 2. Sheet steel thickness: No. 14 United States Standard gauge, minimum.
 3. Sheet steel finish: Galvanized.
 4. Attach covers with bronze, brass, or stainless-steel screws or bolts.
- J. Adhesive: In accordance with conduit manufacturer's recommendations.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Layout and Procedures:
1. Determine routing of exposed conduit when not indicated on drawings.
 2. Ensure that conduit runs and bends are free from kinks, indentations, or flattened surfaces.
- B. Preparation:
1. Remove burrs and sharp corners at ends of metal conduit.
 2. Coat male threads of rigid metal conduit joints with suitable graphite or zinc sealing material before making joints.

- C. Install conduit with necessary fittings and supports.
1. Support conduit as indicated on Standard Drawing 40-D-7094.
 2. Tighten conduit joints securely to ensure electrical continuity and to prevent entrance of moisture or foreign material.
 3. Provide bushings or Chase-type nipples on ends of conduit to protect insulated conductors from abrasion.
 4. Use locknuts and bondnuts to provide tight ground connections between conduit and boxes, panelboards, and cabinets.
- D. Bends:
1. Install smooth, gradual bends to permit pulling insulated conductors without undue stress or damage to conductors or conduit.
 2. Bend conduit cold to prevent damage to protective coating.
- E. Exposed conduit runs:
1. Straight and parallel with each other and with centerlines of room or structure.
 2. Support conduit rigidly from wall or ceiling within 3 feet of each outlet box, junction box, cabinet, or fitting; and at intervals of not more than 5 feet.
 3. Installation includes, where required, drilling holes in bottom, side, or top of enclosures or plates of other electrical equipment.
 4. Tighten conduit securely.
 5. Support conduit rigidly in place.
 6. Make connections to outdoor boxes watertight.
- F. Liquidtight Flexible Metal Conduit (LFMC):
1. Use for connections to motors, actuators, and electrical equipment that is subject to movement, vibration, or misalignment; where flexibility is required; or where shown.
 2. Maximum length of conduit section is not to exceed 6 feet. Lengths exceeding 6 feet shall be approved by COR.
 3. Secure and support conduit sections in accordance with NFPA 70.
- G. Plastic-Coated Rigid Steel (PCRS) Conduit:
1. Joints:
 - a. Watertight.
 - b. Coated or covered in accordance with manufacturer's instructions.
 - c. Tighten securely with plumbers-type strap wrench.

- d. Repair or recover damaged portions of protective coat in accordance with manufacturer's instructions.
 - e. Use solvent when installing fittings and couplings to permanently bond sleeves to plastic coating of conduit.
- 2. Bend plastic-coated conduit in accordance with manufacturer's recommendations.
 - 3. Use factory bends when manufacturer warns of possible damage to conduit or plastic coating when bending larger sizes.

END OF SECTION

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SECTION 26 05 90

WIRING CHECKOUT AND TESTS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Cost:
1. Include in lump sum price offered in the Schedule for General Electrical System.

1.02 REFERENCE STANDARDS

- A. All references below are to be the current adopted edition at the time of solicitation for this contract. If any conflict arises between installing the new equipment in the existing installation, the contractor is to notify the COR. At which point, the CO will give direction on the course of action.
- B. National Fire Protection Association (NFPA)
1. NFPA 70 National Electrical Code (NEC)

1.03 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals and 01 33 26 - Electrical Drawings and Data.
- B. RSN 26 05 90-1, Test Reports
1. See Article 3.02 Field Test Reports.

1.04 RELATED REQUIREMENTS

- A. This section applies, but is not limited to, sections of Division 26 and 35 unless specified otherwise in the individual sections.

1.05 DEFINITIONS

- A. External Wiring: Conductor, wire, or cable external to enclosure of equipment being tested.
- B. Internal Wiring: Conductor, wire, or cable internal to enclosure of the equipment being tested.
- C. Qualified Person: In accordance with Article 100 of NFPA 70.
- D. Wiring Checkout: A complete wire-by-wire, contact-by-contact, terminal-by-terminal inspection of wiring associated with a specific piece of equipment.

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1.06 PERFORMANCE REQUIREMENTS

- A. Test in addition to tests required by Section 26 05 10 - Conductors and Cables.
- B. Furnish materials, including test equipment, required for checkout and testing.
- C. Make wiring changes and drawing revisions to ensure no discrepancies between check prints, and internal equipment wiring and external cabling.
- D. Replace defective equipment and make device setting adjustments as required for operation of electrical systems in accordance with specifications and manufacturer's instructions.
- E. If required during testing, remove and replace wire connections.
- F. Make wiring changes, wire tag revisions, and other repairs or revisions during the testing session the discrepancy is discovered.
- G. Perform wiring checkout, and operational and functional tests in presence of the COR.

1.07 QUALIFICATIONS

- A. Testing Personnel:
 - 1. Competent and qualified person to perform wiring checkout and operational and functional testing.
 - 2. Thorough knowledge of the electrical equipment, installation, and drawings.
 - 3. Qualified to operate equipment being tested.

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PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.01 FIELD QUALITY CONTROL

- A. Notify COR in writing 5 working days in advance of each testing session.
- B. Provide following at time wiring checkouts and tests are performed:
 - 1. Two complete sets of full-size (ANSI D) electrical schematic and schematic diagram check prints for equipment being tested.
 - a. One set to be used and maintained by the COR.

- b. One set to be maintained by Contractor and used to produce final drawings.
 - 2. Operation and maintenance instruction books and test reports for contractor furnished equipment being tested.
- C. Wiring Checkout:
 - 1. Use electrical schematic and schematic diagram check prints to perform:
 - a. Internal wiring check of equipment.
 - b. Wiring check of external cabling.
 - 2. Perform a complete wire-by-wire, contact-by-contact, terminal-by-terminal check of circuit logic.
 - 3. Verify contact configuration and contact numbering.
 - 4. Verify continuity of wiring.
 - 5. Verify proper wire tagging at ends of conductors and cables.
 - 6. Check conductors for insulation damage. Perform additional insulation resistance test of damaged conductors.
- D. Operational and Functional Testing:
 - 1. Perform after completion of wiring checkout.
 - 2. All equipment being powered by the new electrical power system shall undergo operational and functional testing. Coordinate with COR for testing equipment not furnished as part of this contract.
 - 3. If required, adjust and calibrate protective devices in accordance with specifications, and manufacturer's instructions and recommendations.
 - 4. Check electrical systems and controls for proper sequence of operation, correct adjustments and settings, and agreement with schematic diagram.

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3.02 FIELD TEST REPORTS

- A. Provide a letter stating wiring has been checked and tested by the contractor and witnessed by the COR. A signature from the COR is required on this letter.

3.03 SCHEDULE

- A. The following equipment requires a wiring checkout and operational test:
 - 1. Butterfly valve actuators.
 - 2. Fixed-cone valve actuators.
 - 3. Contractor furnished electrical equipment.

END OF SECTION

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SECTION 26 24 24

DISTRIBUTION PANELBOARDS

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

A. Cost:

1. Include in price offered in the schedule for General Electrical System.

1.02 REFERENCE STANDARDS

- A. All references below are to be the current adopted edition at the time of solicitation for this contract. If any conflict arises between installing the new equipment in the existing installation, the contractor is to notify the COR. At which point, the CO will give direction on the course of action.

B. Federal Specifications (FS)

1. FS W-C-375D Circuit Breakers, Molded Case; Branch Circuit and Service

C. Electrical Institute of Electrical and Electronics Engineers (IEEE)

1. IEEE C37.13 Low-Voltage AC Power Circuit Breakers Used in Enclosures
2. IEEE C37.14 Low-Voltage DC Power Circuit Breakers Used in Enclosures
3. IEEE C37.16 Low-Voltage Power Circuit Breakers and AC Power Circuit Protectors - Preferred Ratings, Related Requirements, and Application Recommendations
4. IEEE C37.20.1 Metal-Enclosed Low-Voltage Power Circuit Breaker Switchgear

D. National Electrical Manufacturers Association (NEMA)

1. NEMA 250 Enclosures for Electrical Equipment (1000 Volts Maximum)
2. NEMA AB 3 Molded Case Circuit Breakers and Their Application
3. NEMA PB 1 Panelboards
4. NEMA ST 20 Dry-type Transformers for General Applications

- E. National Fire Protection Association (NFPA)
 - 1. NFPA 70 National Electrical Code (NEC)
- F. Underwriters Laboratories, Inc. (UL)
 - 1. UL 50 Enclosures for Electrical Equipment
 - 2. UL 67 Panelboards
 - 3. UL 489 Molded-Case Circuit Breakers, Molded-Case Switches, and Circuit-Breaker Enclosures

1.03 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals and Section 01 33 26 - Electrical Drawings and Data.
- B. RSN 26 24 24-1, Approval Data:
 - 1. Equipment layout drawings, schematic diagrams, manufacturer's data, bills of material for each device or component of equipment, and circuit directory for each board.
- C. RSN 26 24 24-2, Factory Test Report.
- D. RSN 26 24 24-3, Operation and Maintenance Instruction Book.

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PART 2 PRODUCTS

2.01 SYSTEM DESCRIPTION

- A. Design Requirements: see Specifications Section 26 05 02 - Basic Electrical Materials and Methods.

2.02 DISTRIBUTION PANELBOARDS

- A. Type: Dead-front, rigid, Type 3 in accordance with NEMA 250, indoor, dustproof, with hinged door.
- B. Suitable for use with 3-phase, 4-wire, 240-volt, alternating-current power supply, and solid-neutral bus.
- C. Main Breaker Thermal Rating: per design.
- D. Branch Circuits: per design.
- E. Voltage and System: 240 volts, 3-phase, 60 hertz.
- F. Solid Neutral Bus: None required.

- G. Bus Material: Aluminum or copper.
- H. Mounting: Surface, Wall.
- I. Conform to:
 - 1. IEEE C37.13, C37.14, C37.16, and C37.20.1,
 - 2. NEMA AB 3 and PB 1,
 - 3. UL 50, 67, and 489,
 - 4. NEC,
 - 5. FS W-C-375.

2.03 NAMEPLATES

- A. As indicated on Standard Drawing No. 40-D-6234.
- B. Engraving: per design.

2.04 MAIN BUS

- A. Rating: per design.
- B. Copper, bolted type bars.
- C. Main and Branch Circuit Connections: Unless silver plated, use copper based on 1,000-ampere-per-square-inch current density and contact surfaces with not more than 200 amperes per square inch.
- D. Ampacity: per design.
- E. Equip screws and bolts used for making copper connections with lockwashers to prevent loosening. Do not use riveted bus connections.

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2.05 CIRCUIT BREAKER

- A. Molded-case type.
- B. Thermal-magnetic type with inverse-time trip characteristics on overload and instantaneous trip on short circuit.
- C. Voltage: per design.
- D. Interrupting Rating: per design.
- E. Arc quenchers.
- F. Quick-make and quick-break toggle mechanisms.

- G. Trip-free operating handle.
- H. Single operating handle. Do not use handle ties on multiple-pole circuit breakers.
- I. Connect breakers in branch circuits of direct-current boards to main buses such that adjacent poles of each breaker are opposite in polarity. For three-phase bus, alternating-current boards, phase arrangement shall conform to NFPA 70 article 408.
- J. Mount circuit breaker handles to be operable from front of boards.
- K. Assemble boards so that individual breakers can be removed or added without disturbing adjacent units, without loosening or removing supplemental insulation supplied to meet requirements of UL, and without additional machining, drilling, or tapping.
- L. Assemble boards with molded-case circuit breakers individually mounted on fixed framework behind easily removable covers.
- M. Provide blank spaces for future circuit breakers on each board as indicated on drawings. Cover blank spaces with removable plates.
- N. Padlock attachments for locking breakers in OFF position.
- O. In accordance with FS W-C-375 and NEMA AB-5 for molded-case circuit breakers.
- P. Equip multiple-pole breakers with common trip so an overload on one pole will automatically cause all poles of breaker to open.
- Q. Interrupting rating of circuit breakers for 120- and 240-volt service: per design.
- R. Equip with solderless-type connectors on load side of breakers for conductors No. 8 AWG and smaller. Furnish solder-type or pressure-wire connectors on load side of breakers for conductors larger than No. 8 AWG.
- S. Per design, furnish breakers with factory-installed shunt-trip solenoids that are wired to outgoing terminal blocks.
- T. Solenoid ratings: per design.
- U. Furnish 2 spare breakers of each size.
- V. Number of Poles: per design.

2.06 SOLDERLESS, BOLTED-TYPE LUGS OR TERMINALS

- A. Provide lugs or terminals suitable for copper conductors on breakers for feeder and branch circuit cables, and on bus for supply cables. Size lugs or terminals in accordance with cable sizes per design.

- B. Provide neutral buses with sufficient solderless, bolted-type lugs or terminals to accommodate each single-pole branch circuit breaker, each future breaker, and neutral leads per design.
- C. Provide solderless, bolted-type copper lugs or terminals at rear base of boards for connecting boards to ground system. Connect lugs or terminals to ground system by AWG copper conductor sized per design.

2.07 CONNECTORS BETWEEN BREAKER TERMINALS AND BUSES

- A. Copper bars designed to prevent excessive stress on bus or breaker during expansion due to temperature changes.

2.08 GROUND-FAULT CIRCUIT INTERRUPTER BRANCH CIRCUIT BREAKERS

- A. Provide overload protection.
- B. Short-circuit protection.
- C. Ground-fault protection: In accordance with UL class A ground-fault protection resulting in automatic tripping for fault currents to ground of 6 milliamperes or more.

2.09 PLUG RECEPTACLES

- A. 125 volts, 15 amperes, 2 wire, 3 pole, polarized grounded.
- B. Weatherproof covers with screwcaps.
- C. Equip with matching plugs.

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2.10 TERMINAL BLOCKS

- A. 600 volts and 25 amperes.
- B. Suitable for use with No. 8 AWG wire.
- C. Molded-block type to accommodate ring connectors 1/2 inch wide (outer diameter) at terminal screws.
- D. Binding-head or washer-head screws having serrated or grooved contact surfaces or having lockwashers.
- E. Molded insulating barriers between terminals.
- F. Removable marking strips. Provide marking strips for terminal blocks with conductor designations engraved or printed with a permanent marking fluid. Furnish one spare blank marking strip with each terminal block.

- G. Removable covers to prevent accidental contact with terminal by operating personnel.
- H. Furnish and install heavy-duty terminal blocks with barriers for external supply cables.
- I. Arrange and locate terminal blocks to support incoming and outgoing cables.
- J. Separate adjacent rows of terminal blocks at least 6 inches edge to edge and at least 6 inches from sides, top, or bottom of the cabinet.
- K. Install grommets at holes for wires and cables.
- L. Furnish sufficient terminal blocks to wire and make ready for operation all equipment on boards, and to terminate incoming control cables.
- M. Provide 25 percent spare terminals on each terminal block.
- N. Mounting: Near bottom of boards for bottom entry, and near top for top entry.

2.11 GROUND BUS

- A. Copper.
- B. Length: Width of enclosure.
- C. Install at bottom of enclosure.
- D. Bond mechanically and electrically to enclosure.
- E. Drill and tap every inch for 10-32 screws. Install a 10-32 screw in each tapped hole.

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2.12 CONDUCTOR IDENTIFYING MARKERS

- A. Tag conductors near each terminal to identify them with their respective studs or terminals.
- B. Vinyl-cloth with silicone plastic overcoating.

2.13 CONTROL WIRING FOR ALARM CIRCUITS

- A. Minimum No. 16 AWG, class K stranded copper conductor, with NEC switchboard type SIS insulation rated for 600 volt service.
- B. Make all terminations of wire with heavy-duty, preinsulated pressure-crimp-type terminal connectors with ring tongues.
- C. Connectors:
 - 1. Tin-plated copper.

2. Serrated inner barrel.
 3. 600-volt-rated nylon insulation.
 4. Insulation support sleeve for vibration resistance.
 5. UL listed.
 6. Compatible with conductor size and type on which it is used.
- D. Keep wiring to a minimum.
- E. Form wiring into compact groups, bound together and firmly supported.
- F. Run wiring straight, horizontally or vertically, with short-radius right-angle bends.
- G. Protect wiring from abrasion where it leaves channels or ducts.
- H. Do not splice wiring.
- I. Make connections at device studs or terminal blocks.
- J. Terminate wiring which connects to outgoing control cables at terminal blocks.

2.14 ENCLOSURE

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- A. Type 3 in accordance with NEMA 250.
- B. Construct enclosures with a maximum dimension less than 20 inches of sheet steel not lighter than No. 16 United States Standard gauge. Construct enclosures with a maximum dimension more than 20 inches of sheet steel not lighter than No. 20 United States Standard gauge.
- C. Barriers to separate 48-volt section from 120/240-volt section.
- D. Equip enclosure door with flush lock and two keys.
- E. Welded joints.
- F. Galvanize after fabrication.
- G. Provide barriers to separate voltage sections per design.
- H. Finish: Manufacturer's standard finish.

2.15 FABRICATION

- A. Mount ground bus at bottom of panelboard enclosure. Bond bus to enclosure.
- B. Mount circuit directory on inside of door.

- C. Make equipment, removable covers, terminal blocks, wiring, and wiring gutters accessible from fronts of the boards for operation, maintenance, removal, and replacement so boards are capable of being permanently mounted with rear against wall.
- D. Make provisions for anchoring the backs of floor-mounted boards to walls for additional support.
- E. Equip ventilation openings with grilles or screens having dust filters.
- F. Dimensions:
 - 1. Surface mounted boards sized per design.
 - 2. Floor-mounted boards:
 - a. 6 feet high, minimum, excluding bases.
 - b. Height of centerline of top circuit breaker handle. 6 feet 6 inches, maximum.
- G. Provide gutter space on all sides of all boards in accordance with NFPA 70 Article 408.
- H. Provide boards with easily removable or hinged front panels so incoming cables may be easily connected or disconnected and inspected.
- I. Equip tops of floor-mounted boards, and bottom and top of wall-mounted boards with removable plates for field drilling.
- J. Provide access doors for each board.
- K. Equip each door with locks, latches, and concealed hinges to allow doors to open 105 degrees, minimum, from fully closed position. Key all locks alike with identifying number stamped on each key. Provide five keys.
- L. Circuit Directory:
 - 1. Prepare circuit directory for each board, indicating "as-built" conditions.
 - 2. Include board designation, breaker numbers and descriptions of connected loads indicated in their relative physical location.
 - 3. Show breaker number and connected load.
 - 4. Use machine lettering for directory entries. Do not provide pencil or hand written directory entries.
 - 5. Mount directory in transparent plastic front on inside surface of access doors for each board.
 - 6. Fabricate circuit directory frame such that directory need not be folded when placed in frame, and such that all information on directory is clearly visible and legible.

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- M. Fabricate frames and enclosures of sheet steel with sufficient weight and strength to ensure permanent rigidity and alignment.
- N. Hide butt joints on outside surfaces.
- O. Hide holes or fasteners be from front of boards for rear-of- board-mounted equipment.
- P. Provide boards with manufacturer's standard base provided other requirements of this Section are fulfilled.
- Q. Support bus bars by structural frames. Do not depend on breakers for support.
- R. Back pan:
 - 1. Provide as foundation for mounting bus and breakers.
 - 2. Construct rigidly to ensure against damage during shipment and installation.
 - 3. Provide reinforcement on back pan consisting either of flanging back pan entire length of pan, or securely fastening angle iron to both sides of a flat back pan.
 - 4. Do not use nonreinforced back pans of ordinary flat sheet steel.
- S. Construct board frames and enclosures to provide adequate clearance for incoming cables entering from conduits below and above. Make provisions near bottom of boards for supporting incoming and outgoing cables.

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2.16 CONTRACTOR SOURCE QUALITY TESTING

- A. Dielectric test in factory in compliance with NEMA PB1, as applicable.
- B. Test circuits before shipment to demonstrate correctness of completed wiring connections and proper functioning of devices.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Cables entry into enclosure: From beneath by means of conduit under and from above by means of conduit above panelboards.
- B. Provide anchoring, supporting, and mounting materials as required.
- C. Install ground cable between panelboard enclosure and grounding plate.

END OF SECTION

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SECTION 35 22 25

VALVES

PART 1 GENERAL

1.01 MEASUREMENT AND PAYMENT

- A. Removal of existing two 30-inch fixed-cone valves with motor-operators, 30-inch spool pieces, valve supports, and anchorage.
 - 1. Payment: Lump sum price offered in Price Schedule under Valve Removal and Installation.
- B. Furnishing and installing two 24-inch fixed-cone valves with motor-operators, valve hoods, 30-to-24-inch reducers, valve or reducer supports, anchorage, and 30-inch butterfly valves:
 - 1. Payment: Lump sum price offered in Price Schedule under Valve Removal and Installation.

1.02 REFERENCE STANDARDS

- A. All references below are to be the current adopted edition at the time of solicitation for this contract. If any conflict arises between installing the new equipment in the existing installation, the contractor is to notify the COR. At which point, the CO will give direction on the course of action.
- B. American National Standards Institute (ANSI)
 - 1. ANSI Y14.5 Dimensioning and Tolerancing
- C. American Society of Mechanical Engineers (ASME)
 - 1. ASME BPVC-VIII-1 Boiler and Pressure Vessel Code – Section VIII – Rules for Construction of Pressure Vessels Division 1
 - 2. ASME B16.9 Factory-Made Wrought Buttwelding Fittings
- D. American Society for Testing and Materials (ASTM)
 - 1. ASTM A193/A193M Alloy-Steel and Stainless Steel Bolting for High Temperature or High Pressure Service and Other Special Purpose Applications
 - 2. ASTM A194/A194M Standard Specification for Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High Pressure or High Temperature Service, or Both

3. ASTM A276/A276M Stainless Steel Bars and Shapes
 4. ASTM A479/A479M Stainless Steel Bars and Shapes for Use in Boilers and Other Pressure Vessels
- E. American Water Works Association (AWWA)
1. AWWA C207 Steel Pipe Flanges for Waterworks Service, Sizes 4 In. Through 144 In.
 2. AWWA C208 Dimensions for Fabricated Steel Water Pipe Fittings
 3. AWWA C504 Rubber-Seated Butterfly Valves
 4. AWWA C540 Power-Actuating Devices for Valves and Slide Gates
 5. AWWA C542 Electric Motor Actuators for Valves and Slide Gates
- F. American Welding Society (AWS)
1. AWS D 1.1 Structural Welding Code
- G. National Electrical Manufacturers Association (NEMA)
1. NEMA ICS 5 Industrial Control Systems: Control Circuit and Pilot Devices
 2. NEMA MG 1 Motors and Generators
 3. NEMA 250 Enclosures for Electrical Equipment (1000 Volts Maximum)
- H. National Fire Protection Association (NFPA)
1. NFPA 70 National Electric Code (NEC)

1.03 SUBMITTALS

- A. Submit the following in accordance with Section 01 33 00 - Submittals.
- B. RSN 35 22 25-1, Fixed Cone Valve and Reducer Approval Data:
1. Shop drawings and data showing assembly, type, size, and detail dimensions of fixed-cone valve, hood, and motor operator assembly including all dimensions, materials, all applicable standards specifications, center of gravity, and total weight of complete valve assembly. Include motor operator capacity.
 2. Graph of Discharge Coefficient versus Percent-Open for the valve.
 3. Minimum Percent-Open at 187-ft of head pressure.

4. Shop drawings of reducer including all dimensions, materials, all applicable standards specifications, center of gravity, and total weight.
5. Manufacturer's standard wiring diagrams, modified as needed for this project, showing circuit connections for limit switches supplied with and as part of the motor-operator. This wiring diagram does not need to meet the requirements of wiring diagrams per Section 01 33 26 - Electrical Drawings and Data.
6. Motor data:
 - a. Locked-rotor torque: foot-pounds.
 - b. Full-load torque: foot-pounds.
 - c. Horsepower rating: horsepower.
 - d. Voltage rating: volts.
 - e. Locked-rotor current: amperes.
 - f. Full-load current: amperes.
 - g. Time rating: minutes.
7. Installation procedure. Drawing and or written procedure describing in detail the proposed installation procedure for the valves and reducers. Include weights, loading, support, rigging, and special equipment to be used. Equipment can be installed through the top opening into the valve pit utilizing the government crane or can be flown in through the downstream opening of the valve pit with the use of a contractor provided crane.

C. RSN 35 22 25-2, Butterfly Valve Approval Data

1. Commercial products data, including complete identifying data giving manufacture's name, type, model, size, and dimensions.
2. Valve data:
 - a. Maximum torque required (foot-pounds).
 - b. Test data of rated flow coefficient (Cv). Provide data in graph or table format with data points at a maximum of 10-degree intervals, for disk angles of 10 degree through 90 degree.
3. Motor data:
 - a. Locked rotor torque: foot-pounds.
 - b. Full load torque: foot-pounds.
 - c. Horsepower rating: horsepower.
 - d. Voltage rating: volts.
 - e. Locked rotor amps: amperes.
 - f. Full load amps: amperes.

- g. Time rating: minutes.
- D. RSN 35 22 25-3, Fixed Cone Valve Shop Test Data:
 - 1. Seal and seat leakage and pressure test results.
 - a. Include pressures and times for testing.
 - 2. Operational test results.
- E. RSN 35 22 25-4, Butterfly Valve Shop Test Data:
 - 1. Certification that butterfly valves meet hydrostatic and leak testing requirements.
- F. RSN 35 22 25-5, Final Data:
 - 1. Service manuals for all equipment, including wiring diagrams for the motor-operator.
 - 2. Photographs.
 - a. Oblique front and rear view of both sides of assembled valve.
 - b. Front, rear, side, and top (either vertically downward or right and left views at least 45 degrees above horizontal) views of assembled valve.
 - 3. As-built drawings or, if no modifications were made, a letter stating so.
 - 4. Field tests.
 - a. Seal and seat leakage test results.
 - b. Operational test results.
- G. RSN 35 22 25-6, Training.

1.04 PROJECT CONDITIONS

- A. Government to inspect condition of penstock during outage. Provide 3 days in schedule for Government inspection. Government inspection to be after existing valves have been removed and the pipe has been dewatered and prior to installation of new valves.
- B. Existing valves are 30-inch motor-operated fixed-cone valves.
- C. Supply Power: 240 volts, 3-phase, 60-hertz (Hz).
- D. Access for removing and installing fixed-cone valves are:
 - 1. Through an opening in the control house floor above each valve. Dimensions of each opening are 5-feet by 5-feet; or
 - 2. Through the downstream opening of the valve pit. Dimension of the opening is 6.5-feet tall by 15-feet wide.

- E. Government-owned 5-Ton overhead travelling crane can be used to install equipment through the control house floor.
- F. Access to the Bradbury Dam Outlet Works requires crossing the spillway on a narrow bridge. See Section 01 55 00, Vehicular Access and Parking, for more information.
- G. Leaking water will reach approximately 0.5 inches above the invert of the pipe during valve replacement.

PART 2 PRODUCTS

2.01 BUTTERFLY VALVE

- A. Manufactured by Dezurik/APCO/Hilton, 250 Riverside Ave. North, Sartell MN 56377; Emerson/Keystone, 8000 West Florissant Ave., St. Louis MO 63136; Henry Pratt Company, 401 South Highland Avenue, Aurora IL 60506; or equal, with the following essential characteristics:
 - 1. AWWA C504, Class 150B.
 - 2. Valve seat:
 - a. Material: Acrylonitrile-Butadiene (NBR)
 - b. Permits removal, replacement, and adjustment from the downstream side of the valve without removing the valve from the manifold.
 - c. Located in the valve body (mating with a stainless-steel seating surface located on the valve disc).
 - d. Seats located in the valve body: locked into a recessed cavity in the valve body by means of epoxy injection method.
 - 3. Mating surface for the valve seat:
 - a. Stainless steel type 316.
 - b. Sprayed or plated surfaces are not acceptable.
 - 4. Disc:
 - a. Ductile iron or stainless steel.
 - 5. Shaft:
 - a. Stainless steel.
 - b. Shaft does not penetrate the seat or seating surface.
 - 6. Disc and shaft movement:
 - a. Design that creates a camming effect to allow the disc to swing into and out of the seat to reduce wear between the seat and seating surface.

7. Provide valves which are bubble tight at the rated pressure in either direction and suitable for operation after long periods of inactivity.
8. The valves are required to operate from the fully opened to the fully closed position without damage to the seats, valve body, or motor under the operating conditions described below.
9. Operating conditions for 30-inch diameter valves:
 - a. Maximum pressure = 85 psig.
 - b. Valves are required to function as guard valves when maintenance is needed for the fixed-cone valves.
 - c. They are positioned in either the fully open or fully closed position.
10. Provide valves which are suitable for operation with flow in either direction.
11. Mark the valves with a flow arrow which indicates the manufacturer's recommended flow direction through the valve.
12. Axis of valve leaf: horizontal when the valve is in the installed position.
13. Upstream flanged end designed to fit the existing, non-standard flange on the outlet pipe.
14. Downstream flanged ends suitable for connecting to AWWA Class D flanges.
15. All valves: Suitable for exposed service

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2.02 MOTOR OPERATORS FOR BUTTERFLY VALVES:

- A. Motor operated actuators: Motor operated actuators: Manufactured by EIM Company, Inc., 13840 Pike Road, Missouri City TX 77459; Limitorque Corporation, 5114 Woodall Road, Lynchburg VA 24502; Rotork Controls, Inc., 675 Mile Crossing Blvd., Rochester NY 14624; or equal, with the following essential characteristics:
 1. Open and closing time: Approximately 240 seconds.
 2. Provide NEMA 250, type 4X enclosures for all electrical equipment.
 3. Manufactured in accordance with AWWA C540, except as specified herein.
 4. Reduction gears.
 5. Mechanical position indicator.
 6. Electric motor:
 - a. Suitable for operation with ambient temperatures ranging from -10 degrees C to 50 degrees C.
 - b. Suitable for operation at an elevation of 200-feet.
 - c. Induction type rated 240 volts, 3 phase, 60 hertz, conforming to NEMA MG1 and designed for required service.
 - d. Enclosure: Totally-enclosed.

- e. Minimum time rating: 15 minutes.
 - f. Designed for full-voltage starting and reversing duty.
 - g. Locked-rotor torque: Not less than 250 percent of full-load torque.
 - h. Insulation: Class F or H.
7. Integral motor starter:
- a. Full-voltage, reversing type with mechanical and electrical interlocks.
 - b. Coils: 120 volts, 60 Hz.
 - c. Provide with thermal overload device. Thermal protector integral with motor may be used in lieu of motor starter overload device.
 - d. Control power transformer: Voltage: 240-120 volt, 60 Hz; Provide with primary and secondary fuses.
8. Torque switch mechanism:
- a. Integral part of motor operator which functions to protect valve and motor operator in opening and closing directions.
 - b. Independently adjustable for both travel directions.
 - c. Provide with A600 contact designation in accordance with NEMA ICS-5.
9. Limit switches:
- a. Integral part of motor operator and furnished with necessary gearing.
 - b. Provide with minimum of 8 independent limit switches set to function as follows: Two electrically independent contacts – Close at fully-open position; Two electrically independent contacts – Open at fully-open position; Two electrically independent contacts – Close at fully-closed position; Two electrically independent contacts – Open at fully-closed position.
10. Heavy-duty type.
11. Adjustable to operate at any valve position.
12. Provide with A600 contact designation in accordance with NEMA ICS-5.
13. Manual Operation Handwheel:
- a. Provide a handwheel for manual operation that does not operate during motor operation.
 - b. Failure of the motor shall not prevent manual operation.
 - c. Provide a declutching mechanism to engage the handwheel for manual operation.
14. Pushbuttons:
- a. Momentary contact type.

- b. Provide for OPEN, CLOSE, and STOP functions.
 - c. Operable from outside the enclosure.
- 15. Selector Switch:
 - a. Two position type LOCAL-REMOTE.
 - b. Operable from outside the enclosure.
- 16. Provide terminal blocks for terminating external cables.
 - a. Suitable for connection to a future SCADA system

2.03 REDUCER

- A. Reducers must meet specifications as stated in AWWA C208 or ASME B16.9.
- B. Class:
 - 1. Working pressure of reducer shall meet or exceed the maximum operating head of the valve per Paragraph 2.02 A.2.
- C. Flanges:
 - 1. Meeting requirements of AWWA C207.
- D. Gaskets:
 - 1. Meeting requirements of AWWA C207.
- E. Bolts:
 - 1. Meeting requirements of AWWA C207.
 - 2. Stainless Steel type 304 or 316 per ASTM A193.
- F. Galvanic Isolation Kits:
 - 1. Where there are dissimilar metals, galvanic isolation shall be provided.
- G. Provide lifting lugs for installation and removal of the reducers.
- H. Install all reducers in accordance with manufacturer's instructions.

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2.04 FIXED-CONE VALVE

- A. Valve Design and Fabrication:
 - 1. Suitable for free discharge.
 - 2. Maximum Operating Head: 187 feet.
 - 3. Minimum Operating Head: 77 feet.
 - 4. Diameter: 24 inches.

5. Discharge Coefficient (no hood): >0.82
6. Discharge Coefficient (hood): >0.75
7. Operation range: 0 to 100 percent open without cavitation damage or excessive vibration
8. Operating Time: 5 minutes full travel plus or minus 20 percent.
9. Ability to precisely control the free discharge flow from 0 to 150 cubic feet per second.
10. Body:
 - a. Stainless steel construction with alloy selected for high yield strength, weldability, and corrosion resistance where applicable.
 - 1) To show compliance with specifications, include ASTM standards for all materials submitted for approval under RSN 35 22 25-1.
 - b. Name plate showing year of manufacture permanently attached to valve body.
11. Sleeve:
 - a. Stainless steel construction with alloy selected for high yield strength, weldability, and corrosion resistance.
 - b. Flow is controlled by linear movement of an external stainless steel outer sleeve or cylinder that directs flow between the sleeve and the fixed cone.
 - c. Upstream seal is provided to prevent leakage between sleeve and body.
12. Fixed Cone:
 - a. Stainless steel construction with alloy selected for high yield strength, weldability, and corrosion resistance where applicable.
 - b. Inline and concentric with valve body.
 - c. Supported by ribs between the cone and valve body that are shaped to minimize vibration and to prevent cavitation damage.
13. Hood:
 - a. Stainless steel construction with alloy selected for high yield strength, weldability, and corrosion resistance.
 - b. Designed to reduce the amount of spray and to prevent direct water impingement onto the concrete structure.
 - c. Able to be installed and removed without taking the valve from the installed position.
14. Seat:
 - a. Stainless steel construction with alloy selected for high yield strength, weldability, and corrosion resistance.

- 1) Full depth of seat shall be Nitronic 60, Inconel, Hastaloy or equivalent in accordance with ASTM A479.
 - b. Conical face.
 - c. Drip-tight. Seat when in the closed position shall have a maximum seat leakage allowance of 0.4 fl. oz. per diameter inch per minute.
 - d. Replaceable without having to remove or disassemble the valve.
 - e. Self aligning/adjusting.
 - f. Designed to resist galling, corrosion, and pitting.
15. Sleeve Actuation:
- a. Actuated by twin screws and motor-operator.
 - b. Mount motor-operator above the valve pit in the same location as the existing.
 - c. Provide shafts, supports, bearings, universal joints, and gear reducers as necessary.
 - d. Provide a key for adjustment of twin screws in case of a misaligned sleeve.
16. To ensure smooth flow and avoid possible cavitation damage, it is necessary that water passages be true, smooth, and in proper alignment with no offsets into or out of the flow. Use special care in welding to avoid distortion and to ensure water-tightness.

2.05 MOTOR-OPERATORS FOR FIXED CONE VALVES

- A. Design and manufacture in accordance with AWWA C542, except:
1. Install motor operator on the fixed-cone valve in accordance with AWWA C542 and as shown on drawings.
 2. Handwheel and Declutching Mechanism:
 - a. Provide handwheel for manual operation.
 - 1) Manual operation not prevented by fused motor.
 - 2) Handwheel pull, maximum: 40 pounds when raising or lowering valve under maximum load.
 - 3) Locate centerline of handwheel between 2.5 and 3.5-feet above ground or structure on which valve operator will stand, unless otherwise dimensioned or noted on drawings.
 - b. Declutching lever:
 - 1) Place operator in manual operation by mechanically disconnecting motor drive from gear train.

- 2) Provide positive indication for remote use when handwheel is engaged for manual operation.
 - 3) Automatically return to electric drive position after manual operation.
- B. Enclose gears in a housing.
- C. Provide for lubrication of sliding and rolling surfaces.
- D. Counter or Dial Type Position Indicator:
 1. Show valve opening in inches and or percent open.
 2. Measurement divisions: 1/10-inch and or 1-percent.
- E. Operation: multi-turn shaft.
- F. Torque: Design to be suitable for operating the fixed-cone valve under maximum operating head shown on drawings and listed above.
- G. Operating Speed: 5 minutes full travel plus or minus 20 percent.
- H. Gear box(es), universal joint(s), shafts, bearing(s), and support(s) included as necessary as part of the complete fixed-cone valve.
 1. Provide a spare set of the above parts.
- I. Actuator Control Compartment:
 1. Watertight, dustproof, and O-ring sealed to meet NEMA type 4X requirements of NEMA 250.
 2. Provide thermostatically controlled, anti-condensation heater in compartment.
 3. Compartment to house motor contactors, torque and position switches, and communication modules.
- J. Actuator Termination Compartment:
 1. Provided as separate, sealed compartment.
 2. Terminals; Allow termination of conductors with crimped ring connectors.
 3. Terminal Identification: Integral part of terminal strip.
- K. Conduit Entry: Tapped for threaded conduit termination. Two entries sized to match existing conduit, bottom entry, minimum.
- L. Motor:
 1. Induction type specifically designed for valve actuator service.
 2. Rated Voltage: 240 volts, 3-phase, 60 Hz.

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3. Totally enclosed, non-ventilated (TENV)
4. Time rating, minimum: 15 minutes continuous.
5. Design motor for full-voltage starting, reversing duty.
6. Locked-rotor torque, minimum: 250 percent of full-load torque.
7. Motor stator insulation: Class F or H.
8. Winding Protection: Integral armature winding thermostats.
9. Motor Contactor: Full-voltage, electro-mechanical, reversing type.

M. Actuator Electrical Disconnect:

1. Disconnect switch or circuit breaker supplied by actuator manufacturer. Electrical rating in accordance with NFPA 70.
2. Designed to simultaneously open all phases of power between actuator and incoming electrical power supply.
3. Mounted in a separate weather proof compartment. An electrical disconnect mounted integral to the actuator is acceptable.

N. Torque Switch Mechanism:

1. Integral part of motor operator.
2. Function: Protect gate and motor operator in both opening and closing directions.
3. Torque switch:
 - a. Adjustable settings and independent adjustment for opening and closing directions.
 - b. Electronic torque switches are not permitted.

O. Limit Switch Mechanism:

1. Limit switches and associated gearing: Integral part of motor operator.
2. Contact Rating: 5 amperes, 250 volts, 60 Hz., minimum.
3. Adjustable to operate at any opening.
4. Provide at least eight independent limit switches, set to function as listed below:
 - a. Two electrically independent contacts: Close at fully-open position.
 - b. Two electrically independent contacts: Open at fully-open position.
 - c. Two electrically independent contacts: Close at fully-closed position.
 - d. Two electrically independent contacts: Open at fully-closed position.
5. Provide limit switches in motor contactor control circuit that will stop motor when valve has reached its fully open and fully closed limit.

P. Remote Position Indication:

1. Suitable for use with digital control module.
2. Provide additional contacts for transmission of potentiometer signal to SCADA.

Q. Local Controls:

1. Pushbuttons for OPEN, STOP and CLOSE. Use of control knob or switch in lieu of pushbuttons is acceptable.
2. Selector knob or switch for LOCAL and REMOTE operation.
3. LED type pilot lights to indicate valve OPEN, valve CLOSED, and valve stopped in mid-travel, minimum.
4. Provide additional contacts to operate the valve remotely by SCADA with the selector knob or switch placed in the REMOTE position.

R. Corrosion Protection:

1. All external and internal fasteners and brackets used in actuator are to be made from type 316 stainless steel in accordance with ASTM A193.
2. Printed circuit card interconnects have gold-plated connectors.

2.06 ELECTRICAL EQUIPMENT

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- A. Provide cables, conduits, terminations, and accessories required for interconnecting the equipment included in this section.
- B. Provide cable and wiring in accordance with Section 26 05 10 – Conductors and Cables.
- C. Provide conduit and conduit accessories in accordance with Section 26 05 33 – Raceways and Boxes.
- D. Electrical work and equipment shall be suitable for 240-volt, three phase, 60 hertz power supply.
- E. Drive motors and electrical enclosures shall be grounded and bonded in accordance with Section 26 05 20 – Grounding and Bonding.

2.07 FINISHES

- A. Coat in accordance with Section 09 96 20 - Coatings.

2.08 SOURCE QUALITY CONTROL

A. Shop Assembly:

1. Government Inspector will witness shop assembly, unless otherwise notified.

2. Completely shop assemble and test parts of fixed-cone valve with valve in normal position (as it will be installed) to ensure that mating parts fit properly and function correctly.
3. Check assembled valve for alignment of fluidway surfaces. Grind flush and flair offsets existing at fluidway joints.

B. Shop Tests:

1. Government Inspector will witness all tests, unless otherwise directed.
2. Seal and seat leakage and pressure test.
 - a. Per manufacturer's standard pressure test. If no standard test exists, pressurize the water to 200-percent of working pressure and hold the pressure for one-half hour while examining the valve for leaks.
 - b. If any leakage through the body or welds is found, repair or replace as directed.
 - c. Seat leakage shall not exceed drip-tight requirements per Paragraph 2.04 A.14.c.
 - d. Record results of the seal leakage and pressure test and submit results under RSN 35 22 25-2.
3. Operational test. With valve fully assembled in the normal position and adjusted, operate the valve through at least three full cycles.
 - a. Use a motor operator to operate the valve, and if practical use the motor operator to be furnished with the valve for shop testing.
 - b. Make any required adjustments or corrections until the valve operates properly.
 - c. Submit a record of operating voltages and amperages or torque required to open and close the valve under RSN 35 22 25-3.

2.09 DELIVERY, STORAGE, AND HANDLING

- A. Disassemble valve in shop only as necessary for shipment and installation.
- B. Box or crate all loose parts.
- C. Properly support and secure valve during transport and storage.
- D. Store major parts of valve with operator on timber blocking and provide adequate covers to protect valve, operator, and associated equipment from weather and damage.

PART 3 EXECUTION

3.01 REMOVAL OF EXISTING EQUIPMENT

- A. Existing equipment shall not be removed until all new equipment, parts, and materials required are approved, delivered to site, and accepted.
- B. Remove and dispose of existing fixed-cone valve and appurtenant parts. Refer to demolition drawing, 368-208-60012.

3.02 INSTALLATION

- A. Facility will be performing a penstock inspection after existing valve equipment has been removed. Contractor shall not proceed with installation of new equipment until directed by COR. Assume 3 days in schedule between removal and install of equipment.
- B. Include in installation of valve: Proper alignment and bolting of reducer to existing outlet pipe and touch-up painting as required.
- C. Install transition with gasket, fasteners, and galvanic isolation.
- D. Disassemble the valves as necessary for handling and to install the valve through the access opening through the operating floor. It is the Contractor's responsibility to accurately weigh the parts to be lifted to ensure that they do not exceed the maximum safe handling weight of hoisting equipment used.
- E. Use new gaskets in any joints disconnected. Wipe all bronze, brass, and stainless steel surfaces clean of dirt and foreign matter prior to installation. Take special care to avoid contaminating stainless steel surfaces with carbon steel. Do not use steel wool or wire brushes made of carbon steel for cleaning stainless steel. Protect stainless steel surfaces at all times against weld spatter, iron particles, or being splattered by carbon steel, as presence of carbon steel on surface of stainless steel will impair corrosion resistance.
- F. Install motor operator, shafts, supports, bearings, universal joints, gear reducers and all appurtenant parts provided to make a fully-functional valve.
- G. Electrically connect the motor operator using liquid-tight flexible metal conduit and new conductors per manufacturer's recommendations and in accordance with Sections 26 05 10 – Conductors and Cables and 26 05 33 – Raceways and Boxes.
- H. Lubricate stem and motor operator in accordance with manufacturer's instructions.

3.03 FIELD QUALITY CONTROL

- A. Government Inspector will witness all tests, unless otherwise directed.
- B. Test Scheduling. The Contractor shall notify the COR not less than three working days prior to conducting any test involving releases into the river or involving communication

with the on-site control system. The Contractor shall notify the COR not less than 24 hours prior to all other testing specified by the manufacturer or these Specifications on the valves, adapter pieces, or peripheral equipment.

- C. **Test Planning.** Contractor shall prepare a detailed plan of the required flow performance test including the specific operating points to be used, the duration of flow at each flow point, the estimated water consumed for testing in acre-feet, and the identities, responsibilities, and mobile telephone numbers of key personnel. This plan shall be submitted not less than two weeks prior to the flow test event. Contractor shall participate in a flow test event planning meeting approximately one week prior the test to coordinate the roles and responsibilities among Contractor and Government staff and the Fisheries Division staff of our cooperating agency Cachuma Operation and Maintenance Board.
- D. **Fixed-Cone Valve Testing**
1. Test after valve with operator has been installed.
 2. Seal and seat leakage test. Using available reservoir head and with the valve closed, hold the pressure for one-half hour while examining the valve for leaks. If any leakage is found, repair as directed. Record results of the seal leakage test and submit results.
 3. Operational test. With valve fully assembled and installed and the operator electrically connected and adjusted, operate the valve through at least three full cycles using the furnished motor operator. Measure and record full-travel times. Set position indicator to show exact open and closed positions of valve. Operate the valve using the handwheel sufficiently to ensure proper function. Make any required adjustments or corrections until the valve operates properly. Submit a record of operating voltages and amperages required to open and close the valve against reservoir head and description of any other observations (sounds, vibrations, and signs of abnormal operation).
 - a. Demonstrate proper mechanical function of the valves and actuators by operating the valves through their full range of travel using electrical controls as well as demonstrating proper function of the mechanical hand cranks.
 - b. Verify that the actuator torque and limit switches and position sensors are properly calibrated.
 - c. Verify that the actuator responds appropriately to external control signals to open and close the valve and accurately report valve position to external controls.
 4. Flow Performance Test. Flow testing shall be conducted after Government acceptance of dry operational testing and seal and seat testing. Using available reservoir head, Contractor shall operate the valve at a sufficient number of points, such as 10%, 25%, 50%, 75%, and 100% opening, to characterize actual flow performance of the valve.

- E. Butterfly Valve Testing
 - 1. Test after valve with operator has been installed.
 - 2. Perform 3 open and close cycles under balanced conditions.
 - 3. Perform 3 open and close cycles under unbalanced conditions.
- F. Test Report. Test report shall include the results of the dry test operational test, leakage test, voltages and amperages required to open and close the valve against reservoir head, and description of any additional observations, sounds, vibrations, and signs of abnormal operation, if any. Results of the hydraulic performance testing shall be recorded including lake elevation, head pressure upstream of each valve, and measured flow rates at each operation point. Contractor shall provide flow performance curves updated for observed flows against lake elevation presented in one or more charts.

3.04 TRAINING

- A. Conduct user familiarization and training course on the operation and maintenance of the equipment in this section.
 - 1. Trainings will be no more than four (4) hours for up to ten (10) participants.
 - 2. The Contractor shall provide digital recording of training material with RSN 35 22 25-5, Training.
- B. Coordinate with the COR to provide training at mutually agreeable date and time.
- C. The instructor(s) shall be completely knowledgeable on all installed equipment.
- D. Training Coverage: All pertinent points involved in operating, starting, stopping, and servicing the equipment, including all major elements of the Operation and Maintenance Manuals. Demonstrate all routine maintenance operations such as lubrication, general inspection, and calibration.
- E. Provide written record of trainings with RSN 35 22 25-5, Training.

END OF SECTION

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SECTION 51 00 00 INFORMATION AVAILABLE TO OFFERORS

PART 1 GENERAL

1.01 ORDER OF PRECEDENCE

- A. In case of differences between sections in Division 51 - Information Available to Offerors and sections in other divisions of the specifications, the requirements in the other divisions will govern.
- B. The Government assumes no responsibility for deductions, interpretations, or conclusions made by the Contractor based on information made available by the Government, in accordance with the clause at FAR 52.236-3 “Site Investigations and Conditions Affecting the Work.”

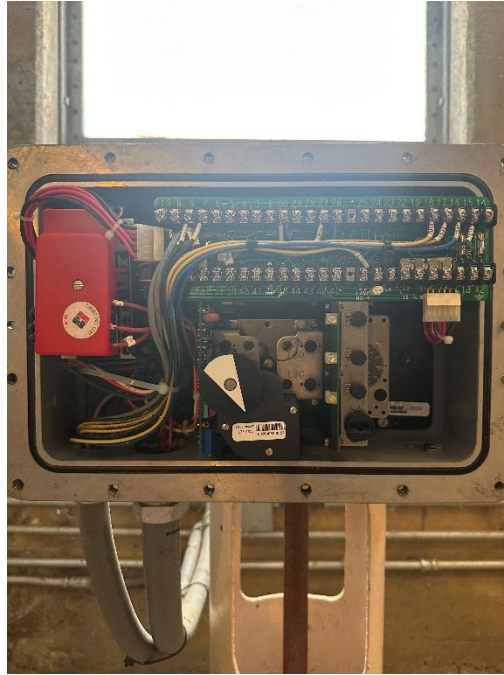
1.02 PHOTOGRAPHS

- A. Bradbury Dam – Outlet Works Building



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- 1. Photo 1 – Floor of Outlet Works Building as seen from catwalk at entrance. Left Cone Valve stem and actuator is in the photo above the partially obscured Right Cone Valve stem and actuator. Control/Data Panel is shown centered in the photo. Power Panel is not shown and located to the left, below the stairs.



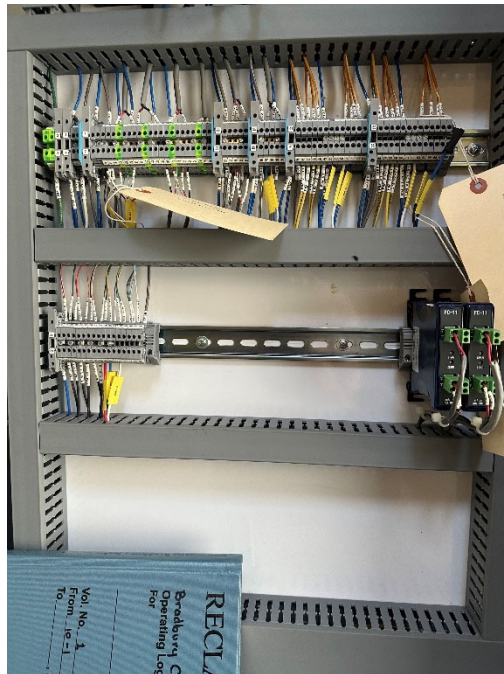
2. Photo 2 – Left Valve actuator interior layout and wiring.



3. Photo 3 – Flexible metal conduit connecting actuator to power and control conduits.



4. Photo 4 – Control Conduit entering Control Panel. Each valve has its own control conduit.



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5. Photo 5 – Control/Data Panel interior.



6. Photo 6 – Power Conduit continues to Power Panel.

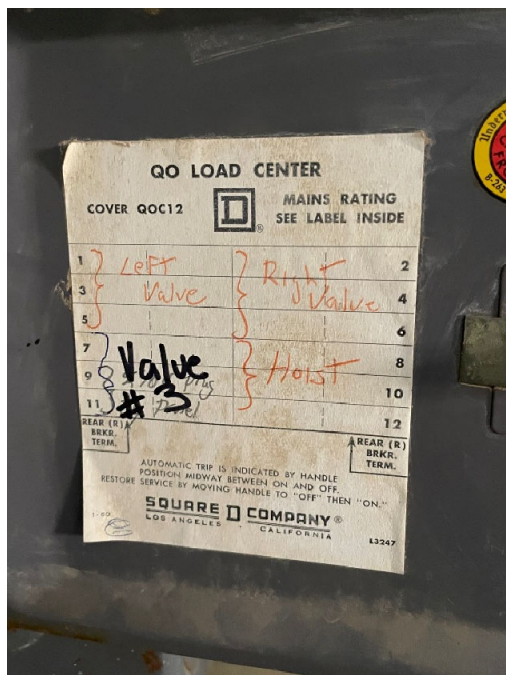


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7. Photo 7 – Power Conduit entering Power Panel (with “HOIST DISCONNECT” sticker).



8. Photo 8 – Power Panel interior.



9. Photo 9 – Power Panel directory inside of door.

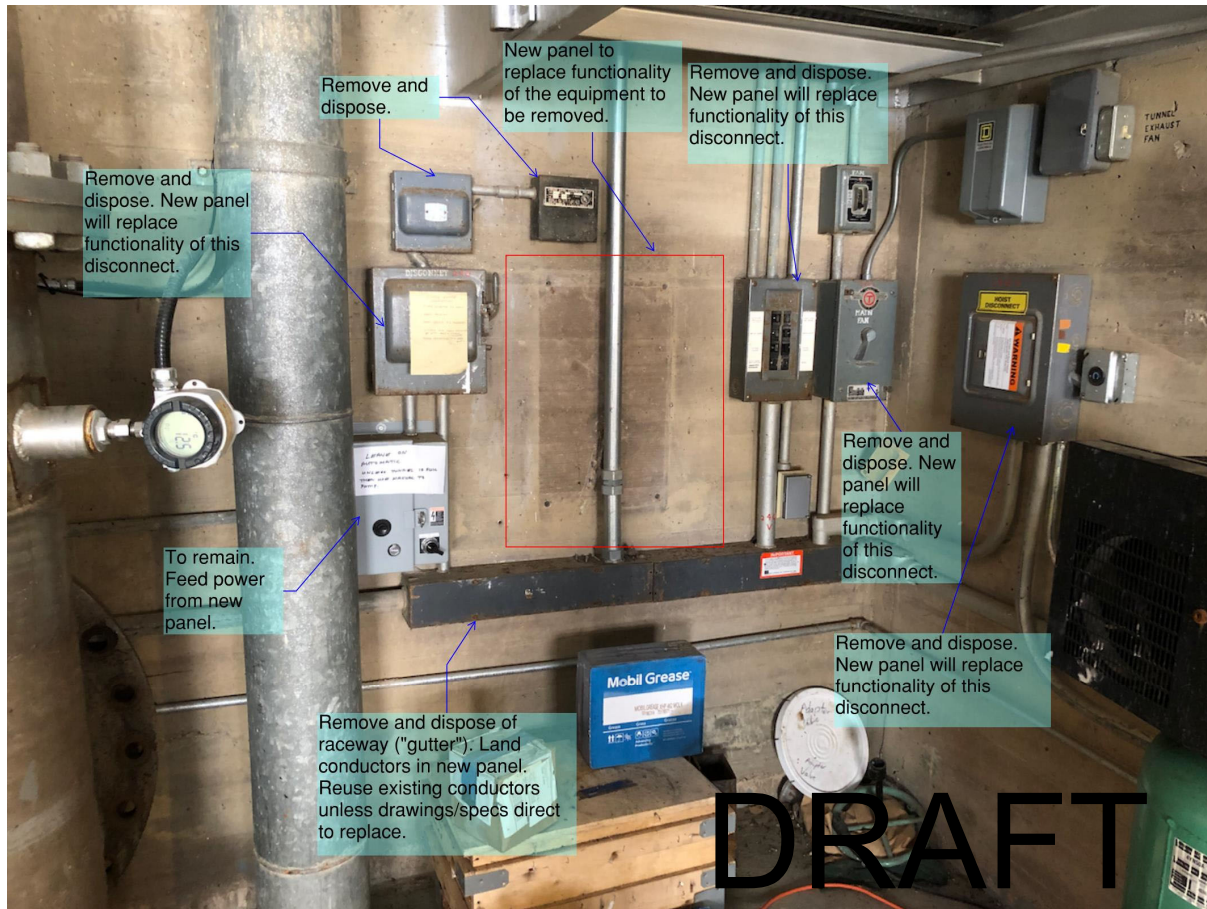
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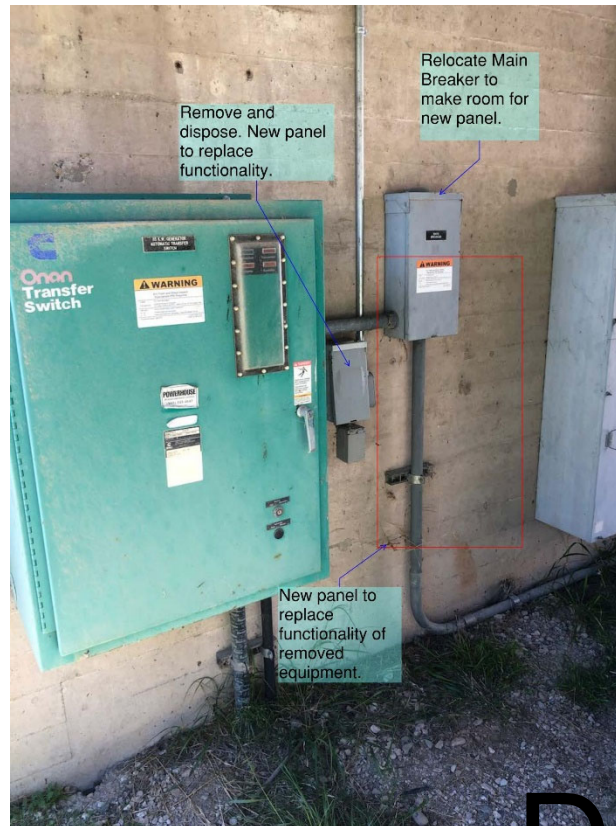
10. Photo 10 – 5-ton government crane controlled by pendant.



11. Photo 11 – Side view of 5-ton government crane controlled by pendant.



12. Photo 12 – Markup of baseline electrical design work showing equipment to be removed and location of new panel inside of Outlet Works building.



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13. Photo 13 – Markup of baseline electrical design work showing equipment to be relocated or removed and location of new panel outside of Outlet Works building.



14. Photo 14 – Markup of baseline electrical design work showing equipment to be removed.



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15. Photo 15 – Markup of baseline electrical design work showing equipment to be removed.



16. Photo 16 – Overhead view of Bradbury Dam and Spillway with potential staging area for the Temporary Bypass Pump System.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

END OF SECTION

SECTION 51 00 50 ELECTRONIC MEDIA RELEASE

U. S. BUREAU OF RECLAMATION ELECTRONIC MEDIA RELEASE

U.S. Bureau of Reclamation, Contract Number/Solicitation Number _____
This release for electronic media is dated the ____ day of _____ (month), ____ (year), between
_____ (RECIPIENT) and U.S. Bureau of Reclamation, for the
exchange of electronic media (CD ROMs, DVD, Flash Drives, External Hard Disk Drives, etc.) containing
information on the _____ project (hereinafter referred to as the PROJECT)
for use by the RECIPIENT.

Therefore, RECIPIENT and U.S Bureau of Reclamation (Reclamation), agree as follows:

1. All data and intellectual property contained on the electronic files remains the property of U.S. Bureau of Reclamation, and may not be copied or distributed to any other party without the written permission of U.S. Bureau of Reclamation.
2. The electronic files provided to RECIPIENT by U.S. Bureau of Reclamation, for the PROJECT may be used or adapted by RECIPIENT for their own use. If RECIPIENT chooses to alter in any way, in whole or in part, the electronic files provided for the PROJECT or any future project(s), RECIPIENT agrees that the use of the altered files shall be without liability or legal exposure to U.S. Bureau of Reclamation.
3. Because information and data provided electronically may be altered, whether inadvertently or otherwise, U.S. Bureau of Reclamation, reserves the right to retain copies of the electronic file(s) and to remove from the electronic files provided to RECIPIENT all identification (such as logo, surveyor's seal, etc.) reflecting the involvement of U.S. Bureau of Reclamation in their preparation.
4. The electronic files are provided solely as a convenience to RECIPIENT by U.S. Bureau of Reclamation, and shall not be considered "Drawings of Record" or "Construction Documents." All documents considered "Drawings of Record" or "Construction Documents" will be hardcopy drawing sheets with signatures or scanned files of the hardcopy drawings with signatures.
5. RECIPIENT is advised to check all electronic media for viruses and malware before loading the files. RECIPIENT is solely responsible for intercepting and disabling viruses, if any, that may be transmitted with the electronic files and hereby agrees to indemnify and hold U.S. Bureau of Reclamation harmless from and against all claims of any type or nature asserted by RECIPIENT or any party as a result of viruses transmitted with the electronic files.
6. Files distributed electronically are subject to data erosion, erasure, and/or alteration, and computer systems and software become obsolete in time. By accepting these electronic files, RECIPIENT acknowledges these risks and agrees to waive all claims against U.S. Bureau of Reclamation, should data erosion, erasure, and/or alteration of these electronic files occur.
7. RECIPIENT agrees to defend, indemnify, and hold U.S. Bureau of Reclamation, harmless from all claims, injuries, damages, losses, expenses, and costs, including attorney's fees, arising out of breach of this agreement, or the modification or reuse of the electronic files.

ACCEPTED FOR RECIPIENT:

By: _____
Title: _____
Date: _____

RELEASED BY U.S. BUREAU OF RECLAMATION:

By: _____
Title: _____
Date: _____

END OF SECTION

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SECTION 52 00 00

DRAWINGS

PART 1 GENERAL

1.01 DISCREPANCIES, ERRORS, OR OMISSIONS

- A. Inform the CO of discrepancies discovered on drawings in accordance with clause at FAR 52.236-21, Specifications and Drawings for Construction.
- B. In accordance with clause at FAR 52.236-21, Specifications and Drawings for Construction, in case of discrepancies, written specifications take precedence over drawings unless otherwise specified.

1.02 PROJECT CONDITIONS

- A. Where there are differences as determined by the CO between details and dimensions shown on drawings and details and dimensions of existing features at jobsite, use details and dimensions of existing features at jobsite.

1.03 SPECIFICATION DRAWINGS

- A. Electronic versions (AutoCAD Format Release 2018) of the following drawings are available to offerors from the COR:
 - 1. 368-208-60011, -60012, -60013, -60014, -60015, -60016, -60017, -60018, and 40-D-7102.

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1.04 INFORMATION DRAWINGS

- A. Drawings marked "For Information Only" in the drawing list are included to show existing features about which knowledge is required to perform work under this contract. These drawings do not show work to be performed under this contract.
- B. If there are differences as determined by the CO between details and dimensions shown on these drawings and those of existing features at jobsite, use details and dimensions of existing features at jobsite.

1.05 STANDARD DRAWINGS

- A. Standard drawings may show details which are not a part of work under this contract. Disregard details shown on these drawings which are not applicable to work under this contract.

1.06 LIST OF DRAWINGS

- A. Drawings listed in Table 52 00 00A - List of Drawings, are made a part of Section C - Description/Specifications.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

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Table 52 00 00A - List of Drawings

Sheet No.	Drawing No.	Title
1	368-208-60008	Fixed-Cone Valve Replacement – Location and Vicinity Maps
2	368-208-60009	Access from Highway 154 to Spillway – General Plan
3	368-208-60010	Access from Spillway to Outlet Works Control House – General Plan
4	368-208-60011	Control House – Plan and Elevation
5	368-208-60012	Demolition – Sections and Detail
6	368-208-60013	Fixed-Cone Valve Installation – Sections and Detail
7	368-208-60014	Reducer Support – Sections and Detail
8	368-208-60015	Valve Support – Sections and Detail
9	368-208-60016	30x24 Reducer – Sections and Detail
10	368-208-60017	Unverified Existing Electrical System – Sketch
11	368-208-60018	Unverified Electrical Baseline Design – Sketch
Information Drawings:		
12	368-D-98	Cachuma Dam – Outlet Works – Plan and Sections
13	368-D-99	Cachuma Dam – Outlet Works – Intake Structure and Gate Chamber Details
14	368-D-100	Cachuma Dam – Outlet Works – Control House Details
15	368-D-115	Cachuma Dam – Outlet Works – 38” Steel Pipe – Plan and Elevation
16	368-D-116	Cachuma Dam – Outlet Works – 38” Steel Pipe – Wye-Branch and Support Ring
17	368-D-122	Cachuma Dam – Outlet Works – Venturi Rings and Piping Installation
18	368-D-205	Cachuma Dam – Outlet Works – Electrical Installation
19	368-208-500	Bradbury Dam – Outlet Works Control House – Electrical Installation – Modification

Table 52 00 00A - List of Drawings

Sheet No.	Drawing No.	Title
20	368-208-401	Cachuma Dam – Outlet Works Control House – Electrical Installation – Hollow Jet Valve Control
21	368-D-1073	Bradbury Dam – Electrical Installation – Spillway Plan View – Buried Cable Layout
22	368-D-1075	Bradbury Dam – Electrical Installation – Buildings Plan & Section Views – Conduit & Equipment Layouts
23	368-D-1077	Bradbury Dam – Electrical Installation – Outlet Works & Dam Tenders Houses – 120/240 Volt Distribution – Three-Line Diagram
24	368-D-1139	Bradbury Dam – Electrical Installation – Power Distribution – Single-Line Diagram
25	61060	EIM Controls – Wiring Diagram
Standard Drawings:		
26	40-D-4334	Electrical Installation – Typical Grounding Details
27	40-D-4335	Electrical Installation – Typical Grounding Details
28	40-D-6234	Standard Nameplates
29	40-D-7094	Electrical Installation – Conduit Clamp and Conduit Supports – Typical Details
30	104-D-757	Pumping Plant – Electrical Installations – Designations and Symbols
31	104-D-1165	Typical Powerplant/Pumping Plant – Electrical Installation – Main Control Board CSA – Panel 7R – Wiring Diagram
32	40-D-7102	Government Drawing Format - Drawing Border, Sample Title Block and Signature Lines

END OF SECTION